ATMOSPHERIC AND OCEANIC FLUID DYNAMICS

Supplementary Material for 2nd Edition

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Contents

Preface

This is a sample document taken from the book *Atmospheric and Oceanic Fluid Dynamics* by G. K. Vallis. This version uses Georgia for text, Cambria (from Microsoft) for math and Lucida Sans for sans serif. The following style file was used to set the fonts. Ignore the last few lines.

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Part I

FUNDAMENTALS OF GEOPHYSICAL FLUID DYNAMICS

If a body is moving in any direction, there is a force, arising from the Earth's rotation, which always deflects it to the right in the northern hemisphere, and to the left in the southern.

William Ferrel, *The influence of the Earth's rotation upon the relative motion of bodies near its surface,* 1858.

CHAPTER TWO

Effects of Rotation and Stratification

THE ATMOSPHERE AND OCEAN are shallow layers of fluid on a sphere in that their thickness or depth is much less than their horizontal extent. Furthermore, their motion is strongly influenced by two effects: rotation and stratification, the latter meaning that there is a mean vertical gradient of (potential) density that is often large compared with the horizontal gradient. Here we consider how the equations of motion are affected by these effects. First, we consider some elementary effects of rotation on a fluid and derive the Coriolis and centrifugal forces, and then we write down the equations of motion appropriate for motion on a sphere. Then we discuss some approximations to the equations of motion that are appropriate for large-scale flow in the ocean and atmosphere, in particular the hydrostatic and geostrophic approximations. Following this we discuss gravity waves, a particular kind of wave motion that is enabled by the presence of stratification, and finally we talk about how rotation leads to the production of certain types of boundary layers — Ekman layers — in rotating fluids.

2.1 THE EQUATIONS OF MOTION IN A ROTATING FRAME OF REFERENCE

Newton's second law of motion, that the acceleration on a body is proportional to the imposed force divided by the body's mass, applies in so-called inertial frames of reference. The Earth rotates with a period of almost 24 hours (23h 56m) relative to the distant stars, and the surface of the Earth manifestly is not, in that sense, an inertial frame. Nevertheless, because the surface of the Earth is moving (in fact at speeds of up to a few hundreds of metres per second) it is very convenient to describe the flow relative to the Earth's surface, rather than in some inertial frame. This necessitates recasting the equations into a form that is appropriate for a rotating frame of reference, and that is the subject of this section.

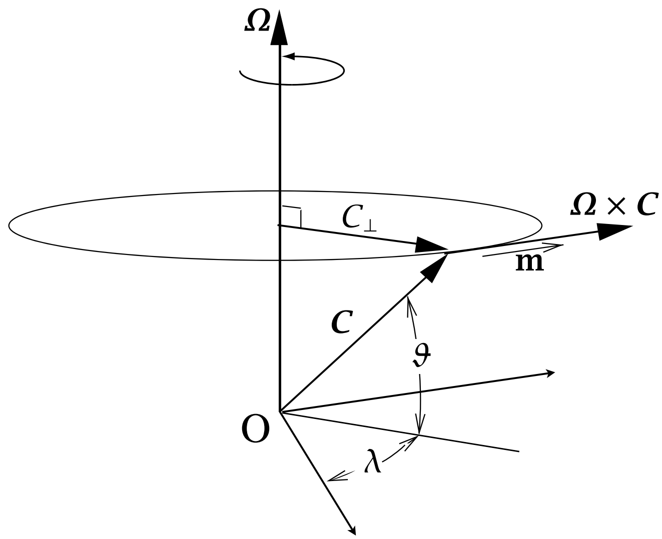

Fig. 2.1 A vector $\boldsymbol{C}$ rotating at an angular velocity $\boldsymbol{\Omega}$. It appears to be a constant vector in the rotating frame, whereas in the inertial frame it evolves according to $(\mathrm{d}\boldsymbol{C}/\mathrm{d}t)_I = \boldsymbol{\Omega} \times \boldsymbol{C}$.

2.1.1 Rate of change of a vector

Consider first a vector $\boldsymbol{C}$ of constant length rotating relative to an inertial frame at a constant angular velocity $\boldsymbol{\Omega}$. Then, in a frame rotating with that same angular velocity it appears stationary and constant. If in a small interval of time δt the vector $\boldsymbol{C}$ rotates through a small angle $\delta\lambda$ then the change in $\boldsymbol{C}$, as perceived in the inertial frame, is given by (see Fig. 2.1)

$$\delta \boldsymbol{C} = |\boldsymbol{C}| \cos\vartheta \, \delta\lambda \, \boldsymbol{m}, \tag{2.1}$$

where the vector $\boldsymbol{m}$ is the unit vector in the direction of change of $\boldsymbol{C}$, which is perpendicular to both $\boldsymbol{C}$ and $\boldsymbol{\Omega}$. But the rate of change of the angle λ is just, by definition, the angular velocity so that $\delta\lambda = |\boldsymbol{\Omega}|\delta t$ and

$$\delta \boldsymbol{C} = |\boldsymbol{C}||\boldsymbol{\Omega}| \sin\hat{\vartheta} \, \boldsymbol{m} \, \delta t = \boldsymbol{\Omega} \times \boldsymbol{C} \, \delta t. \tag{2.2}$$

using the definition of the vector cross product, where $\hat{\vartheta} = (\pi/2 - \vartheta)$ is the angle between $\boldsymbol{\Omega}$ and $\boldsymbol{C}$. Thus

$$\left(\frac{\mathrm{d}\boldsymbol{C}}{\mathrm{d}t}\right)_I = \boldsymbol{\Omega} \times \boldsymbol{C}, \tag{2.3}$$

where the left-hand side is the rate of change of $\boldsymbol{C}$ as perceived in the inertial frame.

Now consider a vector $\boldsymbol{B}$ that changes in the inertial frame. In a small time δt the change in $\boldsymbol{B}$ as seen in the rotating frame is related to the change seen in the inertial frame by

$$(\delta \boldsymbol{B})_I = (\delta \boldsymbol{B})_R + (\delta \boldsymbol{B})_{rot}, \tag{2.4}$$

where the terms are, respectively, the change seen in the inertial frame, the change due to the vector itself changing as measured in the rotating frame, and the change due to the rotation. Using (2.2) $(\delta \boldsymbol{B})_{rot} = \boldsymbol{\Omega} \times \boldsymbol{B} \, \delta t$, and so the rates of change of the vector $\boldsymbol{B}$ in the inertial and rotating frames are related by

$$\boxed{\left(\frac{\mathrm{d}\boldsymbol{B}}{\mathrm{d}t}\right)_I = \left(\frac{\mathrm{d}\boldsymbol{B}}{\mathrm{d}t}\right)_R + \boldsymbol{\Omega} \times \boldsymbol{B}}\,. \tag{2.5}$$

This relation applies to a vector $\boldsymbol{B}$ that, as measured at any one time, is the same in both inertial and rotating frames.

2.1.2 Velocity and acceleration in a rotating frame

The velocity of a body is not measured to be the same in the inertial and rotating frames, so care must be taken when applying (2.5) to velocity. First apply (2.5) to $\boldsymbol{r}$, the position of a particle to obtain

$$\left(\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\right)_I = \left(\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\right)_R + \boldsymbol{\Omega}\times\boldsymbol{r} \tag{2.6}$$

or

$$\boldsymbol{v}_I = \boldsymbol{v}_R + \boldsymbol{\Omega}\times\boldsymbol{r}. \tag{2.7}$$

We refer to $\boldsymbol{v}_R$ and $\boldsymbol{v}_I$ as the relative and inertial velocity, respectively, and (2.7) relates the two. Apply (2.5) again, this time to the velocity $\boldsymbol{v}_R$ to give

$$\left(\frac{\mathrm{d}\boldsymbol{v}_R}{\mathrm{d}t}\right)_I = \left(\frac{\mathrm{d}\boldsymbol{v}_R}{\mathrm{d}t}\right)_R + \boldsymbol{\Omega}\times\boldsymbol{v}_R, \tag{2.8}$$

or, using (2.7)

$$\left(\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{v}_I - \boldsymbol{\Omega}\times\boldsymbol{r})\right)_I = \left(\frac{\mathrm{d}\boldsymbol{v}_R}{\mathrm{d}t}\right)_R + \boldsymbol{\Omega}\times\boldsymbol{v}_R, \tag{2.9}$$

or

$$\left(\frac{\mathrm{d}\boldsymbol{v}_I}{\mathrm{d}t}\right)_I = \left(\frac{\mathrm{d}\boldsymbol{v}_R}{\mathrm{d}t}\right)_R + \boldsymbol{\Omega}\times\boldsymbol{v}_R + \frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t}\times\boldsymbol{r} + \boldsymbol{\Omega}\times\left(\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\right)_I. \tag{2.10}$$

Then, noting that

$$\left(\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\right)_I = \left(\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\right)_R + \boldsymbol{\Omega}\times\boldsymbol{r} = (\boldsymbol{v}_R + \boldsymbol{\Omega}\times\boldsymbol{r}), \tag{2.11}$$

and assuming that the rate of rotation is constant, (2.10) becomes

$$\left(\frac{\mathrm{d}\boldsymbol{v}_R}{\mathrm{d}t}\right)_R = \left(\frac{\mathrm{d}\boldsymbol{v}_I}{\mathrm{d}t}\right)_I - 2\boldsymbol{\Omega}\times\boldsymbol{v}_R - \boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}). \tag{2.12}$$

This equation may be interpreted as follows. The term on the left-hand side is the rate of change of the relative velocity as measured in the rotating frame. The first term on the right-hand side is the rate of change of the inertial velocity as measured in the inertial frame (or, loosely, the inertial acceleration). Thus, by Newton's second law, it is equal to the force on a fluid parcel divided by its mass. The second and third terms on the right-hand side (including the minus signs) are the *Coriolis force* and the *centrifugal force* per unit mass. Neither of these are true forces — they may be thought of as quasi-forces (i.e., 'as if' forces); that is, when a body is observed from a rotating frame it seems to behave as if unseen forces are present that affect its motion. If (2.12) is written, as is common, with the terms $+2\boldsymbol{\Omega}\times\boldsymbol{v}_r$ and $+\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r})$ on the left-hand side then these terms should be referred to as the Coriolis and centrifugal *accelerations*.[1]

Centrifugal force

If $\boldsymbol{r}_\perp$ is the perpendicular distance from the axis of rotation (see Fig. 2.1 and substitute $\boldsymbol{r}$ for $\boldsymbol{C}$), then, because $\boldsymbol{\Omega}$ is perpendicular to $\boldsymbol{r}_\perp$, $\boldsymbol{\Omega}\times\boldsymbol{r} = \boldsymbol{\Omega}\times\boldsymbol{r}_\perp$. Then, using the vector identity $\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}_\perp) = (\boldsymbol{\Omega}\cdot\boldsymbol{r}_\perp)\boldsymbol{\Omega} - (\boldsymbol{\Omega}\cdot\boldsymbol{\Omega})\boldsymbol{r}_\perp$ and noting that the first term is zero, we see that the centrifugal force per unit mass is just given by

$$\boldsymbol{F}_{ce} = -\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}) = \Omega^2\boldsymbol{r}_\perp. \tag{2.13}$$

This may usefully be written as the gradient of a scalar potential,

$$\boldsymbol{F}_{ce} = -\nabla\Phi_{ce}. \tag{2.14}$$

where $\Phi_{ce} = -(\Omega^2 r_\perp^2)/2 = -(\boldsymbol{\Omega}\times\boldsymbol{r}_\perp)^2/2$.

Coriolis force

The Coriolis force per unit mass is:

$$\boldsymbol{F}_{Co} = -2\boldsymbol{\Omega}\times\boldsymbol{v}_R. \tag{2.15}$$

It plays a central role in much of geophysical fluid dynamics and will be considered extensively later on. For now, we just note three basic properties.

(i) There is no Coriolis force on bodies that are stationary in the rotating frame.

(ii) The Coriolis force acts to deflect moving bodies at right angles to their direction of travel.

(iii) The Coriolis force does no work on a body because it is perpendicular to the velocity, and so $\boldsymbol{v}_R\cdot(\boldsymbol{\Omega}\times\boldsymbol{v}_R) = 0$.

2.1.3 Momentum equation in a rotating frame

Since (2.12) simply relates the accelerations of a particle in the inertial and rotating frames, then in the rotating frame of reference the momentum equation may be written

$$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + 2\boldsymbol{\Omega}\times\boldsymbol{v} = -\frac{1}{\rho}\nabla p - \nabla\Phi, \tag{2.16}$$

incorporating the centrifugal term into the potential, Φ. We have dropped the subscript R; henceforth, unless we need to be explicit, all velocities without a subscript will be considered to be relative to the rotating frame.

2.1.4 Mass and tracer conservation in a rotating frame

Let ϕ be a scalar field that, in the inertial frame, obeys

$$\frac{\mathrm{D}\phi}{\mathrm{D}t} + \phi\nabla\cdot\boldsymbol{v}_I = 0. \tag{2.17}$$

Now, observers in both the rotating and inertial frame measure the same value of ϕ. Further, $\mathrm{D}\phi/\mathrm{D}t$ is simply the rate of change of ϕ associated with a material parcel, and therefore is reference frame invariant. Thus,

$$\left(\frac{\mathrm{D}\phi}{\mathrm{D}t}\right)_R = \left(\frac{\mathrm{D}\phi}{\mathrm{D}t}\right)_I, \tag{2.18}$$

where $(\mathrm{D}\phi/\mathrm{D}t)_R = (\partial\phi/\partial t)_R + \boldsymbol{v}_R \cdot \nabla\phi$ and $(\mathrm{D}\phi/\mathrm{D}t)_I = (\partial\phi/\partial t)_I + \boldsymbol{v}_I \cdot \nabla\phi$ and the local temporal derivatives $(\partial\phi/\partial t)_R$ and $(\partial\phi/\partial t)_I$ are evaluated at fixed locations in the rotating and inertial frames, respectively.

Further, since $\boldsymbol{v} = \boldsymbol{v}_I - \boldsymbol{\Omega} \times \boldsymbol{r}$, we have that

$$\nabla \cdot \boldsymbol{v}_I = \nabla \cdot (\boldsymbol{v}_I - \boldsymbol{\Omega} \times \boldsymbol{r}) = \nabla \cdot \boldsymbol{v}_R \tag{2.19}$$

since $\nabla \cdot (\boldsymbol{\Omega} \times \boldsymbol{r}) = 0$. Thus, using (2.18) and (2.19), (2.17) is equivalent to

$$\frac{\mathrm{D}\phi}{\mathrm{D}t} + \phi \nabla \cdot \boldsymbol{v} = 0, \tag{2.20}$$

where all observables are measured in the *rotating* frame. Thus, the equation for the evolution of a scalar whose measured value is the same in rotating and inertial frames is unaltered by the presence of rotation. In particular, the mass conservation equation is unaltered by the presence of rotation.

Although we have taken (2.18) as true a priori, the individual components of the material derivative differ in the rotating and inertial frames. In particular

$$\left(\frac{\partial\phi}{\partial t}\right)_I = \left(\frac{\partial\phi}{\partial t}\right)_R - (\boldsymbol{\Omega} \times \boldsymbol{r}) \cdot \nabla\phi \tag{2.21}$$

because $\boldsymbol{\Omega} \times \boldsymbol{r}$ is the velocity, in the inertial frame, of a uniformly rotating body. Similarly,

$$\boldsymbol{v}_I \cdot \nabla\phi = (\boldsymbol{v}_R + \boldsymbol{\Omega} \times \boldsymbol{r}) \cdot \nabla\phi. \tag{2.22}$$

Adding the last two equations reprises and confirms (2.18).

2.2 EQUATIONS OF MOTION IN SPHERICAL COORDINATES

The Earth is very nearly spherical and it might appear obvious that we should cast our equations in spherical coordinates. Although this does turn out to be true, the presence of a centrifugal force causes some complications that we should first discuss. The reader who is willing ab initio to treat the Earth as a perfect sphere and to neglect the horizontal component of the centrifugal force may skip the next section.

2.2.1◆ The centrifugal force and spherical coordinates

The centrifugal force is a potential force, like gravity, and so we may therefore define an 'effective gravity' equal to the sum of the true, or Newtonian, gravity and the centrifugal force. The Newtonian gravitational force is directed approximately toward the centre of the Earth, with small deviations due mainly to the Earth's oblateness. The line of action of the effective gravity will in general differ slightly from this, and therefore have a component in the 'horizontal' plane, that is the plane perpendicular to the radial direction. The magnitude of the centrifugal force is $\Omega^2 r_\perp$, and so the effective gravity is given by

$$\boldsymbol{g} \equiv \boldsymbol{g}_{eff} = \boldsymbol{g}_{grav} + \Omega^2 \boldsymbol{r}_\perp, \tag{2.23}$$

where $\boldsymbol{g}_{grav}$ is the Newtonian gravitational force due to the gravitational attraction of the Earth and $\boldsymbol{r}_\perp$ is normal to the rotation vector (in the direction $\boldsymbol{C}$ in Fig. 2.2), with $r_\perp = r\cos\vartheta$.

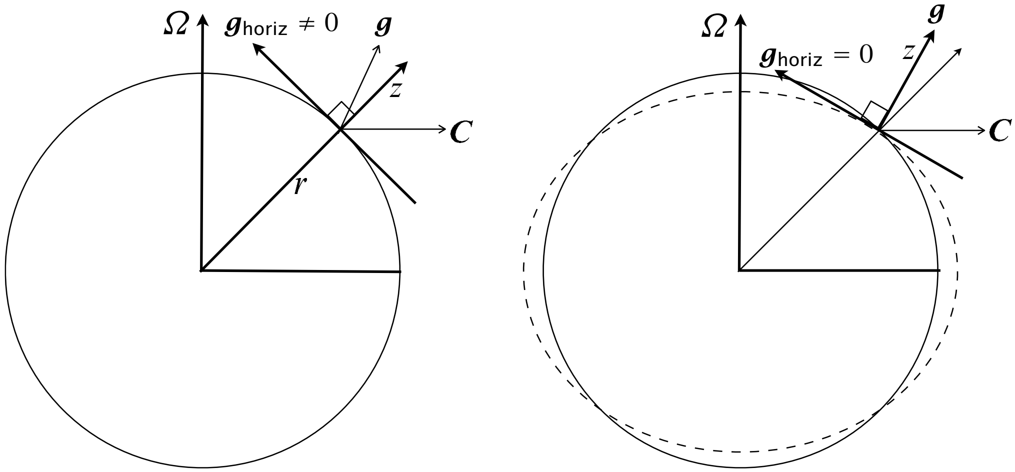

Fig. 2.2 Left: directions of forces and coordinates in true spherical geometry. $\boldsymbol{g}$ is the effective gravity (including the centrifugal force, $\boldsymbol{C}$) and its horizontal component is evidently non-zero. Right: a modified coordinate system, in which the vertical direction is defined by the direction of $\boldsymbol{g}$, and so the horizontal component of $\boldsymbol{g}$ is identically zero. The dashed line schematically indicates a surface of constant geopotential. The differences between the direction of $\boldsymbol{g}$ and the direction of the radial coordinate, and between the sphere and the geopotential surface, are much exaggerated and in reality are similar to the thickness of the lines themselves.

Both gravity and centrifugal force are potential forces and therefore we may define the *geopotential,* Φ, such that

$$\boldsymbol{g} = -\nabla \Phi. \tag{2.24}$$

Surfaces of constant Φ are not quite spherical because $r_\perp$, and hence the centrifugal force, vary with latitude (Fig. 2.2); this has certain ramifications, as we now discuss.

The components of the centrifugal force parallel and perpendicular to the radial direction are $\Omega^2 r \cos^2 \vartheta$ and $\Omega^2 r \cos\vartheta \sin\vartheta$. Newtonian gravity is much larger than either of these, and at the Earth's surface the ratio of centrifugal to gravitational terms is approximately, and no more than,

$$\alpha \approx \frac{\Omega^2 a}{g} \approx \frac{(7.27 \times 10^{-5})^2 \times 6.4 \times 10^6}{10} \approx 3 \times 10^{-3}. \tag{2.25}$$

(Note that at the equator and pole the horizontal component of the centrifugal force is zero and the effective gravity is aligned with Newtonian gravity.) The angle between $\boldsymbol{g}$ and the line to the centre of the Earth is given by a similar expression and so is also small, typically around 3×10^{-3} radians. However, the horizontal component of the centrifugal force is still large compared to the Coriolis force, their ratio in mid-latitudes being given by

$$\frac{\text{horizontal centrifugal force}}{\text{Coriolis force}} \approx \frac{\Omega^2 a \cos\vartheta \sin\vartheta}{2\Omega u} \approx \frac{\Omega a}{4|u|} \approx 10, \tag{2.26}$$

using $u = 10\,\mathrm{m\,s^{-1}}$. The centrifugal term therefore dominates over the Coriolis term, and is largely balanced by a pressure gradient force. Thus, if we adhered to true spherical coordinates, both the horizontal and radial components of the momentum equation would be dominated by a static balance between a pressure gradient and gravity or centrifugal terms. Although in principle there is nothing wrong with writing the equations this way, it obscures the dynamical balances involving the Coriolis force and pressure that determine the large-scale horizontal flow.

A way around this problem is to use the direction of the geopotential force to *define* the vertical direction, and then for all geometric purposes to regard the surfaces of constant Φ as if they were true spheres.[2] The horizontal component of effective gravity is then identically zero, and we have traded a potentially large dynamical error for a very small geometric error. In fact, over time, the Earth has developed an equatorial bulge to compensate for and neutralize the centrifugal force, so that the effective gravity does act in a direction virtually normal to the Earth's surface; that is, the surface of the Earth is an oblate spheroid of nearly constant geopotential. The geopotential Φ is then a function of the vertical coordinate alone, and for many purposes we can just take $\Phi = gz$; that is, the direction normal to geopotential surfaces, the local vertical, is, in this approximation, taken to be the direction of increasing r in spherical coordinates. It is because the oblateness is very small (the polar diameter is about 12 714 km, whereas the equatorial diameter is about 12 756 km) that using spherical coordinates is a very accurate way to map the spheroid. If the angle between effective gravity and a natural direction of the coordinate system were not small then more heroic measures would be called for.

If the solid Earth did not bulge at the equator, the *behaviour* of the atmosphere and ocean would differ significantly from that of the present system. For example, the surface of the ocean is nearly a geopotential surface, and if the solid Earth were exactly spherical then the ocean would perforce become much deeper at low latitudes and the ocean basins would dry out completely at high latitudes. We could still choose to use the spherical coordinate system discussed above to describe the dynamics, but the shape of the surface of the solid Earth would have to be represented by a topography, with the topographic height increasing monotonically polewards nearly everywhere.

2.2.2 Some identities in spherical coordinates

The location of a point is given by the coordinates (λ, ϑ, r) where λ is the angular distance eastwards (i.e., longitude), ϑ is angular distance polewards (i.e., latitude) and r is the radial distance from the centre of the Earth — see Fig. 2.3. (In some other fields of study co-latitude is used as a spherical coordinate.) If a is the radius of the Earth, then we also define $z = r - a$. At a given location we may also define the Cartesian increments $(\delta x, \delta y, \delta z) = (r\cos\vartheta\,\delta\lambda, r\delta\vartheta, \delta r)$.

For a scalar quantity ϕ the material derivative in spherical coordinates is

$$\frac{\mathrm{D}\phi}{\mathrm{D}t} = \frac{\partial \phi}{\partial t} + \frac{u}{r\cos\vartheta}\frac{\partial \phi}{\partial \lambda} + \frac{v}{r}\frac{\partial \phi}{\partial \vartheta} + w\frac{\partial \phi}{\partial r}, \tag{2.27}$$

where the velocity components corresponding to the coordinates (λ, ϑ, r) are

$$(u, v, w) \equiv \left(r\cos\vartheta \frac{\mathrm{D}\lambda}{\mathrm{D}t}, r\frac{\mathrm{D}\vartheta}{\mathrm{D}t}, \frac{\mathrm{D}r}{\mathrm{D}t} \right). \tag{2.28}$$

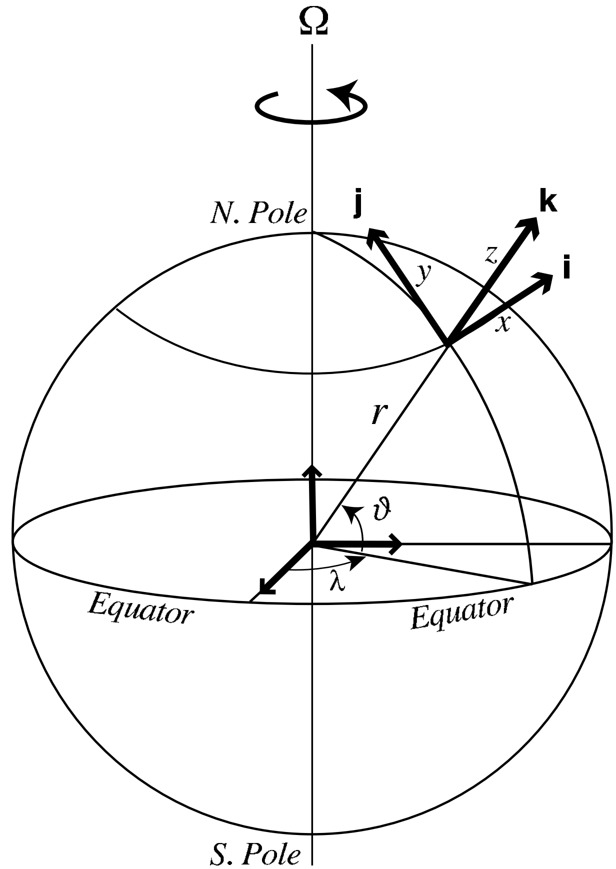

Fig. 2.3 The spherical coordinate system. The orthogonal unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ point in the direction of increasing longitude λ, latitude ϑ, and altitude z. Locally, one may apply a Cartesian system with variables x, y and z measuring distances along $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.

That is, u is the zonal velocity, v is the meridional velocity and w is the vertical velocity. If we define $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ to be the unit vectors in the direction of increasing (λ, ϑ, r) then

$$\boldsymbol{v} = \mathbf{i}u + \mathbf{j}v + \mathbf{k}w. \tag{2.29}$$

Note also that $\mathrm{D}r/\mathrm{D}t = \mathrm{D}z/\mathrm{D}t$.

The divergence of a vector $\boldsymbol{B} = \mathbf{i}\,B^\lambda + \mathbf{j}\,B^\vartheta + \mathbf{k}\,B^r$ is

$$\nabla\cdot\boldsymbol{B} = \frac{1}{\cos\vartheta}\left[\frac{1}{r}\frac{\partial B^\lambda}{\partial\lambda} + \frac{1}{r}\frac{\partial}{\partial\vartheta}(B^\vartheta\cos\vartheta) + \frac{\cos\vartheta}{r^2}\frac{\partial}{\partial r}(r^2 B^r)\right]. \tag{2.30}$$

The vector gradient of a scalar is:

$$\nabla\phi = \mathbf{i}\frac{1}{r\cos\vartheta}\frac{\partial\phi}{\partial\lambda} + \mathbf{j}\frac{1}{r}\frac{\partial\phi}{\partial\vartheta} + \mathbf{k}\frac{\partial\phi}{\partial r}. \tag{2.31}$$

The Laplacian of a scalar is:

$$\nabla^2\phi \equiv \nabla\cdot\nabla\phi = \frac{1}{r^2\cos\vartheta}\left[\frac{1}{\cos\vartheta}\frac{\partial^2\phi}{\partial\lambda^2} + \frac{\partial}{\partial\vartheta}\left(\cos\vartheta\frac{\partial\phi}{\partial\vartheta}\right) + \cos\vartheta\frac{\partial}{\partial r}\left(r^2\frac{\partial\phi}{\partial r}\right)\right]. \tag{2.32}$$

The curl of a vector is:

$$\operatorname{curl}\boldsymbol{B} = \nabla\times\boldsymbol{B} = \frac{1}{r^2\cos\vartheta}\begin{vmatrix} \mathbf{i}\,r\cos\vartheta & \mathbf{j}\,r & \mathbf{k} \\ \partial/\partial\lambda & \partial/\partial\vartheta & \partial/\partial r \\ B^\lambda r\cos\vartheta & B^\vartheta r & B^r \end{vmatrix}. \tag{2.33}$$

The vector Laplacian $\nabla^2\boldsymbol{B}$ (used for example when calculating viscous terms in the momentum equation) may be obtained from the vector identity:

$$\nabla^2\boldsymbol{B} = \nabla(\nabla\cdot\boldsymbol{B}) - \nabla\times(\nabla\times\boldsymbol{B}). \tag{2.34}$$

Only in Cartesian coordinates does this take the simple form:

$$\nabla^2\boldsymbol{B} = \frac{\partial^2\boldsymbol{B}}{\partial x^2} + \frac{\partial^2\boldsymbol{B}}{\partial y^2} + \frac{\partial^2\boldsymbol{B}}{\partial z^2}. \tag{2.35}$$

The expansion in spherical coordinates is, to most eyes, rather uninformative.

Rate of change of unit vectors

In spherical coordinates the defining unit vectors are $\mathbf{i}$, the unit vector pointing eastwards, parallel to a line of latitude; $\mathbf{j}$ is the unit vector pointing polewards, parallel to a meridian; and $\mathbf{k}$, the unit vector pointing radially outward. The directions of these vectors change with location, and in fact this is the case in nearly all coordinate systems, with the notable exception of the Cartesian one, and thus their material derivative is not zero. One way to evaluate this is to consider geometrically how the coordinate axes change with position. Another way, and the way that we shall proceed, is to first obtain the effective rotation rate $\boldsymbol{\Omega}_{flow}$, relative to the Earth, of a unit vector as it moves with the flow, and then apply (2.3). Specifically, let the fluid velocity be $\boldsymbol{v} = (u, v, w)$. The meridional component, v, produces a displacement $r\delta\vartheta = v\delta t$, and this gives rise a local effective vector rotation rate around the local zonal axis of $-(v/r)\mathbf{i}$, the minus sign arising because a displacement in the direction of the north pole is produced by negative rotational displacement around the $\mathbf{i}$ axis. Similarly, the zonal component, u, produces a displacement $\delta\lambda r\cos\vartheta = u\delta t$ and so an effective rotation rate, about the Earth's rotation axis, of $u/(r\cos\vartheta)$. Now, a rotation around the Earth's rotation axis may be written as (see Fig. 2.4)

$$\boldsymbol{\Omega} = \Omega(\mathbf{j}\cos\vartheta + \mathbf{k}\sin\vartheta). \tag{2.36}$$

If the scalar rotation rate is not Ω but is $u/(r\cos\vartheta)$, then the vector rotation rate is

$$\frac{u}{r\cos\vartheta}(\mathbf{j}\cos\vartheta + \mathbf{k}\sin\vartheta) = \mathbf{j}\frac{u}{r} + \mathbf{k}\frac{u\tan\vartheta}{r}. \tag{2.37}$$

Thus, the total rotation rate of a vector that moves with the flow is

$$\boldsymbol{\Omega}_{flow} = -\mathbf{i}\frac{v}{r} + \mathbf{j}\frac{u}{r} + \mathbf{k}\frac{u\tan\vartheta}{r}. \tag{2.38}$$

Applying (2.3) to (2.38), we find

$$\frac{\mathrm{D}\mathbf{i}}{\mathrm{D}t} = \boldsymbol{\Omega}_{flow}\times\mathbf{i} = \frac{u}{r\cos\vartheta}(\mathbf{j}\sin\vartheta - \mathbf{k}\cos\vartheta), \tag{2.39a}$$

$$\frac{\mathrm{D}\mathbf{j}}{\mathrm{D}t} = \boldsymbol{\Omega}_{flow}\times\mathbf{j} = -\mathbf{i}\frac{u}{r}\tan\vartheta - \mathbf{k}\frac{v}{r}, \tag{2.39b}$$

$$\frac{\mathrm{D}\mathbf{k}}{\mathrm{D}t} = \boldsymbol{\Omega}_{flow}\times\mathbf{k} = \mathbf{i}\frac{u}{r} + \mathbf{j}\frac{v}{r}. \tag{2.39c}$$

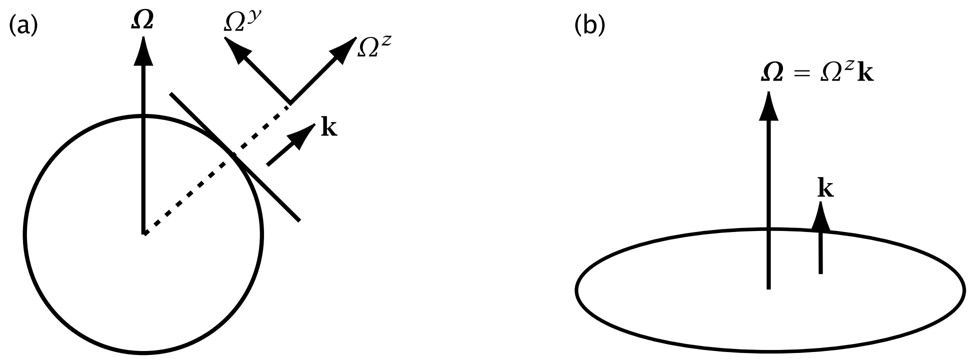

Fig. 2.4 (a) On the sphere the rotation vector $\boldsymbol{\Omega}$ can be decomposed into two components, one in the local vertical and one in the local horizontal, pointing toward the pole. That is, $\boldsymbol{\Omega} = \Omega_y \mathbf{j} + \Omega_z \mathbf{k}$ where $\Omega_y = \Omega \cos\vartheta$ and $\Omega_z = \Omega \sin\vartheta$. In geophysical fluid dynamics, the rotation vector in the local vertical is often the more important component in the horizontal momentum equations. On a rotating disk, (b), the rotation vector $\boldsymbol{\Omega}$ is parallel to the local vertical $\mathbf{k}$.

2.2.3 Equations of motion

Mass Conservation and Thermodynamic Equation

The mass conservation equation, (1.36a), expanded in spherical co-odinates, is

$$\begin{aligned}\frac{\partial \rho}{\partial t} &+ \frac{u}{r\cos\vartheta}\frac{\partial \rho}{\partial \lambda} + \frac{v}{r}\frac{\partial \rho}{\partial \vartheta} + w\frac{\partial \rho}{\partial r} \\ &+ \frac{\rho}{r\cos\vartheta}\left[\frac{\partial u}{\partial \lambda} + \frac{\partial}{\partial \vartheta}(v\cos\vartheta) + \frac{1}{r}\frac{\partial}{\partial r}(wr^2\cos\vartheta)\right] = 0.\end{aligned} \tag{2.40}$$

Equivalently, using the form (1.36b), this is

$$\frac{\partial \rho}{\partial t} + \frac{1}{r\cos\vartheta}\frac{\partial (u\rho)}{\partial \lambda} + \frac{1}{r\cos\vartheta}\frac{\partial}{\partial \vartheta}(v\rho\cos\vartheta) + \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 w\rho) = 0. \tag{2.41}$$

The thermodynamic equation, (1.108), is a tracer advection equation. Thus, using (2.27), its (adiabatic) spherical coordinate form is

$$\frac{\mathrm{D}\theta}{\mathrm{D}t} = \frac{\partial \theta}{\partial t} + \frac{u}{r\cos\vartheta}\frac{\partial \theta}{\partial \lambda} + \frac{v}{r}\frac{\partial \theta}{\partial \vartheta} + w\frac{\partial \theta}{\partial r} = 0, \tag{2.42}$$

and similarly for tracers such as water vapour or salt.

Momentum Equation

Recall that the inviscid momentum equation is:

$$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + 2\boldsymbol{\Omega}\times\boldsymbol{v} = -\frac{1}{\rho}\nabla p - \nabla\Phi, \tag{2.43}$$

where Φ is the geopotential. In spherical coordinates the directions of the coordinate axes change with position and so the component expansion of (2.43) is

$$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} = \frac{\mathrm{D}u}{\mathrm{D}t}\mathbf{i} + \frac{\mathrm{D}v}{\mathrm{D}t}\mathbf{j} + \frac{\mathrm{D}w}{\mathrm{D}t}\mathbf{k} + u\frac{\mathrm{D}\mathbf{i}}{\mathrm{D}t} + v\frac{\mathrm{D}\mathbf{j}}{\mathrm{D}t} + w\frac{\mathrm{D}\mathbf{k}}{\mathrm{D}t} \tag{2.44a}$$

$$= \frac{\mathrm{D}u}{\mathrm{D}t}\mathbf{i} + \frac{\mathrm{D}v}{\mathrm{D}t}\mathbf{j} + \frac{\mathrm{D}w}{\mathrm{D}t}\mathbf{k} + \boldsymbol{\Omega}_{flow} \times \boldsymbol{v}, \tag{2.44b}$$

using (2.39). Using either (2.44a) and the expressions for the rates of change of the unit vectors given in (2.39), or (2.44b) and the expression for $\boldsymbol{\Omega}_{flow}$ given in (2.38), (2.44) becomes

$$\begin{aligned}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} = \mathbf{i}\left(\frac{\mathrm{D}u}{\mathrm{D}t} - \frac{uv\tan\vartheta}{r} + \frac{uw}{r}\right) + \mathbf{j}\left(\frac{\mathrm{D}v}{\mathrm{D}t} + \frac{u^2\tan\vartheta}{r} + \frac{vw}{r}\right) \\ + \mathbf{k}\left(\frac{\mathrm{D}w}{\mathrm{D}t} - \frac{u^2+v^2}{r}\right).\end{aligned} \tag{2.45}$$

Using the definition of a vector cross product the Coriolis term is:

$$\begin{aligned}2\boldsymbol{\Omega} \times \boldsymbol{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2\Omega\cos\vartheta & 2\Omega\sin\vartheta \\ u & v & w \end{vmatrix} \\ &= \mathbf{i}\,(2\Omega w\cos\vartheta - 2\Omega v\sin\vartheta) + \mathbf{j}\,2\Omega u\sin\vartheta - \mathbf{k}\,2\Omega u\cos\vartheta.\end{aligned} \tag{2.46}$$

Using (2.45) and (2.46), and the gradient operator given by (2.31), the momentum equation (2.43) becomes:

$$\frac{\mathrm{D}u}{\mathrm{D}t} - \left(2\Omega + \frac{u}{r\cos\vartheta}\right)(v\sin\vartheta - w\cos\vartheta) = -\frac{1}{\rho r\cos\vartheta}\frac{\partial p}{\partial\lambda}, \tag{2.47a}$$

$$\frac{\mathrm{D}v}{\mathrm{D}t} + \frac{wv}{r} + \left(2\Omega + \frac{u}{r\cos\vartheta}\right)u\sin\vartheta = -\frac{1}{\rho r}\frac{\partial p}{\partial\vartheta}, \tag{2.47b}$$

$$\frac{\mathrm{D}w}{\mathrm{D}t} - \frac{u^2+v^2}{r} - 2\Omega u\cos\vartheta = -\frac{1}{\rho}\frac{\partial p}{\partial r} - g. \tag{2.47c}$$

The terms involving Ω are called Coriolis terms, and the quadratic terms on the left-hand sides involving $1/r$ are often called metric terms.

2.2.4 The primitive equations

The so-called *primitive equations* of motion are simplifications of the above equations frequently used in atmospheric and oceanic modelling.[3] Three related approximations are involved.

(i) The hydrostatic approximation. In the vertical momentum equation the gravitational term is assumed to be balanced by the pressure gradient term, so that

$$\frac{\partial p}{\partial z} = -\rho g. \tag{2.48}$$

The advection of vertical velocity, the Coriolis terms, and the metric term $(u^2+v^2)/r$ are all neglected.

(ii) The shallow-fluid approximation. We write $r = a + z$ where the constant a is the radius of the Earth and z increases in the radial direction. The coordinate r is then replaced by a except where it is used as the differentiating argument. Thus, for example,

$$\frac{1}{r^2}\frac{\partial(r^2 w)}{\partial r} \to \frac{\partial w}{\partial z}. \tag{2.49}$$

(iii) The traditional approximation. Coriolis terms in the horizontal momentum equations involving the vertical velocity, and the still smaller metric terms uw/r and vw/r, are neglected.

The second and third of these approximations should be taken, or not, together, the underlying reason being that they both relate to the presumed small aspect ratio of the motion, so the approximations succeed or fail together. If we make one approximation but not the other then we are being asymptotically inconsistent, and angular momentum and energy conservation are not assured (see section 2.2.7 and problem 2.12). The hydrostatic approximation also depends on the small aspect ratio of the flow, but in a slightly different way. For large-scale flow in the terrestrial atmosphere and ocean all three approximations are in fact all very accurate approximations. We defer a more complete treatment until section 2.7, in part because a treatment of the hydrostatic approximation is done most easily in the context of the Boussinesq equations, derived in section 2.4.

Making these approximations, the momentum equations become

$$\frac{\mathrm{D}u}{\mathrm{D}t} - 2\Omega \sin\vartheta v - \frac{uv}{a}\tan\vartheta = -\frac{1}{a\rho\cos\vartheta}\frac{\partial p}{\partial\lambda}, \tag{2.50a}$$

$$\frac{\mathrm{D}v}{\mathrm{D}t} + 2\Omega \sin\vartheta u + \frac{u^2\tan\vartheta}{a} = -\frac{1}{\rho a}\frac{\partial p}{\partial\vartheta}, \tag{2.50b}$$

$$0 = -\frac{1}{\rho}\frac{\partial p}{\partial z} - g, \tag{2.50c}$$

where

$$\frac{\mathrm{D}}{\mathrm{D}t} = \left(\frac{\partial}{\partial t} + \frac{u}{a\cos\vartheta}\frac{\partial}{\partial\lambda} + \frac{v}{a}\frac{\partial}{\partial\vartheta} + w\frac{\partial}{\partial z}\right). \tag{2.51}$$

We note the ubiquity of the factor $2\Omega\sin\vartheta$, and take the opportunity to define the *Coriolis parameter*, $f \equiv 2\Omega\sin\vartheta$.

The corresponding mass conservation equation for a shallow fluid layer is:

$$\begin{aligned}\frac{\partial\rho}{\partial t} &+ \frac{u}{a\cos\vartheta}\frac{\partial\rho}{\partial\lambda} + \frac{v}{a}\frac{\partial\rho}{\partial\vartheta} + w\frac{\partial\rho}{\partial z}\\ &+ \rho\left[\frac{1}{a\cos\vartheta}\frac{\partial u}{\partial\lambda} + \frac{1}{a\cos\vartheta}\frac{\partial}{\partial\vartheta}(v\cos\vartheta) + \frac{\partial w}{\partial z}\right] = 0,\end{aligned} \tag{2.52}$$

or equivalently,

$$\frac{\partial\rho}{\partial t} + \frac{1}{a\cos\vartheta}\frac{\partial(u\rho)}{\partial\lambda} + \frac{1}{a\cos\vartheta}\frac{\partial}{\partial\vartheta}(v\rho\cos\vartheta) + \frac{\partial(w\rho)}{\partial z} = 0. \tag{2.53}$$

2.2.5 Primitive equations in vector form

The primitive equations on a sphere may be written in a compact vector form provided we make a slight reinterpretation of the material derivative of the coordinate axes. Instead of (2.39) we take the material derivative of the unit vectors to be

$$\frac{\mathrm{D}\mathbf{i}}{\mathrm{D}t} = \widetilde{\boldsymbol{\Omega}}_{flow} \times \mathbf{i} = \mathbf{j}\frac{u\tan\vartheta}{a}, \tag{2.54a}$$

$$\frac{\mathrm{D}\mathbf{j}}{\mathrm{D}t} = \widetilde{\boldsymbol{\Omega}}_{flow} \times \mathbf{j} = -\mathbf{i}\frac{u\tan\vartheta}{a}, \tag{2.54b}$$

where $\widetilde{\boldsymbol{\Omega}}_{flow} = \mathbf{k}u\tan\vartheta/a$, which is the vertical component of (2.38) with r replaced by a. Given (2.54), the primitive equations (2.50a) and (2.50b) may be written as

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} + \boldsymbol{f}\times\boldsymbol{u} = -\frac{1}{\rho}\nabla_z p, \tag{2.55}$$

where $\boldsymbol{u} = u\mathbf{i} + v\mathbf{j} + 0\,\mathbf{k}$ is the horizontal velocity, $\nabla_z p = [(a\cos\vartheta)^{-1}\partial p/\partial\lambda,\ a^{-1}\partial p/\partial\vartheta]$ is the gradient operator at constant z, and $\boldsymbol{f} = f\mathbf{k} = 2\Omega\sin\vartheta\mathbf{k}$. In (2.55) the material derivative of the horizontal velocity is given by

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} = \mathbf{i}\frac{\mathrm{D}u}{\mathrm{D}t} + \mathbf{j}\frac{\mathrm{D}v}{\mathrm{D}t} + u\frac{\mathrm{D}\mathbf{i}}{\mathrm{D}t} + v\frac{\mathrm{D}\mathbf{j}}{\mathrm{D}t}, \tag{2.56}$$

The advection of the horizontal wind $\boldsymbol{u}$ is still by the three-dimensional velocity $\boldsymbol{v}$.

The vertical momentum equation is the hydrostatic equation, (2.50c), and the mass conservation equation is

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} + \rho\nabla\cdot\boldsymbol{v} = 0 \qquad \text{or} \qquad \frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\boldsymbol{v}) = 0, \tag{2.57}$$

where $\mathrm{D}/\mathrm{D}t$ is given by (2.51), and the second expression is written out in full in (2.53).

2.2.6 The vector invariant form of the momentum equation

The 'vector invariant' form of the momentum equation is so-called because it appears to take the same form in all coordinate systems — there is no advective derivative of the coordinate system to worry about. With the aid of the identity $(\boldsymbol{v}\cdot\nabla)\boldsymbol{v} = -\boldsymbol{v}\times\boldsymbol{\omega} + \nabla(\boldsymbol{v}^2/2)$, where $\boldsymbol{\omega} \equiv \nabla\times\boldsymbol{v}$ is the relative vorticity, the three-dimensional momentum equation, (2.16), may be written:

$$\frac{\partial\boldsymbol{v}}{\partial t} + (2\boldsymbol{\Omega} + \boldsymbol{\omega})\times\boldsymbol{v} = -\frac{1}{\rho}\nabla p - \frac{1}{2}\nabla\boldsymbol{v}^2 + \boldsymbol{g}, \tag{2.58}$$

and this is the vector invariant momentum equation. In spherical coordinates the relative vorticity is given by:

$$\begin{aligned}\boldsymbol{\omega} \;=\; \nabla\times\boldsymbol{v} &= \frac{1}{r^2\cos\vartheta}\begin{vmatrix}\mathbf{i}\,r\cos\vartheta & \mathbf{j}\,r & \mathbf{k}\\ \partial/\partial\lambda & \partial/\partial\vartheta & \partial/\partial r\\ ur\cos\vartheta & rv & w\end{vmatrix}\\ &= \mathbf{i}\,\frac{1}{r}\left(\frac{\partial w}{\partial\vartheta} - \frac{\partial(rv)}{\partial r}\right) - \mathbf{j}\,\frac{1}{r\cos\vartheta}\left(\frac{\partial w}{\partial\lambda} - \frac{\partial}{\partial r}(ur\cos\vartheta)\right)\\ &\quad + \mathbf{k}\,\frac{1}{r\cos\vartheta}\left(\frac{\partial v}{\partial\lambda} - \frac{\partial}{\partial\vartheta}(u\cos\vartheta)\right).\end{aligned} \tag{2.59}$$

We can write the horizontal momentum equations of the primitive equations in a similar way. Making the traditional and shallow fluid approximations, the horizontal components of (2.58) become

$$\frac{\partial\boldsymbol{u}}{\partial t} + (\boldsymbol{f} + \mathbf{k}\zeta)\times\boldsymbol{u} + w\frac{\partial\boldsymbol{u}}{\partial z} = -\frac{1}{\rho}\nabla_z p - \frac{1}{2}\nabla\boldsymbol{u}^2, \tag{2.60}$$

where $\boldsymbol{u} = (u, v, 0)$, $\boldsymbol{f} = \mathbf{k}\,2\Omega\sin\vartheta$, ∇_z is the horizontal gradient operator (the gradient at a constant value of z), and using (2.59), ζ is given by

$$\zeta = \frac{1}{a\cos\vartheta}\frac{\partial v}{\partial\lambda} - \frac{1}{a\cos\vartheta}\frac{\partial}{\partial\vartheta}(u\cos\vartheta) = \frac{1}{a\cos\vartheta}\frac{\partial v}{\partial\lambda} - \frac{1}{a}\frac{\partial u}{\partial\vartheta} + \frac{u}{a}\tan\vartheta. \tag{2.61}$$

The separate components of the momentum equation are given by:

$$\frac{\partial u}{\partial t} - (f + \zeta)v + w\frac{\partial u}{\partial z} = -\frac{1}{a\cos\vartheta}\left(\frac{1}{\rho}\frac{\partial p}{\partial \lambda} + \frac{1}{2}\frac{\partial \boldsymbol{u}^2}{\partial \lambda}\right), \tag{2.62}$$

and

$$\frac{\partial v}{\partial t} + (f + \zeta)u + w\frac{\partial v}{\partial z} = -\frac{1}{a}\left(\frac{1}{\rho}\frac{\partial p}{\partial \vartheta} + \frac{1}{2}\frac{\partial \boldsymbol{u}^2}{\partial \vartheta}\right). \tag{2.63}$$

Related expressions are given in problem 2.3, and we treat vorticity at greater length in chapter 4.

2.2.7 Angular momentum

The zonal momentum equation can be usefully expressed as a statement about axial angular momentum; that is, angular momentum about the rotation axis. The zonal angular momentum per unit mass is the component of angular momentum in the direction of the axis of rotation and it is given by, without making any shallow atmosphere approximation,

$$m = (u + \Omega r\cos\vartheta)r\cos\vartheta. \tag{2.64}$$

The evolution equation for this quantity follows from the zonal momentum equation and has the simple form

$$\frac{\mathrm{D}m}{\mathrm{D}t} = -\frac{1}{\rho}\frac{\partial p}{\partial \lambda}, \tag{2.65}$$

where the material derivative is

$$\frac{\mathrm{D}}{\mathrm{D}t} = \frac{\partial}{\partial t} + \frac{u}{r\cos\vartheta}\frac{\partial}{\partial \lambda} + \frac{v}{r}\frac{\partial}{\partial \vartheta} + w\frac{\partial}{\partial r}. \tag{2.66}$$

Using the mass continuity equation, (2.65) can be written as

$$\frac{\mathrm{D}\rho m}{\mathrm{D}t} + \rho m\nabla\cdot\boldsymbol{v} = -\frac{\partial p}{\partial \lambda} \tag{2.67}$$

or

$$\frac{\partial \rho m}{\partial t} + \frac{1}{r\cos\vartheta}\frac{\partial(\rho u m)}{\partial \lambda} + \frac{1}{r\cos\vartheta}\frac{\partial}{\partial \vartheta}(\rho v m\cos\vartheta) + \frac{1}{r^2}\frac{\partial}{\partial r}(\rho m w r^2) = -\frac{\partial p}{\partial \lambda}. \tag{2.68}$$

This is an angular momentum conservation equation.

If the fluid is confined to a shallow layer near the surface of a sphere, then we may replace r, the radial coordinate, by a, the radius of the sphere, in the definition of m, and we define $\widetilde{m} \equiv (u + \Omega a\cos\vartheta)a\cos\vartheta$. Then (2.65) is replaced by

$$\frac{\mathrm{D}\widetilde{m}}{\mathrm{D}t} = -\frac{1}{\rho}\frac{\partial p}{\partial \lambda}, \tag{2.69}$$

where now

$$\frac{\mathrm{D}}{\mathrm{D}t} = \frac{\partial}{\partial t} + \frac{u}{a\cos\vartheta}\frac{\partial}{\partial \lambda} + \frac{v}{a}\frac{\partial}{\partial \vartheta} + w\frac{\partial}{\partial z}. \tag{2.70}$$

In the shallow fluid approximation (2.68) becomes

$$\frac{\partial \rho m}{\partial t} + \frac{1}{a\cos\vartheta}\frac{\partial(\rho u m)}{\partial \lambda} + \frac{1}{a\cos\vartheta}\frac{\partial}{\partial \vartheta}(\rho v m\cos\vartheta) + \frac{\partial}{\partial z}(\rho m w) = -\frac{\partial p}{\partial \lambda}. \tag{2.71}$$

which is an angular momentum conservation equation for a shallow atmosphere.

◆ *From angular momentum to the spherical component equations*

An alternative way of deriving the three components of the momentum equation in spherical polar coordinates is to *begin* with (2.65) and the principle of conservation of energy. That is, we take the equations for conservation of angular momentum and energy as true a priori and demand that the forms of the momentum equation be constructed to satisfy these. Expanding the material derivative in (2.65), noting that $\mathrm{D}r/\mathrm{D}t = w$ and $\mathrm{D}\cos\vartheta/\mathrm{D}t = -(v/r)\sin\vartheta$, immediately gives (2.47a). Multiplication by u then yields

$$u\frac{\mathrm{D}u}{\mathrm{D}t} - 2\Omega uv\sin\vartheta + 2\Omega uw\cos\vartheta - \frac{u^2 v\tan\vartheta}{r} + \frac{u^2 w}{r} = -\frac{u}{\rho r\cos\vartheta}\frac{\partial p}{\partial\lambda}. \tag{2.72}$$

Now suppose that the meridional and vertical momentum equations are of the form

$$\frac{\mathrm{D}v}{\mathrm{D}t} + \text{Coriolis and metric terms} = -\frac{1}{\rho r}\frac{\partial p}{\partial\vartheta} \tag{2.73a}$$

$$\frac{\mathrm{D}w}{\mathrm{D}t} + \text{Coriolis and metric terms} = -\frac{1}{\rho}\frac{\partial p}{\partial r}, \tag{2.73b}$$

but that we do not know what form the Coriolis and metric terms take. To determine that form, construct the kinetic energy equation by multiplying (2.73) by v and w, respectively. Now, the metric terms must vanish when we sum the resulting equations along with (2.72), so that (2.73a) must contain the Coriolis term $2\Omega u\sin\vartheta$ as well as the metric term $u^2\tan\vartheta/r$, and (2.73b) must contain the term $-2\Omega u\cos\phi$ as well as the metric term u^2/r. But if (2.73b) contains the term u^2/r it must also contain the term v^2/r by isotropy, and therefore (2.73a) must also contain the term vw/r. In this way, (2.47) is precisely reproduced, although the sceptic might argue that the uniqueness of the form has not been demonstrated.

A particular advantage of this approach arises in determining the appropriate momentum equations that conserve angular momentum and energy in the shallow-fluid approximation. We begin with (2.69) and expand to obtain (2.50a). Multiplying by u gives

$$u\frac{\mathrm{D}u}{\mathrm{D}t} - 2\Omega uv\sin\vartheta - \frac{u^2 v\tan\vartheta}{a} = -\frac{u}{\rho a\cos\vartheta}\frac{\partial p}{\partial\lambda}. \tag{2.74}$$

To ensure energy conservation, the meridional momentum equation must contain the Coriolis term $2\Omega u\sin\vartheta$ and the metric term $u^2\tan\vartheta/a$, but the vertical momentum equation must have neither of the metric terms appearing in (2.47c). Thus we deduce the following equations:

$$\frac{\mathrm{D}u}{\mathrm{D}t} - \left(2\Omega\sin\vartheta + \frac{u\tan\vartheta}{a}\right)v = -\frac{1}{\rho a\cos\vartheta}\frac{\partial p}{\partial\lambda}, \tag{2.75a}$$

$$\frac{\mathrm{D}v}{\mathrm{D}t} + \left(2\Omega\sin\vartheta + \frac{u\tan\vartheta}{a}\right)u = -\frac{1}{\rho a}\frac{\partial p}{\partial\vartheta}, \tag{2.75b}$$

$$\frac{\mathrm{D}w}{\mathrm{D}t} = -\frac{1}{\rho}\frac{\partial p}{\partial r} - g. \tag{2.75c}$$

This equation set, when used in conjunction with the thermodynamic and mass continuity equations, conserves appropriate forms of angular momentum and energy. In the hydrostatic approximation the material derivative of w in (2.75c) is *additionally* neglected. Thus, the hydrostatic approximation is mathematically and physically consistent with the shallow-fluid approximation, but it is an additional approximation with slightly different requirements that one may choose, rather than being required, to make. From an asymptotic perspective, the difference lies in the small parameter necessary for either approximation to hold, namely:

$$\text{shallow fluid and traditional approximations:} \qquad \gamma \equiv \frac{H}{a} \ll 1, \tag{2.76a}$$

$$\text{small aspect ratio for hydrostatic approximation:} \qquad \alpha \equiv \frac{H}{L} \ll 1, \tag{2.76b}$$

where L is the horizontal scale of the motion and a is the radius of the Earth. For hemispheric or global scale phenomena $L \sim a$ and the two approximations coincide. (Requirement (2.76b) for the hydrostatic approximation will be derived in section 2.7.)

2.3 CARTESIAN APPROXIMATIONS: THE TANGENT PLANE

2.3.1 The f-plane

Although the rotation of the Earth is central for many dynamical phenomena, the sphericity of the Earth is not always so. This is especially true for phenomena on a scale somewhat smaller than global where the use of spherical coordinates becomes awkward, and it is more convenient to use a locally Cartesian representation of the equations. Referring to Fig. 2.4 we will define a plane tangent to the surface of the Earth at a latitude ϑ_0, and then use a Cartesian coordinate system (x, y, z) to describe motion on that plane. For small excursions on the plane, $(x, y, z) \approx (a\lambda \cos\vartheta_0, a(\vartheta - \vartheta_0), z)$. Consistently, the velocity is $\boldsymbol{v} = (u, v, w)$, so that u,v and w are the components of the velocity *in the tangent plane,* in approximately in the east–west, north–south and vertical directions, respectively.

The momentum equations for flow in this plane are then

$$\frac{\partial u}{\partial t} + (\boldsymbol{v}\cdot\nabla)u + 2(\Omega^y w - \Omega^z v) = -\frac{1}{\rho}\frac{\partial p}{\partial x}, \tag{2.77a}$$

$$\frac{\partial v}{\partial t} + (\boldsymbol{v}\cdot\nabla)v + 2(\Omega^z u - \Omega^x w) = -\frac{1}{\rho}\frac{\partial p}{\partial y}, \tag{2.77b}$$

$$\frac{\partial w}{\partial t} + (\boldsymbol{v}\cdot\nabla)w + 2(\Omega^x v - \Omega^y u) = -\frac{1}{\rho}\frac{\partial p}{\partial z} - g, \tag{2.77c}$$

where the rotation vector $\boldsymbol{\Omega} = \Omega^x\mathbf{i} + \Omega^y\mathbf{j} + \Omega^z\mathbf{k}$ and $\Omega^x = 0$, $\Omega^y = \Omega\cos\vartheta_0$ and $\Omega^z = \Omega\sin\vartheta_0$. If we make the traditional approximation, and so ignore the components of $\boldsymbol{\Omega}$ not in the direction of the local vertical, then

$$\frac{\mathrm{D}u}{\mathrm{D}t} - f_0 v = -\frac{1}{\rho}\frac{\partial p}{\partial x}, \tag{2.78a}$$

$$\frac{\mathrm{D}v}{\mathrm{D}t} + f_0 u = -\frac{1}{\rho}\frac{\partial p}{\partial y}, \tag{2.78b}$$

$$\frac{\mathrm{D}w}{\mathrm{D}t} = -\frac{1}{\rho}\frac{\partial p}{\partial z} - g. \tag{2.78c}$$

where $f_0 = 2\Omega^z = 2\Omega\sin\vartheta_0$. Defining the horizontal velocity vector $\boldsymbol{u} = (u, v, 0)$, the first two equations may also be written as

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} + \boldsymbol{f}_0 \times \boldsymbol{u} = -\frac{1}{\rho}\nabla_z p, \tag{2.79}$$

where $\mathrm{D}\boldsymbol{u}/\mathrm{D}t = \partial \boldsymbol{u}/\partial t + \boldsymbol{v}\cdot\nabla\boldsymbol{u}$, $\boldsymbol{f}_0 = 2\Omega\sin\vartheta_0\mathbf{k} = f_0\mathbf{k}$, and $\mathbf{k}$ is the direction perpendicular to the plane (it does not change its orientation with latitude). These equations are, evidently, exactly the same as the momentum equations in a system in which the rotation vector is aligned with the local vertical, as illustrated in the right-hand panel in Fig. 2.4 (on page 63). They will describe flow on the surface of a rotating sphere to a good approximation provided the flow is of limited latitudinal extent so that the effects of sphericity are unimportant; we have made what is known as the *f-plane* approximation since the Coriolis parameter is a constant. We may in addition make the hydrostatic approximation, in which case (2.78c) becomes the familiar $\partial p/\partial z = -\rho g$.

2.3.2 The beta-plane approximation

The magnitude of the vertical component of rotation varies with latitude, and this has important dynamical consequences. We can approximate this effect by allowing the effective rotation vector to vary. Thus, noting that, for small variations in latitude,

$$f = 2\Omega\sin\vartheta \approx 2\Omega\sin\vartheta_0 + 2\Omega(\vartheta-\vartheta_0)\cos\vartheta_0, \tag{2.80}$$

then on the tangent plane we may mimic this by allowing the Coriolis parameter to vary as

$$\boxed{f = f_0 + \beta y}\,, \tag{2.81}$$

where $f_0 = 2\Omega\sin\vartheta_0$ and $\beta = \partial f/\partial y = (2\Omega\cos\vartheta_0)/a$. This important approximation is known as the *beta-plane*, or *β-plane,* approximation; it captures the the most important *dynamical* effects of sphericity, without the complicating *geometric* effects, which are not essential to describe many phenomena. The momentum equations (2.78) are unaltered except that f_0 is replaced by $f_0+\beta y$ to represent a varying Coriolis parameter. Thus, sphericity combined with rotation is dynamically equivalent to a *differentially rotating* system. For future reference, we write down the β-plane horizontal momentum equations:

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} + \boldsymbol{f}\times\boldsymbol{u} = -\frac{1}{\rho}\nabla_z p, \tag{2.82}$$

where $\boldsymbol{f} = (f_0+\beta y)\widehat{\mathbf{k}}$. In component form this equation becomes

$$\frac{\mathrm{D}u}{\mathrm{D}t} - fv = -\frac{1}{\rho}\frac{\partial p}{\partial x}, \qquad \frac{\mathrm{D}v}{\mathrm{D}t} + fu = -\frac{1}{\rho}\frac{\partial p}{\partial y}. \tag{2.83a,b}$$

The mass conservation, thermodynamic and hydrostatic equations in the β-plane approximation are the same as the usual Cartesian, f-plane, forms of those equations.

2.4 EQUATIONS FOR A STRATIFIED OCEAN: THE BOUSSINESQ APPROXIMATION

The density variations in the ocean are quite small compared to the mean density, and we may exploit this to derive somewhat simpler but still quite accurate equations of motion. Let us first examine how much density does vary in the ocean.

2.4.1 Variation of density in the ocean

The variations of density in the ocean are due to three effects: the compression of water by pressure (which we denote as $\Delta_p \rho$), the thermal expansion of water if its temperature changes ($\Delta_T \rho$), and the haline contraction if its salinity changes ($\Delta_S \rho$). How big are these? An appropriate equation of state to approximately evaluate these effects is the linear one

$$\rho = \rho_0 \left[1 - \beta_T (T - T_0) + \beta_S (S - S_0) + \frac{p}{\rho_0 c_s^2} \right], \tag{2.84}$$

where $\beta_T \approx 2 \times 10^{-4}\,\mathrm{K}^{-1}$, $\beta_S \approx 10^{-3}\,\mathrm{psu}^{-1}$ and $c_s \approx 1500\,\mathrm{m\,s}^{-1}$ (see the table on page 35). The three effects may then be evaluated as follows.

Pressure compressibility. We have $\Delta_p \rho \approx \Delta p / c_s^2 \approx \rho_0 g H / c_s^2$. where H is the depth and the pressure change is quite accurately evaluated using the hydrostatic approximation. Thus,

$$\frac{|\Delta_p \rho|}{\rho_0} \ll 1 \qquad \text{if} \qquad \frac{gH}{c_s^2} \ll 1, \tag{2.85}$$

or if $H \ll c_s^2/g$. The quantity $c_s^2/g \approx 200\,\mathrm{km}$ is the density scale height of the ocean. Thus, the pressure at the bottom of the ocean (say $H = 10\,\mathrm{km}$ in the deep trenches), enormous as it is, is insufficient to compress the water enough to make a significant change in its density. Changes in density due to dynamical variations of pressure are small if the Mach number is small, and this is also the case.

Thermal expansion. We have $\Delta_T \rho \approx -\beta_T \rho_0 \Delta T$ and therefore

$$\frac{|\Delta_T \rho|}{\rho_0} \ll 1 \qquad \text{if} \qquad \beta_T \Delta T \ll 1. \tag{2.86}$$

For $\Delta T = 20\,\mathrm{K}$, $\beta_T \Delta T \approx 4 \times 10^{-3}$, and evidently we would require temperature differences of order β_T^{-1}, or 5000 K to obtain order one variations in density.

Saline contraction. We have $\Delta_S \rho \approx \beta_S \rho_0 \Delta S$ and therefore

$$\frac{|\Delta_S \rho|}{\rho_0} \ll 1 \qquad \text{if} \qquad \beta_S \Delta S \ll 1. \tag{2.87}$$

As changes in salinity in the ocean rarely exceed 5 psu, for which $\beta_S \Delta S = 5 \times 10^{-3}$, the fractional change in the density of seawater is correspondingly very small.

Evidently, fractional density changes in the ocean are very small.

2.4.2 The Boussinesq equations

The *Boussinesq equations* are a set of equations that exploit the smallness of density variations in many liquids.[4] To set notation we write

$$\rho = \rho_0 + \delta\rho(x, y, z, t) \tag{2.88a}$$
$$= \rho_0 + \widehat{\rho}(z) + \rho'(x, y, z, t) \tag{2.88b}$$
$$= \widetilde{\rho}(z) + \rho'(x, y, z, t), \tag{2.88c}$$

where ρ_0 is a constant and we assume that

$$|\widehat{\rho}|, |\rho'|, |\delta\rho| \ll \rho_0. \tag{2.89}$$

We need not assume that $|\rho'| \ll |\widehat{\rho}|$, but this is often the case in the ocean. To obtain the Boussinesq equations we will just use (2.88a), but (2.88c) will be useful for the anelastic equations considered later.

Associated with the reference density is a reference pressure that is defined to be in hydrostatic balance with it. That is,

$$p = p_0(z) + \delta p(x, y, z, t) \tag{2.90a}$$
$$= \widetilde{p}(z) + p'(x, y, z, t), \tag{2.90b}$$

where $|\delta p| \ll p_0$, $|p'| \ll \widetilde{p}$ and

$$\frac{\mathrm{d}p_0}{\mathrm{d}z} \equiv -g\rho_0, \qquad \frac{\mathrm{d}\widetilde{p}}{\mathrm{d}z} \equiv -g\widetilde{\rho}. \tag{2.91a,b}$$

Note that $\nabla_z p = \nabla_z p' = \nabla_z \delta p$ and that $p_0 \approx \widetilde{p}$ if $|\widehat{\rho}| \ll \rho_0$.

Momentum equations

To obtain the Boussinesq equations we use $\rho = \rho_0 + \delta\rho$, and assume $\delta\rho/\rho_0$ is small. Without approximation, the momentum equation can be written as

$$(\rho_0 + \delta\rho)\left(\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + 2\boldsymbol{\Omega} \times \boldsymbol{v}\right) = -\nabla\delta p - \frac{\partial p_0}{\partial z}\mathbf{k} - g(\rho_0 + \delta\rho)\mathbf{k}, \tag{2.92}$$

and using (2.91a) this becomes, again without approximation,

$$(\rho_0 + \delta\rho)\left(\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + 2\boldsymbol{\Omega} \times \boldsymbol{v}\right) = -\nabla\delta p - g\delta\rho\mathbf{k}. \tag{2.93}$$

If density variations are small this becomes

$$\boxed{\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + 2\boldsymbol{\Omega} \times \boldsymbol{v} = -\nabla\phi + b\mathbf{k}}, \tag{2.94}$$

where $\phi = \delta p/\rho_0$ and $b = -g\,\delta\rho/\rho_0$ is the *buoyancy*. Note that we should not and do not neglect the term $g\,\delta\rho$, for there is no reason to believe it to be small ($\delta\rho$ may be small, but g is big). Equation (2.94) is the momentum equation in the Boussinesq approximation, and it is common to say that the Boussinesq approximation ignores all variations of density of a fluid in the momentum equation, except when associated with the gravitational term.

For most large-scale motions in the ocean the *deviation* pressure and density fields are also approximately in hydrostatic balance, and in that case the vertical component of (2.94) becomes

$$\frac{\partial\phi}{\partial z} = b. \tag{2.95}$$

A condition for (2.95) to hold is that vertical accelerations are small *compared to $g\,\delta\rho/\rho_0$, and not compared to the acceleration due to gravity itself.* For more discussion of this point, see section 2.7.

Mass Conservation

The unapproximated mass conservation equation is

$$\frac{\mathrm{D}\delta\rho}{\mathrm{D}t} + (\rho_0 + \delta\rho)\nabla\cdot\boldsymbol{v} = 0. \tag{2.96}$$

Provided that time scales advectively — that is to say that $\mathrm{D}/\mathrm{D}t$ scales in the same way as $\boldsymbol{v}\cdot\nabla$ — then we may approximate this equation by

$$\boxed{\nabla\cdot\boldsymbol{v} = 0}, \tag{2.97}$$

which is the same as that for a constant density fluid. This *absolutely does not* allow one to go back and use (2.96) to say that $D\delta\rho/Dt = 0$; the evolution of density is given by the thermodynamic equation in conjunction with an equation of state, and this should not be confused with the mass conservation equation. Note also that in eliminating the time-derivative of density we eliminate the possibility of sound waves.

Thermodynamic equation and equation of state

The Boussinesq equations are closed by the addition of an equation of state, a thermodynamic equation and, as appropriate, a salinity equation. Neglecting salinity for the moment, a useful starting point is to write the thermodynamic equation, (1.116), as

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} - \frac{1}{c_s^2}\frac{\mathrm{D}p}{\mathrm{D}t} = \frac{\dot{Q}}{(\partial\eta/\partial\rho)_p T} \approx -\dot{Q}\left(\frac{\rho_0\beta_T}{c_p}\right) \tag{2.98}$$

using $(\partial\eta/\partial\rho)_p = (\partial\eta/\partial T)_p(\partial T/\partial\rho)_p \approx -c_p/(T\rho_0\beta_T)$. Given the expansions (2.88a) and (2.90a), (2.98) can be written to a good approximation as

$$\frac{\mathrm{D}\delta\rho}{\mathrm{D}t} - \frac{1}{c_s^2}\frac{\mathrm{D}p_0}{\mathrm{D}t} = -\dot{Q}\left(\frac{\rho_0\beta_T}{c_p}\right), \tag{2.99}$$

or, using (2.91a),

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\delta\rho + \frac{\rho_0 g}{c_s^2}z\right) = -\dot{Q}\left(\frac{\rho_0\beta_T}{c_p}\right), \tag{2.100}$$

as in (1.119). The severest approximation to this is to neglect the second term in brackets on the left-hand side, and noting that $b = -g\delta\rho/\rho_0$ we obtain

$$\boxed{\frac{\mathrm{D}b}{\mathrm{D}t} = \dot{b}}, \tag{2.101}$$

where $\dot{b} = g\beta_T\dot{Q}/c_p$. The momentum equation (2.94), mass continuity equation (2.97) and thermodynamic equation (2.101) then form a closed set, called the *simple Boussinesq equations*.

A somewhat more accurate approach is to include the compressibility of the fluid that is due to the hydrostatic pressure. From (2.100), the potential density is given by $\delta\rho_{\mathrm{pot}} = \delta\rho + \rho_0 g z/c_s^2$, and so the *potential buoyancy*, that is the buoyancy based on potential density, is given by

$$b_\sigma \equiv -g\frac{\delta\rho_{\mathrm{pot}}}{\rho_0} = -\frac{g}{\rho_0}\left(\delta\rho + \frac{\rho_0 g z}{c_s^2}\right) = b - g\frac{z}{H_\rho}, \tag{2.102}$$

where $H_\rho = c_s^2/g$. The thermodynamic equation, (2.100), may then be written

$$\frac{\mathrm{D}b_\sigma}{\mathrm{D}t} = \dot{b}_\sigma, \tag{2.103}$$

where $\dot{b}_\sigma = \dot{b}$. Buoyancy itself is obtained from b_σ by the 'equation of state', $b = b_\sigma + gz/H_\rho$.

In many applications we may need to use a still more accurate equation of state. In that case (and see section 1.6.2) we replace (2.101) by the thermodynamic equations

$$\boxed{\frac{\mathrm{D}\theta}{\mathrm{D}t} = \dot{\theta}, \qquad \frac{\mathrm{D}S}{\mathrm{D}t} = \dot{S}}, \tag{2.104a,b}$$

where θ is the potential temperature and S is salinity, along with an equation of state. The equation of state has the general form $b = b(\theta, S, p)$, but to be consistent with the level of approximation in the other Boussinesq equations we replace p by the hydrostatic pressure calculated with the reference density, that is by $-\rho_0 gz$, and the equation of state then takes the general form

$$\boxed{b = b(\theta, S, z)}. \tag{2.105}$$

An example of (2.105) is (1.156), taken with the definition of buoyancy $b = -g\delta\rho/\rho_0$. The closed set of equations (2.94), (2.97), (2.104) and (2.105) are the *general Boussinesq equations*. Using an accurate equation of state and the Boussinesq approximation is the procedure used in many comprehensive ocean general circulation models. The Boussinesq equations, which with the hydrostatic and traditional approximations are often considered to be the oceanic primitive equations, are summarized in the shaded box above.

◆ *Mean stratification and the buoyancy frequency*

The processes that cause density to vary in the vertical often differ from those that cause it to vary in the horizontal. For this reason it is sometimes useful to write $\rho = \rho_0 + \widehat{\rho}(z) + \rho'(x, y, z, t)$ and define $\widetilde{b}(z) \equiv -g\widehat{\rho}/\rho_0$ and $b' \equiv -g\rho'/\rho_0$. Using the hydrostatic equation to evaluate pressure, the thermodynamic equation (2.98) becomes, to a good approximation,

$$\frac{\mathrm{D}b'}{\mathrm{D}t} + N^2 w = 0, \tag{2.106}$$

where $\mathrm{D}/\mathrm{D}t$ remains a three-dimensional operator and

$$N^2(z) = \left(\frac{\mathrm{d}\widetilde{b}}{\mathrm{d}z} - \frac{g^2}{c_s^2}\right) = \frac{\mathrm{d}\widetilde{b}_\sigma}{\mathrm{d}z}, \tag{2.107}$$

where $\widetilde{b}_\sigma = \widetilde{b} - gz/H_\rho$. The quantity N^2 is a measure of the mean stratification of the fluid, and is equal to the vertical gradient of the mean potential buoyancy. N is known as the buoyancy frequency, something we return to in section 2.9. Equations (2.106) and (2.107) also hold in the simple Boussinesq equations, but with $c_s^2 = \infty$.

Summary of Boussinesq Equations

The simple Boussinesq equations are, for an inviscid fluid:

momentum equations: $$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + \boldsymbol{f} \times \boldsymbol{v} = -\nabla\phi + b\mathbf{k}, \tag{B.1}$$

mass conservation: $$\nabla \cdot \boldsymbol{v} = 0, \tag{B.2}$$

buoyancy equation: $$\frac{\mathrm{D}b}{\mathrm{D}t} = \dot{b}. \tag{B.3}$$

A more general form replaces the buoyancy equation by:

thermodynamic equation: $$\frac{\mathrm{D}\theta}{\mathrm{D}t} = \dot{\theta}, \tag{B.4}$$

salinity equation: $$\frac{\mathrm{D}S}{\mathrm{D}t} = \dot{S}, \tag{B.5}$$

equation of state: $$b = b(\theta, S, z). \tag{B.6}$$

If the equation of state is $b = b(\theta, S, \phi)$ then energy conservation is not assured.

2.4.3 Energetics of the Boussinesq system

In a uniform gravitational field but with no other forcing or dissipation, we write the simple Boussinesq equations as

$$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + 2\boldsymbol{\Omega} \times \boldsymbol{v} = b\mathbf{k} - \nabla\phi, \qquad \nabla \cdot \boldsymbol{v} = 0, \qquad \frac{\mathrm{D}b}{\mathrm{D}t} = 0. \tag{2.108a,b,c}$$

From (2.108a) and (2.108b) the kinetic energy density evolution (cf. section 1.10) is given by

$$\frac{1}{2}\frac{\mathrm{D}v^2}{\mathrm{D}t} = bw - \nabla \cdot (\phi \boldsymbol{v}), \tag{2.109}$$

where the constant reference density ρ_0 is omitted. Let us now define the potential $\Phi \equiv -z$, so that $\nabla\Phi = -\mathbf{k}$ and

$$\frac{\mathrm{D}\Phi}{\mathrm{D}t} = \nabla \cdot (\boldsymbol{v}\Phi) = -w, \tag{2.110}$$

and using this and (2.108c) gives

$$\frac{\mathrm{D}}{\mathrm{D}t}(b\Phi) = -wb. \tag{2.111}$$

Adding (2.111) to (2.109) and expanding the material derivative gives

$$\frac{\partial}{\partial t}\left(\frac{1}{2}\boldsymbol{v}^2 + b\Phi\right) + \nabla \cdot \left[\boldsymbol{v}\left(\frac{1}{2}\boldsymbol{v}^2 + b\Phi + \phi\right)\right] = 0. \tag{2.112}$$

This constitutes an energy equation for the Boussinesq system, and may be compared to (1.186). (Also see problem 2.13.) The energy density (divided by ρ_0) is just $\boldsymbol{v}^2/2 + b\Phi$. What does the term $b\Phi$ represent? Its integral, multiplied by ρ_0, is the potential energy of the flow minus that of the basic state, or $\int g(\rho - \rho_0)z\,\mathrm{d}z$. If there were a heating term on the right-hand side of (2.108c) this would directly provide a source of potential energy, rather than internal energy as in the compressible system. Because the fluid is incompressible, there is no conversion from kinetic and potential energy into internal energy.

◆ *Energetics with a general equation of state*

Now consider the energetics of the general Boussinesq equations. Suppose first that we allow the equation of state to be a function of pressure; the equations of motion are then (2.108) except that (2.108c) is replaced by

$$\frac{\mathrm{D}\theta}{\mathrm{D}t} = 0, \qquad \frac{\mathrm{D}S}{\mathrm{D}t} = 0, \qquad b = b(\theta, S, \phi). \tag{2.113a,b,c}$$

A little algebraic experimentation will reveal that no energy conservation law of the form (2.112) generally exists for this system! The problem arises because, by requiring the fluid to be incompressible, we eliminate the proper conversion of internal energy to kinetic energy. However, if we use the approximation $b = b(\theta, S, z)$, the system does conserve an energy, as we now show.[5]

Define the potential, Π, as the integral of b at constant potential temperature and salinity; that is

$$\Pi(\theta, S, z) \equiv -\int_a^z b\,\mathrm{d}z', \tag{2.114}$$

where a is any constant, so that $\partial\Pi/\partial z = -b$. Taking the material derivative of the left-hand side gives

$$\frac{\mathrm{D}\Pi}{\mathrm{D}t} = \left(\frac{\partial \Pi}{\partial \theta}\right)_{S,z}\frac{\mathrm{D}\theta}{\mathrm{D}t} + \left(\frac{\partial \Pi}{\partial S}\right)_{\theta,z}\frac{\mathrm{D}S}{\mathrm{D}t} + \left(\frac{\partial \Pi}{\partial z}\right)_{\theta,S}\frac{\mathrm{D}z}{\mathrm{D}t} = -bw, \tag{2.115}$$

using (2.113a,b). Combining (2.115) and (2.109) gives

$$\frac{\partial}{\partial t}\left(\frac{1}{2}\boldsymbol{v}^2 + \Pi\right) + \nabla\cdot\left[\boldsymbol{v}\left(\frac{1}{2}\boldsymbol{v}^2 + \Pi + \phi\right)\right] = 0. \tag{2.116}$$

Thus, energetic consistency is maintained with an arbitrary equation of state, provided the pressure is replaced by a function of z. If b is not an explicit function of z in the equation of state, the conservation law is identical to (2.112).

2.5 EQUATIONS FOR A STRATIFIED ATMOSPHERE: THE ANELASTIC APPROXIMATION

2.5.1 Preliminaries

In the atmosphere the density varies significantly, especially in the vertical. However deviations of both ρ and p from a statically balanced state are often quite small, and the relative vertical variation of potential temperature is also small. We can usefully exploit these observations to give a somewhat simplified set of equations, useful both for theoretical and numerical analyses because sound waves are eliminated by way of an 'anelastic' approximation.[6] To begin we set

$$\rho = \widetilde{\rho}(z) + \delta\rho(x, y, z, t), \qquad p = \widetilde{p}(z) + \delta p(x, y, z, t), \tag{2.117a,b}$$

where we assume that $|\delta\rho| \ll |\widetilde{\rho}|$ and we define $\widetilde{p}$ such that

$$\frac{\partial \widetilde{p}}{\partial z} \equiv -g\widetilde{\rho}(z). \tag{2.118}$$

The notation is similar to that for the Boussinesq case except that, importantly, the density basic state is now a (given) function of vertical coordinate. As with the Boussinesq case, the idea is to ignore dynamic variations of density (i.e., of $\delta\rho$) except where associated with gravity. First recall a couple of ideal gas relationships involving potential temperature, θ, and entropy s (divided by c_p, so $s \equiv \log\theta$), namely

$$s \equiv \log\theta = \log T - \frac{R}{c_p}\log p = \frac{1}{\gamma}\log p - \log\rho, \tag{2.119}$$

where $\gamma = c_p/c_v$, implying

$$\delta s = \frac{1}{\theta}\delta\theta = \frac{1}{\gamma}\frac{\delta p}{p} - \frac{\delta\rho}{\rho} \approx \frac{1}{\gamma}\frac{\delta p}{\widetilde{p}} - \frac{\delta\rho}{\widetilde{\rho}}. \tag{2.120}$$

Further, if $\widetilde{s} \equiv \gamma^{-1}\log\widetilde{p} - \log\widetilde{\rho}$ then

$$\frac{\mathrm{d}\widetilde{s}}{\mathrm{d}z} = \frac{1}{\gamma\widetilde{p}}\frac{\mathrm{d}\widetilde{p}}{\mathrm{d}z} - \frac{1}{\widetilde{\rho}}\frac{\mathrm{d}\widetilde{\rho}}{\mathrm{d}z} = -\frac{g\widetilde{\rho}}{\gamma\widetilde{p}} - \frac{1}{\widetilde{\rho}}\frac{\mathrm{d}\widetilde{\rho}}{\mathrm{d}z}. \tag{2.121}$$

In the atmosphere, the left-hand side is, typically, much smaller than either of the two terms on the right-hand side.

2.5.2 The momentum equation

The exact inviscid horizontal momentum equation is

$$(\widetilde{\rho} + \delta\rho)\left(\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} + \boldsymbol{f}\times\boldsymbol{u}\right) = -\nabla_z\,\delta p. \tag{2.122}$$

Neglecting $\delta\rho$ where it appears with $\widetilde{\rho}$ leads to

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} + \boldsymbol{f}\times\boldsymbol{u} = -\nabla_z\,\phi, \tag{2.123}$$

where $\phi = \delta p/\widetilde{\rho}$, and this is similar to the corresponding equation in the Boussinesq approximation.

The vertical component of the inviscid momentum equation is, without approximation,

$$(\widetilde{\rho} + \delta\rho)\frac{\mathrm{D}w}{\mathrm{D}t} = -\frac{\partial\widetilde{p}}{\partial z} - \frac{\partial\delta p}{\partial z} - g\widetilde{\rho} - g\delta\rho = -\frac{\partial\delta p}{\partial z} - g\delta\rho. \tag{2.124}$$

using (2.118). Neglecting $\delta\rho$ on the left-hand side we obtain

$$\frac{\mathrm{D}w}{\mathrm{D}t} = -\frac{1}{\widetilde{\rho}}\frac{\partial\delta p}{\partial z} - g\frac{\delta\rho}{\widetilde{\rho}} = -\frac{\partial}{\partial z}\left(\frac{\delta p}{\widetilde{\rho}}\right) - \frac{\delta p}{\widetilde{\rho}^2}\frac{\partial\widetilde{\rho}}{\partial z} - g\frac{\delta\rho}{\widetilde{\rho}}. \tag{2.125}$$

This is not a useful form for a gaseous atmosphere, since the variation of the mean density cannot be ignored. However, we may eliminate $\delta\rho$ in favour of δs using (2.120) to give

$$\frac{\mathrm{D}w}{\mathrm{D}t} = g\delta s - \frac{\partial}{\partial z}\left(\frac{\delta p}{\widetilde{\rho}}\right) - \frac{g}{\gamma}\frac{\delta p}{\widetilde{p}} - \frac{\delta p}{\widetilde{\rho}^2}\frac{\partial \widetilde{\rho}}{\partial z}, \tag{2.126}$$

and using (2.121) gives

$$\frac{\mathrm{D}w}{\mathrm{D}t} = g\delta s - \frac{\partial}{\partial z}\left(\frac{\delta p}{\widetilde{\rho}}\right) + \frac{\mathrm{d}\widetilde{s}}{\mathrm{d}z}\frac{\delta p}{\widetilde{\rho}}. \tag{2.127}$$

What have these manipulations gained us? Two things:

(i) The gravitational term now involves δs rather than $\delta\rho$ which enables a more direct connection with the thermodynamic equation.

(ii) The potential temperature scale height (~ 100 km) in the atmosphere is much larger than the density scale height (~ 10 km), and so the last term in (2.127) is small.

The second item thus suggests that we choose our reference state to be one of constant potential temperature (see also problem 2.18). The term $\mathrm{d}\widetilde{s}/\mathrm{d}z$ then vanishes and the vertical momentum equation becomes

$$\boxed{\frac{\mathrm{D}w}{\mathrm{D}t} = g\delta s - \frac{\partial \phi}{\partial z}}, \tag{2.128}$$

where $\phi = \delta p/\widetilde{\rho}$ and $\delta s = \delta\theta/\theta_0$, where θ_0 is a constant. If we define a buoyancy by $b_a \equiv g\delta s = g\delta\theta/\theta_0$, then (2.123) and (2.128) have the same form as the Boussinesq momentum equations, but with a slightly different definition of buoyancy.

2.5.3 Mass conservation

Using (2.117a) the mass conservation equation may be written, without approximation, as

$$\frac{\partial \delta\rho}{\partial t} + \nabla\cdot[(\widetilde{\rho} + \delta\rho)\boldsymbol{v}] = 0. \tag{2.129}$$

We neglect $\delta\rho$ where it appears with $\widetilde{\rho}$ in the divergence term. Further, the local time derivative will be small if time itself is scaled advectively (i.e., $T \sim L/U$ and sound waves do not dominate), giving

$$\nabla\cdot\boldsymbol{u} + \frac{1}{\widetilde{\rho}}\frac{\partial}{\partial z}(\widetilde{\rho}w) = 0. \tag{2.130}$$

It is here that the eponymous anelastic approximation arises: the elastic compressibility of the fluid is neglected, and this serves to eliminate sound waves. For reference, in spherical coordinates the equation is

$$\frac{1}{a\cos\vartheta}\frac{\partial u}{\partial\lambda} + \frac{1}{a\cos\vartheta}\frac{\partial}{\partial\vartheta}(v\cos\vartheta) + \frac{1}{\widetilde{\rho}}\frac{\partial(w\widetilde{\rho})}{\partial z} = 0. \tag{2.131}$$

In an ideal gas, the choice of constant potential temperature determines how the reference density $\widetilde{\rho}$ varies with height. In some circumstances it is convenient to let $\widetilde{\rho}$ be a constant, ρ_0 (effectively choosing a different equation of state), in which case the anelastic equations become identical to the Boussinesq equations, albeit with the buoyancy interpreted in terms of potential temperature in the former and density in the latter. Because of their similarity, the Boussinesq and anelastic approximations are sometimes referred to as the strong and weak Boussinesq approximations, respectively.

2.5.4 Thermodynamic equation

The thermodynamic equation for an ideal gas may be written

$$\frac{\mathrm{D}\ln\theta}{\mathrm{D}t} = \frac{\dot{Q}}{Tc_p}. \tag{2.132}$$

In the anelastic equations, $\theta = \theta_0 + \delta\theta$, where θ_0 is constant, and the thermodynamic equation is

$$\frac{\mathrm{D}\delta s}{\mathrm{D}t} = \frac{\widetilde{\theta}}{Tc_p}\dot{Q}. \tag{2.133}$$

Summarizing, the complete set of anelastic equations, with rotation but with no dissipation or diabatic terms, is

$$\boxed{\begin{gathered} \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + 2\boldsymbol{\Omega}\times\boldsymbol{v} = \mathbf{k}b_a - \nabla\phi \\ \frac{\mathrm{D}b_a}{\mathrm{D}t} = 0 \\ \nabla\cdot(\widetilde{\rho}\boldsymbol{v}) = 0 \end{gathered}}, \tag{2.134a,b,c}$$

where $b_a = g\delta s = g\delta\theta/\theta_0$. The main difference between the anelastic and Boussinesq sets of equations is in the mass continuity equation, and when $\widetilde{\rho} = \rho_0 =$ constant the two equation sets are formally identical. However, whereas the Boussinesq approximation is a very good one for ocean dynamics, the anelastic approximation is much less so for large-scale atmosphere flow: the constancy of the reference potential temperature state is not a particularly good approximation, and the deviations in density from its reference profile are not especially small, leading to inaccuracies in the momentum equation. Nevertheless, the anelastic equations have been used very productively in limited area 'large-eddy simulations' where one does not wish to make the hydrostatic approximation but where sound waves are unimportant.[7] The equations also provide a good jumping-off point for theoretical studies and for the still simpler models of chapter 5.

2.5.5 ◆ Energetics of the anelastic equations

Conservation of energy follows in much the same way as for the Boussinesq equations, except that $\widetilde{\rho}$ enters. Take the dot product of (2.134a) with $\widetilde{\rho}\boldsymbol{v}$ to obtain

$$\widetilde{\rho}\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{1}{2}\boldsymbol{v}^2\right) = -\nabla\cdot(\phi\widetilde{\rho}\boldsymbol{v}) + b_a\widetilde{\rho}w. \tag{2.135}$$

Now, define a potential $\Phi(z)$ such that $\nabla\Phi = -\mathbf{k}$, and so

$$\widetilde{\rho}\frac{\mathrm{D}\Phi}{\mathrm{D}t} = -w\widetilde{\rho}. \tag{2.136}$$

Combining this with the thermodynamic equation (2.134b) gives

$$\widetilde{\rho}\frac{\mathrm{D}(b_a\Phi)}{\mathrm{D}t} = -wb_a\widetilde{\rho}. \tag{2.137}$$

Adding this to (2.135) gives

$$\widetilde{\rho}\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{1}{2}\boldsymbol{v}^2 + b_a\Phi\right) = -\nabla\cdot(\phi\widetilde{\rho}\boldsymbol{v}), \tag{2.138}$$

or, expanding the material derivative,

$$\frac{\partial}{\partial t}\left[\widetilde{\rho}\left(\frac{1}{2}\boldsymbol{v}^2 + b_a\Phi\right)\right] + \nabla\cdot\left[\widetilde{\rho}\boldsymbol{v}\left(\frac{1}{2}\boldsymbol{v}^2 + b_a\Phi + \phi\right)\right] = 0. \tag{2.139}$$

This equation has the form

$$\frac{\partial E}{\partial t} + \nabla\cdot[\boldsymbol{v}(E + \widetilde{\rho}\phi)] = 0, \tag{2.140}$$

where $E = \widetilde{\rho}(\boldsymbol{v}^2/2 + b_a\Phi)$ is the energy density of the flow. This is a consistent energetic equation for the system, and when integrated over a closed domain the total energy is evidently conserved. The total energy density comprises the kinetic energy and a term $\widetilde{\rho}b_a\Phi$, which is analogous to the potential energy of a simple Boussinesq system. However, it is not exactly equal to potential energy because b_a is the buoyancy based on potential temperature, not density; rather, the term combines contributions from both the internal energy and the potential energy into an enthalpy-like quantity.

2.6 CHANGING VERTICAL COORDINATE

Although using z as a vertical coordinate is a natural choice given our Cartesian worldview, it is not the only option, nor is it always the most useful one. Any variable that has a one-to-one correspondence with z in the vertical, so any variable that varies monotonically with z, could be used; pressure and, perhaps surprisingly, entropy, are common choices. In the atmosphere pressure almost always falls monotonically with height, and using it instead of z provides a useful simplification of the mass conservation and geostrophic relations, as well as a more direct connection with observations, which are often taken at fixed values of pressure. (In the ocean pressure coordinates are essentially almost the same as height coordinates because density is almost constant.) Entropy seems an exotic vertical coordinate, but it is very useful in adiabatic flow and we consider it in chapter 3.

2.6.1 General relations

First consider a general vertical coordinate, ξ. Any variable Ψ that is a function of the coordinates (x, y, z, t) may be expressed instead in terms of (x, y, ξ, t) by considering z to be function of the independent variables (x, y, ξ, t); that is, we let $\Psi(x, y, \xi, t) = \Psi(x, y, z(x, y, \xi, t), t)$. Derivatives with respect to z and ξ are related by

$$\frac{\partial\Psi}{\partial\xi} = \frac{\partial\Psi}{\partial z}\frac{\partial z}{\partial\xi} \qquad \text{and} \qquad \frac{\partial\Psi}{\partial z} = \frac{\partial\Psi}{\partial\xi}\frac{\partial\xi}{\partial z}. \tag{2.141a,b}$$

Horizontal derivatives in the two coordinate systems are related by the chain rule,

$$\left(\frac{\partial\Psi}{\partial x}\right)_\xi = \left(\frac{\partial\Psi}{\partial x}\right)_z + \left(\frac{\partial z}{\partial x}\right)_\xi\frac{\partial\Psi}{\partial z}, \tag{2.142}$$

and similarly for time.

The material derivative in ξ coordinates may be derived by transforming the original expression in z coordinates using the chain rule, but because (x, y, t, ξ) are independent coordinates, and noting that the 'vertical velocity' in ξ coordinates is just $\dot{\xi}$ (i.e., $\mathrm{D}\xi/\mathrm{D}t$, just as the vertical velocity in z coordinates is $w = \mathrm{D}z/\mathrm{D}t$), we can write down

$$\frac{\mathrm{D}\Psi}{\mathrm{D}t} = \left.\frac{\partial \Psi}{\partial t}\right|_{x,y,\xi} + \boldsymbol{u} \cdot \nabla_\xi \Psi + \dot{\xi}\frac{\partial \Psi}{\partial \xi}, \tag{2.143}$$

where ∇_ξ is the gradient operator at constant ξ. The operator $\mathrm{D}/\mathrm{D}t$ is physically the same in z or ξ coordinates because it is the total derivative of some property of a fluid parcel, and this is independent of the coordinate system. However, the individual terms within it will differ between coordinate systems.

2.6.2 Pressure coordinates

In pressure coordinates the analogue of the vertical velocity is $\omega \equiv Dp/Dt$, and the advective derivative itself is given by

$$\frac{\mathrm{D}}{\mathrm{D}t} = \frac{\partial}{\partial t} + \boldsymbol{u} \cdot \nabla_p + \omega\frac{\partial}{\partial p}. \tag{2.144}$$

Note, though, that advective derivative is the same operator as it is in height coordinates, since it is just the total derivative of a given fluid parcel; it is just written with different coordinates.

To obtain an expression for the pressure force, now let $\xi = p$ in (2.142) and apply the relationship to p itself to give

$$0 = \left(\frac{\partial p}{\partial x}\right)_z + \left(\frac{\partial z}{\partial x}\right)_p \frac{\partial p}{\partial z}, \tag{2.145}$$

which, using the hydrostatic relationship, gives

$$\left(\frac{\partial p}{\partial x}\right)_z = \rho\left(\frac{\partial \Phi}{\partial x}\right)_p, \tag{2.146}$$

where $\Phi = gz$ is the *geopotential*. Thus, the horizontal pressure force in the momentum equations is

$$\frac{1}{\rho}\nabla_z p = \nabla_p \Phi, \tag{2.147}$$

where the subscripts on the gradient operator indicate that the horizontal derivatives are taken at constant z or constant p. The horizontal momentum equation thus becomes

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} + \boldsymbol{f} \times \boldsymbol{u} = -\nabla_p \Phi, \tag{2.148}$$

where $\mathrm{D}/\mathrm{D}t$ is given by (2.144). The hydrostatic equation in height coordinates is $\partial p/\partial z = -\rho g$ and in pressure coordinates this becomes

$$\frac{\partial \Phi}{\partial p} = -\alpha \qquad \text{or} \qquad \frac{\partial \Phi}{\partial p} = -\frac{p}{RT} \tag{2.149}$$

The mass conservation equation simplifies attractively in pressure coordinates, if the hydrostatic approximation is used. Recall that the mass conservation equation can be derived from the material form

$$\frac{\mathrm{D}}{\mathrm{D}t}(\rho\,\delta V) = 0, \tag{2.150}$$

where $\delta V = \delta x\,\delta y\,\delta z$ is a volume element. But by the hydrostatic relationship $\rho\delta z = -(1/g)\delta p$ and thus

$$\frac{\mathrm{D}}{\mathrm{D}t}(\delta x\,\delta y\,\delta p) = 0. \tag{2.151}$$

This is completely analogous to the expression for the material conservation of volume in an incompressible fluid, (1.15). Thus, without further ado, we write the mass conservation in pressure coordinates as

$$\nabla_p \cdot \boldsymbol{u} + \frac{\partial \omega}{\partial p} = 0, \tag{2.152}$$

where the horizontal derivative is taken at constant pressure.

The (adiabatic) thermodynamic equation is, of course, still $\mathrm{D}\theta/\mathrm{D}t = 0$, and θ may be related to pressure and temperature using its definition and the ideal gas equation to complete the equation set. However, because the hydrostatic equation is written in terms of temperature and not potential temperature it is convenient to write the thermodynamic equation accordingly. To do this we begin with the thermodynamic equation in the form of (1.97b), namely $c_p\mathrm{D}T/\mathrm{D}t - \alpha\mathrm{D}p/\mathrm{D}t = 0$. Since $\omega \equiv \mathrm{D}p/\mathrm{D}t$ this equation is simply

$$c_p\frac{\mathrm{D}T}{\mathrm{D}t} - \frac{RT}{p}\omega = 0, \tag{2.153}$$

which is an appropriate thermodynamic equation in pressure coordinates. It is sometimes useful to write this as

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} - \omega S_p = 0, \tag{2.154}$$

where

$$S_p = \frac{\kappa T}{p} - \frac{\partial T}{\partial p} = -\frac{T}{\theta}\frac{\partial \theta}{\partial p} \tag{2.155}$$

using the ideal gas equation and the definition of potential temperature, with $\kappa = R/c_p$. Evidently, S_p is an appropriate measure of static stability in pressure coordinates and it is closely related to the buoyancy frequency N, as we see in the next subsection.

The main practical difficulty with the pressure-coordinate equations is the lower boundary condition. Using

$$w \equiv \frac{\mathrm{D}z}{\mathrm{D}t} = \frac{\partial z}{\partial t} + \boldsymbol{u}\cdot\nabla_p z + \omega\frac{\partial z}{\partial p}, \tag{2.156}$$

and (2.149), the boundary condition of $w = 0$ at $z = z_s$ becomes

$$\frac{\partial \Phi}{\partial t} + \boldsymbol{u}\cdot\nabla_p\Phi - \alpha\omega = 0 \tag{2.157}$$

at $p(x, y, z_s, t)$. In theoretical studies, it is common to assume that the lower boundary is in fact a constant pressure surface and simply assume that $\omega = 0$, or sometimes the condition $\omega = -\alpha^{-1}\partial\Phi/\partial t$ is used. For realistic studies (with general circulation models, say) the fact that the level $z = 0$ is not a coordinate surface must be properly accounted for. For this

Equations of Motion in Pressure and Log-pressure Coordinates

The adiabatic, inviscid primitive equations in pressure coordinates are:

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} + \boldsymbol{f} \times \boldsymbol{u} = -\nabla_p \Phi, \tag{P.1}$$

$$\frac{\partial \Phi}{\partial p} = \frac{-RT}{p}, \tag{P.2}$$

$$\nabla_p \cdot \boldsymbol{u} + \frac{\partial \omega}{\partial p} = 0, \tag{P.3}$$

$$c_p \frac{\mathrm{D}T}{\mathrm{D}t} - \frac{RT}{p}\omega = 0 \quad \text{or} \quad \frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} - \omega S_p = 0. \tag{P.4}$$

where $S_p = \kappa T/p - \partial T/\partial p$. The above equations are, respectively, the horizontal momentum equation, the hydrostatic equation, the mass continuity equation and the thermodynamic equation.

The corresponding equations in log-pressure coordinates are

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} + \boldsymbol{f} \times \boldsymbol{u} = -\nabla_Z \Phi, \tag{P.5}$$

$$\frac{\partial \Phi}{\partial Z} = \frac{RT}{H}, \tag{P.6}$$

$$\nabla_Z \cdot \boldsymbol{u} + \frac{1}{\rho_R}\frac{\partial \rho_R W}{\partial z} = 0, \tag{P.7}$$

$$c_p \frac{\mathrm{D}T}{\mathrm{D}t} + W\frac{RT}{H} = 0 \quad \text{or} \quad \frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} + W S_Z = 0. \tag{P.8}$$

where $\rho_R = \rho_0 \exp(-Z/H)$ and $S_Z = \kappa T/H + \partial T/\partial Z$. The thermodynamic equation may also be written as

$$\frac{\partial}{\partial t}\frac{\partial \Phi}{\partial Z} + u\frac{\partial}{\partial x}\frac{\partial \Phi}{\partial z} + v\frac{\partial}{\partial y}\frac{\partial \Phi}{\partial z} + WN_*^2 = 0, \tag{P.9}$$

where $N_*^2 = (R/H)S_Z$.

reason, and especially if the lower boundary is uneven because of the presence of topography, so-called *sigma coordinates* are sometimes used, in which the vertical coordinate is chosen so that the lower boundary is a coordinate surface. Sigma coordinates may use height itself as a vertical measure (typical in oceanic applications) or use pressure (typical in atmospheric applications). In the latter case the vertical coordinate is $\sigma = p/p_s$ where $p_s(x, y, t)$ is the surface pressure. The difficulty of applying (2.157) is replaced by a prognostic equation for the surface pressure, derived from the mass conservation equation (problem 2.23).

Finally, we note that the pressure coordinate equations (collected together in the shaded box on the preceding page) are isomorphic to the hydrostatic general Boussinesq equations (see the shaded box on page 74) with $z \Leftrightarrow -p$, $w \Leftrightarrow -\omega$, $\phi \Leftrightarrow \Phi$, $b \Leftrightarrow \alpha$, and an equation of state $b = b(\theta, z) \Leftrightarrow \alpha = \alpha(\theta, p)$. In an ideal gas, for example, $\alpha = (\theta R/p_R)(p_R/p)^{1/\gamma}$. Any conservation laws that hold in one system can therefore be expected to have an analog in the other.

2.6.3 Log-pressure coordinates

A variant of pressure coordinates is *log-pressure* coordinates, in which the vertical coordinate is $Z = -H \ln(p/p_R)$ where p_R is a reference pressure (say 1000 mb) and H is a constant (for example a scale height RT_0/g where T_0 is a constant) so that Z has units of length. (Uppercase letters are conventionally used for some variables in log-pressure coordinates, and these are not to be confused with scaling parameters.) The 'vertical velocity' for the system is now

$$W \equiv \frac{\mathrm{D}Z}{\mathrm{D}t}, \tag{2.158}$$

and the advective derivative is

$$\frac{\mathrm{D}}{\mathrm{D}t} \equiv \frac{\partial}{\partial t} + \boldsymbol{u} \cdot \nabla_p + W\frac{\partial}{\partial Z}. \tag{2.159}$$

The horizontal momentum equation is unaltered from (2.148), although we use (2.159) to evaluate the advective derivative. It is straightforward to show (problem 2.24) that the hydrostatic equation becomes

$$\frac{\partial \Phi}{\partial Z} = \frac{RT}{H} \tag{2.160}$$

The mass continuity equation (2.152) becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial W}{\partial Z} - \frac{W}{H} = 0. \tag{2.161}$$

which may be written as

$$\nabla_Z \cdot \boldsymbol{u} + \frac{1}{\rho_R}\frac{\partial(\rho_R W)}{\partial z} = 0 \tag{2.162}$$

where $\rho_R = \rho_0 \exp(-Z/H)$, so giving a form similar to the mass conservation equation in the anelastic equations. (The value of the constant ρ_0 may be set to one.)

As with pressure coordinates, it is convenient to write the thermodynamic equation in terms of temperature and not potential temperature, and in an analogous procedure to that leading to (2.153) we straightforwardly obtain

$$c_p\frac{\mathrm{D}T}{\mathrm{D}t} + W\frac{RT}{H} = 0. \tag{2.163}$$

This may be written as

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} + WS_Z = 0 \tag{2.164}$$

where

$$S_Z = \frac{\kappa T}{H} + \frac{\partial T}{\partial Z}. \tag{2.165}$$

and we may note that $S_Z = S_p p/H$. Using the hydrostatic equation we may write (2.164) as

$$\frac{\partial}{\partial t}\frac{\partial \Phi}{\partial Z} + u\frac{\partial}{\partial x}\frac{\partial \Phi}{\partial z} + v\frac{\partial}{\partial y}\frac{\partial \Phi}{\partial z} + WN_*^2 = 0, \tag{2.166}$$

where $N_*^2 = (R/H)S_Z$. The quantity N_* is not exactly equal to the square of the buoyancy frequency as normally defined (for an ideal gas $N^2 = (g/\theta)\partial\theta/\partial z$), but the two can be shown to be related by $N_*/N = p/(\rho g H) = RT/gH$, and are equal for an isothermal atmosphere.[8]

Finally, we note that integrating the hydrostatic equation between two pressure levels gives, with $\Phi = gz$,

$$z(p_2) - z(p_1) = -\frac{R}{g}\int_{p_1}^{p_2} T\,\mathrm{d}\ln p. \tag{2.167}$$

Thus, the thickness of the layer is proportional to the average temperature of the layer.

2.7 SCALING FOR HYDROSTATIC BALANCE

We first encountered hydrostatic balance in section 1.3.4 and we imposed it when using pressure coordinates; we now look in more detail at the conditions required for it to hold. Along with geostrophic balance, considered in the next section, it is one of the most fundamental balances in geophysical fluid dynamics. The corresponding states, hydrostasy and geostrophy, are not exactly realized, but their approximate satisfaction has profound consequences on the behaviour of the atmosphere and ocean.

2.7.1 Preliminaries

Consider the relative sizes of terms in (2.77c):

$$\frac{W}{T} + \frac{UW}{L} + \frac{W^2}{H} + \Omega U \sim \left|\frac{1}{\rho}\frac{\partial p}{\partial z}\right| + g. \tag{2.168}$$

For most large-scale motion in the atmosphere and ocean the terms on the right-hand side are orders of magnitude larger than those on the left, and therefore must be approximately equal. Explicitly, suppose $W \sim 1\,\mathrm{cm\,s^{-1}}$, $L \sim 10^5\,\mathrm{m}$, $H \sim 10^3\,\mathrm{m}$, $U \sim 10\,\mathrm{m\,s^{-1}}$, $T = L/U$. Then by substituting into (2.168) it seems that the pressure term is the only one which could balance the gravitational term, and we are led to approximate (2.77c) by,

$$\frac{\partial p}{\partial z} = -\rho g. \tag{2.169}$$

This equation, which is a vertical momentum equation, is known as *hydrostatic balance*.

However, (2.169) is not always a useful equation! Let us suppose that the density is a constant, ρ_0 . We can then write the pressure as

$$p(x, y, z, t) = p_0(z) + p'(x, y, z, t), \tag{2.170}$$

where

$$\frac{\partial p_0}{\partial z} \equiv -\rho_0 g. \tag{2.171}$$

That is, p_0 and ρ_0 are in hydrostatic balance. On the f-plane, the inviscid vertical momentum equation becomes, without approximation,

$$\frac{\mathrm{D}w}{\mathrm{D}t} = -\frac{1}{\rho_0}\frac{\partial p'}{\partial z}. \tag{2.172}$$

Thus, *for constant density fluids the gravitational term has no dynamical effect:* there is no buoyancy force, and the pressure term in the horizontal momentum equations can be replaced by p'. Hydrostatic balance, and in particular (2.171), is certainly not an appropriate vertical momentum equation in this case. If the fluid is stratified, we should therefore subtract off the hydrostatic pressure associated with the mean density before we can determine whether hydrostasy is a useful *dynamical* approximation, accurate enough to determine the horizontal pressure gradients. This is automatic in the Boussinesq equations, where the vertical momentum equation is

$$\frac{\mathrm{D}w}{\mathrm{D}t} = -\frac{\partial \phi}{\partial z} + b, \tag{2.173}$$

and the hydrostatic balance of the basic state is already subtracted out. In the more general equation,

$$\frac{\mathrm{D}w}{\mathrm{D}t} = -\frac{1}{\rho}\frac{\partial p}{\partial z} - g, \tag{2.174}$$

we need to compare the advective term on the left-hand side with the pressure variations arising from horizontal flow in order to determine whether hydrostasy is an appropriate vertical momentum equation. Nevertheless, if we only need to determine the pressure for use in an equation of state then we simply need to compare the sizes of the dynamical terms in (2.77c) with g itself, in order to determine whether a hydrostatic approximation will suffice.

2.7.2 Scaling and the aspect ratio

In a Boussinesq fluid we write the horizontal and vertical momentum equations as

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} + \boldsymbol{f} \times \boldsymbol{u} = -\nabla_z \phi, \qquad \frac{\mathrm{D}w}{\mathrm{D}t} = -\frac{\partial \phi}{\partial z} + b. \tag{2.175a,b}$$

With $\boldsymbol{f} = 0$, (2.175a) implies the scaling

$$\phi \sim U^2. \tag{2.176}$$

If we use mass conservation, $\nabla_z \cdot \boldsymbol{u} + \partial w/\partial z = 0$, to scale vertical velocity then

$$w \sim W = \frac{H}{L}U = \alpha U, \tag{2.177}$$

where $\alpha \equiv H/L$ is the aspect ratio. The advective terms in the vertical momentum equation all scale as

$$\frac{\mathrm{D}w}{\mathrm{D}t} \sim \frac{UW}{L} = \frac{U^2 H}{L^2}. \tag{2.178}$$

Using (2.176) and (2.178) the ratio of the advective term to the pressure gradient term in the vertical momentum equations then scales as

$$\frac{|\mathrm{D}w/\mathrm{D}t|}{|\partial\phi/\partial z|} \sim \frac{U^2 H/L^2}{U^2/H} \sim \left(\frac{H}{L}\right)^2. \tag{2.179}$$

Thus, the condition for hydrostasy, that $|\mathrm{D}w/\mathrm{D}t|/|\partial\phi/\partial z| \ll 1$, is:

$$\boxed{\alpha^2 \equiv \left(\frac{H}{L}\right)^2 \ll 1}. \tag{2.180}$$

The advective term in the vertical momentum may then be neglected. Thus, hydrostatic balance is a *small aspect ratio approximation*.

We can obtain the same result more formally by non-dimensionalizing the momentum equations. Using uppercase symbols to denote scaling values we write

$$
\begin{gathered}
(x, y) = L(\hat{x}, \hat{y}), \qquad z = H\hat{z}, \qquad \boldsymbol{u} = U\hat{\boldsymbol{u}}, \qquad w = W\hat{w} = \frac{HU}{L}\hat{w}, \\
t = T\hat{t} = \frac{L}{U}\hat{t}, \qquad \phi = \Phi\hat{\phi} = U^2\hat{\phi}, \qquad b = B\hat{b} = \frac{U^2}{H}\hat{b},
\end{gathered}
\tag{2.181}
$$

where the hatted variables are non-dimensional and the scaling for w is suggested by the mass conservation equation, $\nabla_z \cdot \boldsymbol{u} + \partial w/\partial z = 0$. Substituting (2.181) into (2.175) (with $\boldsymbol{f} = 0$) gives us the non-dimensional equations

$$
\frac{\mathrm{D}\hat{\boldsymbol{u}}}{\mathrm{D}\hat{t}} = -\nabla\hat{\phi}, \qquad \alpha^2 \frac{\mathrm{D}\hat{w}}{\mathrm{D}\hat{t}} = -\frac{\partial\hat{\phi}}{\partial\hat{z}} + \hat{b},
\tag{2.182a,b}
$$

where $\mathrm{D}/\mathrm{D}\hat{t} = \partial/\partial\hat{t} + \hat{u}\partial/\partial\hat{x} + \hat{v}\partial/\partial\hat{y} + \hat{w}\partial/\partial\hat{z}$ and we use the convention that when ∇ operates on non-dimensional quantities the operator itself is non-dimensional. From (2.182b) it is clear that hydrostatic balance pertains when $\alpha^2 \ll 1$.

2.7.3 ♦ Effects of stratification on hydrostatic balance

To include the effects of stratification we need to involve the thermodynamic equation, so let us first write down the complete set of non-rotating dimensional equations:

$$
\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} = -\nabla_z \phi, \qquad \frac{\mathrm{D}w}{\mathrm{D}t} = -\frac{\partial\phi}{\partial z} + b',
\tag{2.183a,b}
$$

$$
\frac{\mathrm{D}b'}{\mathrm{D}t} + wN^2 = 0, \qquad \nabla \cdot \boldsymbol{v} = 0.
\tag{2.184a,b}
$$

We have written, without approximation, $b = b'(x, y, z, t) + \widetilde{b}(z)$, with $N^2 = \mathrm{d}\widetilde{b}/\mathrm{d}z$; this separation is useful because the horizontal and vertical buoyancy variations may scale in different ways, and often N^2 may be regarded as given. (We have also redefined ϕ by subtracting off a static component in hydrostatic balance with $\widetilde{b}$.) We non-dimensionalize (2.184) by first writing

$$
\begin{gathered}
(x, y) = L(\hat{x}, \hat{y}), \qquad z = H\hat{z}, \qquad \boldsymbol{u} = U\hat{\boldsymbol{u}}, \quad w = W\hat{w} = \epsilon\frac{HU}{L}\hat{w}, \\
t = T\hat{t} = \frac{L}{U}\hat{t}, \qquad \phi = U^2\hat{\phi}, \qquad b' = \Delta b\hat{b}' = \frac{U^2}{H}\hat{b}', \qquad N^2 = \overline{N}^2\hat{N}^2,
\end{gathered}
\tag{2.185}
$$

where ϵ is, for the moment, undetermined, $\overline{N}$ is a representative constant value of the buoyancy frequency and Δb scales only the horizontal buoyancy variations. Substituting (2.185) into (2.183) and (2.184) gives

$$
\frac{\mathrm{D}\hat{\boldsymbol{u}}}{\mathrm{D}\hat{t}} = -\nabla_z\hat{\phi}, \qquad \epsilon\alpha^2 \frac{\mathrm{D}\hat{w}}{\mathrm{D}t} = -\frac{\partial\hat{\phi}}{\partial\hat{z}} + \hat{b}'
\tag{2.186a,b}
$$

$$
\frac{U^2}{\overline{N}^2 H^2}\frac{\mathrm{D}\hat{b}'}{\mathrm{D}\hat{t}} + \epsilon\hat{w}\hat{N}^2 = 0, \qquad \nabla \cdot \hat{\boldsymbol{u}} + \epsilon\frac{\partial\hat{w}}{\partial\hat{z}} = 0.
\tag{2.187a,b}
$$

where now $\mathrm{D}/\mathrm{D}\hat{t} = \partial/\partial\hat{t} + \hat{\boldsymbol{u}}\cdot\nabla_z + \epsilon\hat{w}\partial/\partial\hat{z}$. To obtain a non-trivial balance in (2.187a) we choose $\epsilon = U^2/(\overline{N}^2H^2) \equiv Fr^2$, where Fr is the *Froude number,* a measure of the stratification of the flow. A strong stratification corresponds to a small Froude number. From (2.185), the vertical velocity then scales as

$$W = \frac{Fr^2\,UH}{L} \tag{2.188}$$

and if the flow is highly stratified the vertical velocity will be even smaller than a pure aspect ratio scaling might suggest. (There must, therefore, be some cancellation in horizontal divergence in the mass continuity equation; that is, $|\nabla_z\cdot\boldsymbol{u}| \ll U/L$.) With this choice of ϵ the non-dimensional Boussinesq equations may be written:

$$\frac{\mathrm{D}\hat{\boldsymbol{u}}}{\mathrm{D}\hat{t}} = -\nabla_z\,\hat{\phi}, \qquad Fr^2\alpha^2\frac{\mathrm{D}\hat{w}}{\mathrm{D}\hat{t}} = -\frac{\partial\hat{\phi}}{\partial\hat{z}} + \hat{b}' \tag{2.189a,b}$$

$$\frac{\mathrm{D}\hat{b}'}{\mathrm{D}\hat{t}} + \hat{w}\hat{N}^2 = 0, \qquad \nabla\cdot\hat{\boldsymbol{u}} + Fr^2\frac{\partial\hat{w}}{\partial\hat{z}} = 0. \tag{2.190a,b}$$

The non-dimensional parameters in the system are the aspect ratio and the Froude number (in addition to $\hat{N}$, but by construction this is just an order one function of z). From (2.189b) condition for hydrostatic balance to hold is evidently that

$$\boxed{Fr^2\alpha^2 \ll 1}, \tag{2.191}$$

so generalizing the aspect ratio condition (2.180) to a stratified fluid. Because Fr is a measure of stratification, (2.191) formalizes our intuitive expectation that the more stratified a fluid the more vertical motion is suppressed and therefore the more likely hydrostatic balance is to hold. Also note that (2.191) is equivalent to $U^2/(L^2\overline{N}^2) \ll 1$.

Suppose we solve the hydrostatic equations; that is, we omit the advective derivative in the vertical momentum equation, and by numerical integration we obtain $\boldsymbol{u}$, w and b. This flow is the solution of the non-hydrostatic equations in the small aspect ratio limit. The solution never violates the scaling assumptions, even if w seems large, because we can always rescale the variables in order that condition (2.191) is satisfied.

Why bother with any of this scaling? Why not just say that hydrostatic balance holds when $|Dw/Dt| \ll |\partial\phi/\partial z|$? One reason is that we do not have a good idea of the value of w from direct measurements, and it may change significantly in different oceanic and atmospheric parameter regimes. On the other hand the Froude number and the aspect ratio are familiar non-dimensional parameters with a wide applicability in other contexts, and which we can control in a laboratory setting or estimate in the ocean or atmosphere. Still, when equations are scaled, ascertaining which parameters are to be regarded as given and which should be derived is often a choice, rather than being set a priori.

2.7.4 Hydrostasy in the ocean and atmosphere

Is the hydrostatic approximation in fact a good one in the ocean and atmosphere?

In the ocean

For the large-scale ocean circulation, let $N \sim 10^{-2}\,\mathrm{s}^{-1}$, $U \sim 0.1\,\mathrm{m\,s}^{-1}$ and $H \sim 1\,\mathrm{km}$. Then $Fr = U/(NH) \sim 10^{-2} \ll 1$. Thus, $Fr^2\alpha^2 \ll 1$ even for unit aspect-ratio motion. In fact, for larger scale flow the aspect ratio is also small; for basin-scale flow $L \sim 10^6\,\mathrm{m}$ and $Fr^2\alpha^2 \sim 0.01^2 \times 0.001^2 = 10^{-10}$ and hydrostatic balance is an extremely good approximation.

For intense convection, for example in the Labrador Sea, the hydrostatic approximation may be less appropriate, because the intense descending plumes may have an aspect ratio (H/L) of one or greater and the stratification is very weak. The hydrostatic condition then often becomes the requirement that the Froude number is small. Representative orders of magnitude are $U \sim W \sim 0.1\,\mathrm{m\,s}^{-1}$, $H \sim 1\,\mathrm{km}$ and $N \sim 10^{-3}\,\mathrm{s}^{-1}$ to $10^{-4}\,\mathrm{s}^{-1}$. For these values Fr ranges between 0.1 and 1, and at the upper end of this range hydrostatic balance is violated.

In the atmosphere

Over much of the troposphere $N \sim 10^{-2}\,\mathrm{s}^{-1}$ so that with $U = 10\,\mathrm{m\,s}^{-1}$ and $H = 1\,\mathrm{km}$ we find $Fr \sim 1$. Hydrostasy is then maintained because the aspect ratio H/L is much less than unity. For larger scale synoptic activity a larger vertical scale is appropriate, and with $H = 10\,\mathrm{km}$ both the Froude number and the aspect ratio are much smaller than one; indeed with $L = 1000\,\mathrm{km}$ we find $Fr^2\alpha^2 \sim 0.1^2 \times 0.01^2 = 10^{-6}$ and the flow is hydrostatic to a very good approximation indeed. However, for smaller scale atmospheric motions associated with fronts and, especially, convection, there can be little expectation that hydrostatic balance will be a good approximation.

For large-scale flows in both atmosphere and ocean, the conceptual simplifications afforded by the hydrostatic approximation can hardly be overemphasized.

2.8 GEOSTROPHIC AND THERMAL WIND BALANCE

We now consider the dominant dynamical balance in the horizontal components of the momentum equation. In the horizontal plane (meaning along geopotential surfaces) we find that the Coriolis term is much larger than the advective terms and the dominant balance is between it and the horizontal pressure force. This balance is called *geostrophic balance,* and it occurs when the Rossby number is small, as we now investigate.

2.8.1 The Rossby number

The *Rossby number* characterizes the importance of rotation in a fluid.[9] It is, essentially, the ratio of the magnitude of the relative acceleration to the Coriolis acceleration, and it is of fundamental importance in geophysical fluid dynamics. It arises from a simple scaling of the horizontal momentum equation, namely

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{v}\cdot\nabla)\boldsymbol{u} + \boldsymbol{f}\times\boldsymbol{u} = -\frac{1}{\rho}\nabla_z p, \tag{2.192a}$$

$$U^2/L \qquad fU \tag{2.192b}$$

where U is the approximate magnitude of the horizontal velocity and L is a typical length scale over which that velocity varies. (We assume that $W/H \lesssim U/L$, so that vertical advection does

Table 2.1 Scales of large-scale flow in atmosphere and ocean. The choices given are representative of large-scale mid-latitude eddying motion in both systems.

Variable	Scaling symbol	Meaning	Atmos. value	Ocean value
(x, y)	L	Horizontal length scale	10^6 m	10^5 m
t	T	Time scale	1 day (10^5 s)	10 days (10^6 s)
(u, v)	U	Horizontal velocity	$10\,\mathrm{m\,s^{-1}}$	$0.1\,\mathrm{m\,s^{-1}}$
	Ro	Rossby number, U/fL	0.1	0.01

not dominate the advection.) The ratio of the sizes of the advective and Coriolis terms is defined to be the Rossby number,

$$\boxed{Ro \equiv \frac{U}{fL}}. \tag{2.193}$$

If the Rossby number is small then rotation effects are important, and as the values in Table 2.1 indicate this is the case for large-scale flow in both ocean and atmosphere.

Another intuitive way to think about the Rossby number is in terms of time scales. The Rossby number based on a time scale is

$$Ro_T \equiv \frac{1}{fT}, \tag{2.194}$$

where T is a time scale associated with the dynamics at hand. If the time scale is an advective one, meaning that $T \sim L/U$, then this definition is equivalent to (2.193). Now, $f = 2\Omega \sin\vartheta$, where Ω is the angular velocity of the rotating frame and equal to $2\pi/T_p$ where T_p is the period of rotation (24 hours). Thus,

$$Ro_T = \frac{T_p}{4\pi T \sin\vartheta} = \frac{T_i}{T}, \tag{2.195}$$

where $T_i = 1/f$ is the 'inertial time scale', about three hours in mid-latitudes. Thus, for phenomena with time scales much longer than this, such as the motion of the Gulf Stream or a mid-latitude atmospheric weather system, the effects of the Earth's rotation can be expected to be important, whereas a short-lived phenomena, such as a cumulus cloud or tornado, may be oblivious to such rotation. The expressions (2.193) and (2.194) are, of course, just approximate measures of the importance of rotation.

2.8.2 Geostrophic balance

If the Rossby number is sufficiently small in (2.192a) then the rotation term will dominate the nonlinear advection term, and if the time period of the motion scales advectively then the rotation term also dominates the local time derivative. The only term that can then balance

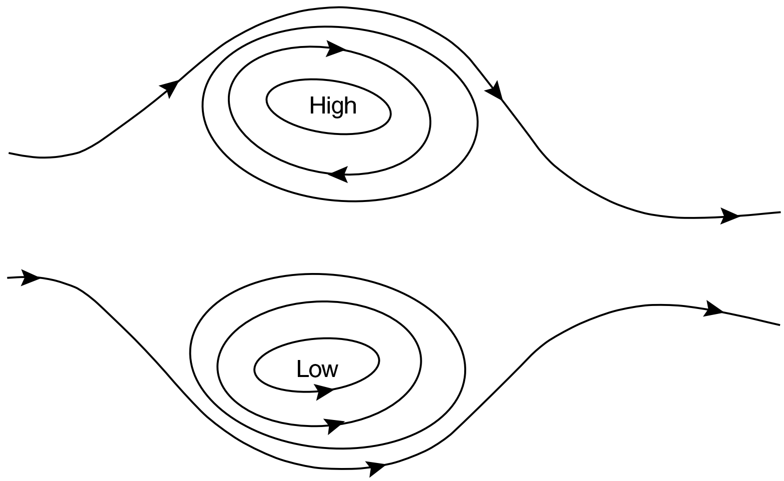

Fig. 2.5 Schematic of geostrophic flow with a positive value of the Coriolis parameter f. Flow is parallel to the lines of constant pressure (isobars). Cyclonic flow is anticlockwise around a low pressure region and anticyclonic flow is clockwise around a high. If f were negative, as in the Southern Hemisphere, (anti)cyclonic flow would be (anti)clockwise.

the rotation term is the pressure term, and therefore we must have

$$\boldsymbol{f} \times \boldsymbol{u} \approx -\frac{1}{\rho}\nabla_z p, \tag{2.196}$$

or, in Cartesian component form

$$fu \approx -\frac{1}{\rho}\frac{\partial p}{\partial y}, \qquad fv \approx \frac{1}{\rho}\frac{\partial p}{\partial x}. \tag{2.197}$$

This balance is known as *geostrophic balance,* and its consequences are profound, giving geophysical fluid dynamics a special place in the broader field of fluid dynamics. We *define* the geostrophic velocity by

$$\boxed{fu_g \equiv -\frac{1}{\rho}\frac{\partial p}{\partial y}, \qquad fv_g \equiv \frac{1}{\rho}\frac{\partial p}{\partial x}}, \tag{2.198}$$

and for low Rossby number flow $u \approx u_g$ and $v \approx v_g$. In spherical coordinates the geostrophic velocity is

$$fu_g = -\frac{1}{\rho a}\frac{\partial p}{\partial \vartheta}, \qquad fv_g = \frac{1}{a\rho\cos\vartheta}\frac{\partial p}{\partial \lambda}, \tag{2.199}$$

where $f = 2\Omega\sin\vartheta$. Geostrophic balance has a number of immediate ramifications:

⋆ Geostrophic flow is parallel to lines of constant pressure (isobars). If $f > 0$ the flow is anticlockwise round a region of low pressure and clockwise around a region of high pressure (see Fig. 2.5).

- ⋆ If the Coriolis force is constant and if the density does not vary in the horizontal the geostrophic flow is horizontally non-divergent and

$$\boxed{\nabla_z \cdot \boldsymbol{u}_g = \frac{\partial u_g}{\partial x} + \frac{\partial v_g}{\partial y} = 0}. \tag{2.200}$$

We may define the *geostrophic streamfunction,* ψ, by

$$\psi \equiv \frac{p}{f_0 \rho_0}, \tag{2.201}$$

whence

$$u_g = -\frac{\partial \psi}{\partial y}, \qquad v_g = \frac{\partial \psi}{\partial x}. \tag{2.202}$$

The vertical component of vorticity, ζ, is then given by

$$\zeta = \mathbf{k} \cdot \nabla \times \boldsymbol{v} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = \nabla_z^2 \psi. \tag{2.203}$$

- ⋆ If the Coriolis parameter is not constant, then cross-differentiating (2.198) gives, for constant density geostrophic flow,

$$v_g \frac{\partial f}{\partial y} + f \nabla_z \cdot \boldsymbol{u}_g = 0, \tag{2.204}$$

which, using the mass continuity equation $\nabla_z \cdot \boldsymbol{u}_g = -\partial w / \partial z$,

$$\beta v_g = f \frac{\partial w}{\partial z}. \tag{2.205}$$

where $\beta \equiv \partial f / \partial y = 2\Omega \cos \vartheta / a$. This geostrophic vorticity balance is sometimes known as 'Sverdrup balance', although that expression is better restricted to the case when the vertical velocity from a wind stress, as considered in chapter 19.

2.8.3 Taylor–Proudman effect

If $\beta = 0$, then (2.205) implies that the vertical velocity is not a function of height. In fact, in that case none of the components of velocity vary with height if density is also constant. To show this, in the limit of zero Rossby number we first write the three-dimensional momentum equation as

$$\boldsymbol{f}_0 \times \boldsymbol{v} = -\nabla \phi - \nabla \chi, \tag{2.206}$$

where $\boldsymbol{f}_0 = 2\boldsymbol{\Omega} = 2\Omega\mathbf{k}$, $\phi = p/\rho_0$, and $\nabla \chi$ represents other potential forces. If $\chi = gz$ then the vertical component of this equation represents hydrostatic balance, and the horizontal components represent geostrophic balance. On taking the curl of this equation, the terms on the right-hand side vanish and the left-hand side becomes

$$(\boldsymbol{f}_0 \cdot \nabla)\boldsymbol{v} - \boldsymbol{f}_0 \nabla \cdot \boldsymbol{v} - (\boldsymbol{v} \cdot \nabla)\boldsymbol{f}_0 + \boldsymbol{v} \nabla \cdot \boldsymbol{f}_0 = 0. \tag{2.207}$$

But $\nabla \cdot \boldsymbol{v} = 0$ by mass conservation, and because $\boldsymbol{f}_0$ is constant both $\nabla \cdot \boldsymbol{f}_0$ and $(\boldsymbol{v} \cdot \nabla)\boldsymbol{f}_0$ vanish. Equation (2.207) thus reduces to

$$(\boldsymbol{f}_0 \cdot \nabla)\boldsymbol{v} = 0, \tag{2.208}$$

which, since $\boldsymbol{f}_0 = f_0\mathbf{k}$, implies $f_0 \partial \boldsymbol{v}/\partial z = 0$, and in particular we have

$$\frac{\partial u}{\partial z} = 0, \quad \frac{\partial v}{\partial z} = 0, \quad \frac{\partial w}{\partial z} = 0. \tag{2.209}$$

A different presentation of this argument proceeds as follows. If the flow is exactly in geostrophic and hydrostatic balance then

$$v = \frac{1}{f_0}\frac{\partial \phi}{\partial x}, \qquad u = -\frac{1}{f_0}\frac{\partial \phi}{\partial y}, \qquad \frac{\partial \phi}{\partial z} = -g. \tag{2.210a,b,c}$$

Differentiating (2.210a,b) with respect to z, and using (2.210c) yields

$$\frac{\partial v}{\partial z} = \frac{-1}{f_0}\frac{\partial g}{\partial x} = 0, \quad \frac{\partial u}{\partial z} = \frac{1}{f_0}\frac{\partial g}{\partial y} = 0. \tag{2.211}$$

Noting that the geostrophic velocities are horizontally non-divergent ($\nabla_z \cdot \boldsymbol{u} = 0$), and using mass continuity then gives $\partial w/\partial z = 0$, as before.

If there is a solid horizontal boundary anywhere in the fluid, for example at the surface, then $w = 0$ at that surface and thus $w = 0$ everywhere. Hence the motion occurs in planes that lie perpendicular to the axis of rotation, and the flow is effectively two dimensional. This result is known as the *Taylor–Proudman effect,* namely that for constant density flow in geostrophic and hydrostatic balance the vertical derivatives of the horizontal and the vertical velocities are zero.[10] At zero Rossby number, if the vertical velocity is zero somewhere in the flow, it is zero everywhere in that vertical column; furthermore, the horizontal flow has no vertical shear, and the fluid moves like a slab. The effects of rotation have provided a *stiffening* of the fluid in the vertical.

In neither the atmosphere nor the ocean do we observe precisely such vertically coherent flow, mainly because of the effects of stratification. However, it is typical of geophysical fluid dynamics that the assumptions underlying a derivation are not fully satisfied, yet there are manifestations of it in real flow. Thus, one might have naïvely expected, because $\partial w/\partial z = -\nabla_z \cdot \boldsymbol{u}$, that the scales of the various variables would be related by $W/H \sim U/L$. However, if the flow is rapidly rotating we expect that the horizontal flow will be in near geostrophic balance and therefore nearly divergence free; thus $\nabla_z \cdot \boldsymbol{u} \ll U/L$, and $W \ll HU/L$.

2.8.4 Thermal wind balance

Thermal wind balance arises by combining the geostrophic and hydrostatic approximations, and this is most easily done in the context of the anelastic (or Boussinesq) equations, or in pressure coordinates. For the anelastic equations, geostrophic balance may be written

$$-fv_g = -\frac{\partial \phi}{\partial x} = -\frac{1}{a\cos\vartheta}\frac{\partial \phi}{\partial \lambda}, \qquad fu_g = -\frac{\partial \phi}{\partial y} = -\frac{1}{a}\frac{\partial \phi}{\partial \vartheta}. \tag{2.212a,b}$$

Combining these relations with hydrostatic balance, $\partial\phi/\partial z = b$, gives

$$\boxed{\begin{aligned} -f\frac{\partial v_g}{\partial z} &= -\frac{\partial b}{\partial x} = -\frac{1}{a\cos\lambda}\frac{\partial b}{\partial\lambda} \\ f\frac{\partial u_g}{\partial z} &= -\frac{\partial b}{\partial y} = -\frac{1}{a}\frac{\partial b}{\partial\vartheta} \end{aligned}}\ . \tag{2.213a,b}$$

These equations represent *thermal wind balance,* and the vertical derivative of the geostrophic wind is the 'thermal wind'. Eq. (2.213b) may be written in terms of the zonal angular momentum as

$$\frac{\partial m_g}{\partial z} = -\frac{a}{2\Omega\tan\vartheta}\frac{\partial b}{\partial y}, \tag{2.214}$$

where $m_g = (u_g + \Omega a\cos\vartheta)a\cos\vartheta$. Potentially more accurate than geostrophic balance is the so-called cyclostrophic balance (related to gradient wind balance discussed in section **??**), which retains a centrifugal term in the momentum equation. Thus, we omit only the material derivative in the meridional momentum equation (2.50b) and obtain

$$2u\Omega\sin\vartheta + \frac{u^2}{a}\tan\vartheta \approx -\frac{\partial\phi}{\partial y} = -\frac{1}{a}\frac{\partial\phi}{\partial\vartheta}. \tag{2.215}$$

For large-scale flow this only differs significantly from geostrophic balance very close to the equator. Taking the vertical derivative of (2.215) and using the hydrostatic relation $\partial\phi/\partial z = b$ gives a modified thermal wind relation, and this may be put in the simple form

$$\frac{\partial m^2}{\partial z} = -\frac{a^3\cos^3\vartheta}{\sin\vartheta}\frac{\partial b}{\partial y}, \tag{2.216}$$

where $m = (u + \Omega a\cos\vartheta)a\cos\vartheta$ is the annular angular momentum.

If the density or buoyancy is constant then there is no shear and (2.213) or (2.216) give the Taylor–Proudman result. But suppose that the temperature falls in the poleward direction. Then thermal wind balance implies that the (eastward) wind will increase with height — just as is observed in the atmosphere! In general, a vertical shear of the horizontal wind is associated with a horizontal temperature gradient, and this is one of the most simple and far-reaching effects in geophysical fluid dynamics. The underlying physical mechanism is illustrated in Fig. 2.6.

Pressure coordinates

In pressure coordinates geostrophic balance is just

$$\boldsymbol{f} \times \boldsymbol{u}_g = -\nabla_p \Phi, \tag{2.217}$$

where Φ is the geopotential and ∇_p is the gradient operator taken at constant pressure. If f iss constant, it follows from (2.217) that the geostrophic wind is non-divergent on pressure surfaces. Taking the vertical derivative of (2.217) (that is, its derivative with respect to p) and using the hydrostatic equation, $\partial\Phi/\partial p = -\alpha$, gives the thermal wind equation

$$\boldsymbol{f} \times \frac{\partial \boldsymbol{u}_g}{\partial p} = \nabla_p\,\alpha = \frac{R}{p}\nabla_p T, \tag{2.218}$$

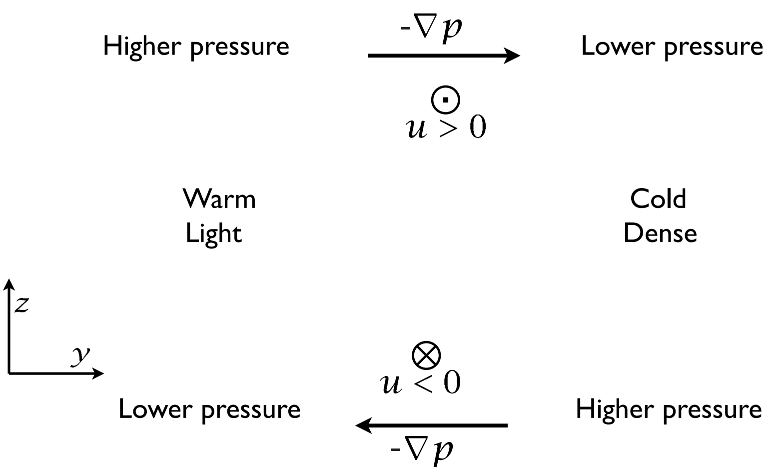

Fig. 2.6 The mechanism of thermal wind. A cold fluid is denser than a warm fluid, so by hydrostasy the vertical pressure gradient is greater where the fluid is cold. Thus, the pressure gradients form as shown, where 'higher' and 'lower' mean relative to the average at that height. The horizontal pressure gradients are balanced by the Coriolis force, producing (for $f > 0$) the horizontal winds shown ($\otimes$ into the paper, and $\odot$ out of the paper). Only the wind *shear* is given by the thermal wind.

where the last equality follows using the ideal gas equation and because the horizontal derivative is at constant pressure. In component form this is

$$-f\frac{\partial v_g}{\partial p} = \frac{R}{p}\frac{\partial T}{\partial x}, \qquad f\frac{\partial u_g}{\partial p} = \frac{R}{p}\frac{\partial T}{\partial y}. \tag{2.219}$$

In log-pressure coordinates, with $Z = -H\ln(p/p_R)$, thermal wind is

$$\boldsymbol{f}\times\frac{\partial \boldsymbol{u}_g}{\partial Z} = -\frac{R}{H}\nabla_Z T. \tag{2.220}$$

The physical meaning in all these cases is the same: a horizontal temperature gradient, or a temperature gradient along an isobaric surface, is accompanied by a vertical shear of the horizontal wind.

2.8.5 ♦ Effects of rotation on hydrostatic balance

Because rotation inhibits vertical motion, we might expect it to affect the requirements for hydrostasy. The simplest setting in which to see this is the rotating Boussinesq equations, (2.175). Let us non-dimensionalize these by writing

$$\begin{gathered}(x,y) = L(\hat{x},\hat{y}), \qquad z = H\hat{z}, \qquad \boldsymbol{u} = U\hat{\boldsymbol{u}}, \qquad t = T\hat{t} = \frac{L}{U}\hat{t}, \qquad \boldsymbol{f} = f_0\hat{\boldsymbol{f}}, \\ w = \frac{\beta HU}{f_0}\hat{w} = \hat{\beta}\frac{HU}{L}\hat{w}, \qquad \phi = \Phi\hat{\phi} = f_0UL\hat{\phi}, \qquad b = B\hat{b} = \frac{f_0UL}{H}\hat{b},\end{gathered} \tag{2.221}$$

where $\widehat{\beta} \equiv \beta L/f_0$. (If $\boldsymbol{f}$ is constant, then $\widehat{\boldsymbol{f}}$ is a unit vector in the vertical direction.) These relations are the same as (2.181), except for the scaling for w, which is suggested by (2.205), and the scaling for ϕ and b', which are suggested by geostrophic and thermal wind balance. Substituting these into (2.175) we obtain the following scaled momentum equations:

$$\boxed{Ro\,\frac{\mathrm{D}\widehat{\boldsymbol{u}}}{\mathrm{D}\widehat{t}} + \widehat{\boldsymbol{f}} \times \widehat{\boldsymbol{u}} = -\nabla\widehat{\phi}, \qquad Ro\,\widehat{\beta}\alpha^2\frac{\mathrm{D}\widehat{w}}{\mathrm{D}\widehat{t}} = -\frac{\partial\widehat{\phi}}{\partial\widehat{z}} - \widehat{b}}\,. \tag{2.222a,b}$$

Here, $\mathrm{D}/\mathrm{D}\widehat{t} = \partial/\partial\widehat{t} + \widehat{\boldsymbol{u}}\cdot\nabla_z + \widehat{\beta}\widehat{w}\partial/\partial\widehat{z}$ and $Ro = U/(f_0 L)$. There are two notable aspects to these equations. First and most obviously, when $Ro \ll 1$, (2.222a) reduces to geostrophic balance, $\boldsymbol{f}\times\boldsymbol{u} = -\nabla\widehat{\phi}$. Second, the material derivative in (2.222b) is multiplied by three non-dimensional parameters, and we can understand the appearance of each as follows.

(i) The aspect ratio dependence (α^2) arises in the same way as for non-rotating flows — that is, because of the presence of w and z in the vertical momentum equation as opposed to (u, v) and (x, y) in the horizontal equations.

(ii) The Rossby number dependence (Ro) arises because in rotating flow the pressure gradient is balanced by the Coriolis force, which is Rossby number larger than the advective terms.

(iii) The factor $\widehat{\beta}$ arises because in rotating flow w is smaller than u by $\widehat{\beta}$ times the aspect ratio.

The factor $Ro\,\widehat{\beta}\alpha^2$ is very small for large-scale flow; the reader is invited to calculate representative values. Evidently, a rapidly rotating fluid is more likely to be in hydrostatic balance than a non-rotating fluid, other conditions being equal. The combined effects of rotation and stratification are, not surprisingly, quite subtle and we leave that topic for chapter 5.

2.9 STATIC INSTABILITY AND THE PARCEL METHOD

In this and the next couple of sections we consider how a fluid might oscillate if it were perturbed away from a resting state. Our focus is on vertical displacements, and the restoring force is gravity, and we will neglect the effects of rotation, and indeed initially we will neglect horizontal motion entirely. Given that, the simplest and most direct way to approach the problem is to consider from first principles the pressure and gravitational forces on a displaced parcel. To this end, consider a fluid initially at rest in a constant gravitational field, and therefore in hydrostatic balance. Suppose that a small parcel of the fluid is adiabatically displaced upwards by the small distance δz, without altering the overall pressure field; that is, the fluid parcel instantly assumes the pressure of its environment. If after the displacement the parcel is lighter than its environment, it will accelerate upwards, because the upward pressure gradient force is now greater than the downward gravity force on the parcel; that is, the parcel is *buoyant* (a manifestation of Archimedes' principle) and the fluid is *statically unstable*. If on the other hand the fluid parcel finds itself heavier than its surroundings, the downward gravitational force will be greater than the upward pressure force and the fluid will sink back towards its original position and an oscillatory motion will develop. Such an equilibrium is *statically stable*. Using such simple 'parcel' arguments we will now develop criteria for the stability of the environmental profile.

2.9.1 A simple special case: a density-conserving fluid

Consider first the simple case of an incompressible fluid in which the density of the displaced parcel is conserved, that is $\mathrm{D}\rho/\mathrm{D}t = 0$ (and refer to Fig. 2.7 setting $\rho_\theta = \rho$). If the environmental profile is $\widetilde{\rho}(z)$ and the density of the parcel is ρ then a parcel displaced from a level z [where its density is $\widetilde{\rho}(z)$] to a level $z + \delta z$ [where the density of the parcel is still $\widetilde{\rho}(z)$] will find that its density then differs from its surroundings by the amount

$$\delta\rho = \rho(z+\delta z) - \widetilde{\rho}(z+\delta z) = \widetilde{\rho}(z) - \widetilde{\rho}(z+\delta z) = -\frac{\partial\widetilde{\rho}}{\partial z}\delta z. \tag{2.223}$$

The parcel will be heavier than its surroundings, and therefore the parcel displacement will be stable, if $\partial\widetilde{\rho}/\partial z < 0$. Similarly, it will be unstable if $\partial\widetilde{\rho}/\partial z > 0$. The upward force (per unit volume) on the displaced parcel is given by

$$F = -g\delta\rho = g\frac{\partial\widetilde{\rho}}{\partial z}\delta z, \tag{2.224}$$

and thus Newton's second law implies that the motion of the parcel is determined by

$$\rho(z)\frac{\partial^2 \delta z}{\partial t^2} = g\frac{\partial\widetilde{\rho}}{\partial z}\delta z, \tag{2.225}$$

or

$$\frac{\partial^2 \delta z}{\partial t^2} = \frac{g}{\widetilde{\rho}}\frac{\partial\widetilde{\rho}}{\partial z}\delta z = -N^2\delta z, \tag{2.226}$$

where

$$N^2 = -\frac{g}{\widetilde{\rho}}\frac{\partial\widetilde{\rho}}{\partial z} \tag{2.227}$$

is the *buoyancy frequency*, or the *Brunt–Väisälä frequency,* for this problem. If $N^2 > 0$ then a parcel displaced upward is heavier than its surroundings, and thus experiences a restoring force; the density profile is said to be stable and N is the frequency at which the fluid parcel oscillates. If $N^2 < 0$, the density profile is unstable and the parcel continues to ascend and convection ensues. In liquids it is often a good approximation to replace $\widetilde{\rho}$ by ρ_0 in the denominator of (2.227).

2.9.2 The general case: using potential density

More generally, in an adiabatic displacement it is *potential density,* ρ_θ, and not density itself that is materially conserved. Consider a parcel that is displaced adiabatically a vertical distance from z to $z + \delta z$; the parcel preserves its potential density, and let us use the pressure at level $z + \delta z$ as the reference level. The *in situ* density of the parcel at $z + \delta z$, namely $\rho(z + \delta z)$, is then equal to its potential density $\rho_\theta(z + \delta z)$ and, because ρ_θ is conserved, this is equal to the potential density of the environment at z, $\widetilde{\rho}_\theta(z)$. The difference in *in situ* density between the parcel and the environment at $z + \delta z$, $\delta\rho$, is thus equal to the difference between the potential density of the environment at z and at $z + \delta z$. Putting this together (and see Fig. 2.7) we have

$$\begin{aligned}\delta\rho &= \rho(z+\delta z) - \widetilde{\rho}(z+\delta z) = \rho_\theta(z+\delta z) - \widetilde{\rho}_\theta(z+\delta z)\\ &= \rho_\theta(z) - \widetilde{\rho}_\theta(z+\delta z) = \widetilde{\rho}_\theta(z) - \widetilde{\rho}_\theta(z+\delta z),\end{aligned} \tag{2.228}$$

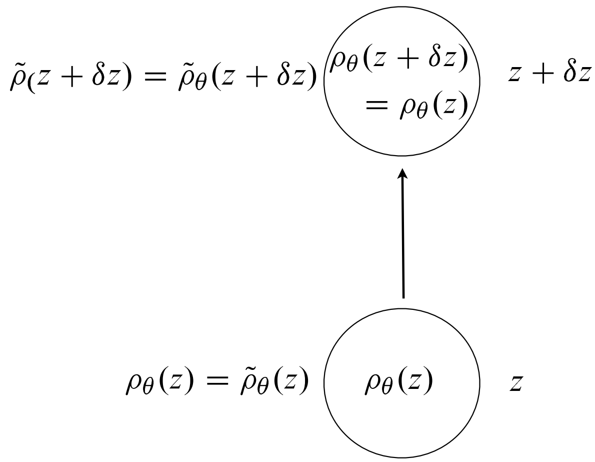

Fig. 2.7 A parcel is adiabatically displaced upward from level z to $z + \delta z$, preserving its potential density, which it takes from the environment at level z. If $z + \delta z$ is the reference level, the potential density there is equal to the actual density. The parcel's stability is determined by the difference between its density and the environmental density [see (2.228)]; if the difference is positive the displacement is stable, and conversely.

and therefore

$$\delta\rho = -\frac{\partial \tilde{\rho}_\theta}{\partial z}\delta z, \tag{2.229}$$

where the derivative on the right-hand side is the environmental gradient of potential density. If the right-hand side is positive, the parcel is heavier than its surroundings and the displacement is stable. Thus, the conditions for stability are:

$$\boxed{\begin{aligned} &\text{stability}: &\frac{\partial \tilde{\rho}_\theta}{\partial z} < 0 \\ &\text{instability}: &\frac{\partial \tilde{\rho}_\theta}{\partial z} > 0 \end{aligned}}\,. \tag{2.230a,b}$$

That is, *the stability of a parcel of fluid is determined by the gradient of the locally-referenced potential density*. The equation of motion of the fluid parcel is

$$\frac{\partial^2 \delta z}{\partial t^2} = \frac{g}{\rho}\left(\frac{\partial \tilde{\rho}_\theta}{\partial z}\right)\delta z = -N^2 \delta z, \tag{2.231}$$

where, noting that $\rho(z) = \tilde{\rho}_\theta(z)$ to within $O(\delta z)$,

$$\boxed{N^2 = -\frac{g}{\tilde{\rho}_\theta}\left(\frac{\partial \tilde{\rho}_\theta}{\partial z}\right)}\,. \tag{2.232}$$

This is a general expression for the buoyancy frequency, true in both liquids and gases. It is important to realize that the quantity $\tilde{\rho}_\theta$ is the *locally-referenced* potential density of the environment, as will become more clear below.

An ideal gas

In the atmosphere potential density is related to potential temperature by $\rho_\theta = p_R/(\theta R)$. Using this in (2.232) gives

$$\boxed{N^2 = \frac{g}{\tilde{\theta}}\left(\frac{\partial \tilde{\theta}}{\partial z}\right)}\,, \tag{2.233}$$

where $\widetilde{\theta}$ refers to the environmental profile of potential temperature. The reference value p_R does not appear, and we are free to choose this value arbitrarily — the surface pressure is a common choice. The conditions for stability, (2.230), then correspond to $N^2 > 0$ for stability and $N^2 < 0$ for instability. In the troposphere (the lowest several kilometres of the atmosphere) the average N is about $0.01\,\mathrm{s}^{-1}$, with a corresponding period, $(2\pi/N)$, of about 10 minutes. In the stratosphere (which lies above the troposphere) N^2 is a few times higher than this.

A liquid ocean

No simple, accurate, analytic expression is available for computing static stability in the ocean. If the ocean had no salt, then the potential density referenced to the surface would generally be a measure of the sign of stability of a fluid column, if not of the buoyancy frequency. However, in the presence of salinity, the surface-referenced potential density is not necessarily even a measure of the sign of stability, because the coefficients of compressibility β_T and β_S vary in different ways with pressure. To see this, suppose two neighbouring fluid elements at the surface have the same potential density, but different salinities and temperatures, and displace them both adiabatically to the deep ocean. Although their potential densities (referenced to the surface) are still equal, we can say little about their actual densities, and hence their stability relative to each other, without doing a detailed calculation because they will each have been compressed by different amounts. It is the profile of the *locally-referenced* potential density that determines the stability.

An approximate expression for stability that is sometimes useful arises by noting that in an adiabatic displacement

$$\delta\rho_\theta = \delta\rho - \frac{1}{c_s^2}\delta p = 0. \tag{2.234}$$

If the fluid is hydrostatic $\delta p = -\rho g \delta z$ so that if a parcel is displaced adiabatically its density changes according to

$$\left(\frac{\partial\rho}{\partial z}\right)_{\rho_\theta} = -\frac{\rho g}{c_s^2}. \tag{2.235}$$

If a parcel is displaced a distance δz upwards then the density difference between it and its new surroundings is

$$\delta\rho = -\left[\left(\frac{\partial\rho}{\partial z}\right)_{\rho_\theta} - \left(\frac{\partial\widetilde{\rho}}{\partial z}\right)\right]\delta z = \left[\frac{\rho g}{c_s^2} + \left(\frac{\partial\widetilde{\rho}}{\partial z}\right)\right]\delta z, \tag{2.236}$$

where the tilde again denotes the environmental field. It follows that the stratification is given by

$$\boxed{N^2 = -g\left[\frac{g}{c_s^2} + \frac{1}{\widetilde{\rho}}\left(\frac{\partial\widetilde{\rho}}{\partial z}\right)\right]}. \tag{2.237}$$

This expression holds for both liquids and gases, and for ideal gases it is precisely the same as (2.233) (problem 2.9). In liquids, a good approximation is to use a reference value ρ_0 for the undifferentiated density in the denominator, whence (2.237) becomes equal to the Boussinesq expression (2.107). Typical values of N in the upper ocean where the density is changing most rapidly (i.e., in the pycnocline — 'pycno' for density, 'cline' for changing) are about $0.01\,\mathrm{s}^{-1}$, falling to $0.001\,\mathrm{s}^{-1}$ in the more homogeneous abyssal ocean. These frequencies correspond to periods of about 10 and 100 minutes, respectively.

◆ *Cabbeling*

Cabbeling is an instability that arises because of the nonlinear equation of state of seawater. From Fig. 1.3 we see that the contours are slightly convex, bowing upwards, especially in the plot at sea level. Suppose we mix two parcels of water, each with the same density ($\sigma_\theta = 28$, say), but with different initial values of temperature and salinity. Then the resulting parcel of water will have a temperature and a salinity equal to the average of the two parcels, but its density will be *higher* than either of the two original parcels. In the appropriate circumstances such mixing may thus lead to a convective instability; this may, for example, be an important source of 'bottom water' formation in the Weddell Sea, off Antarctica.[11]

2.9.3 Lapse rates in dry and moist atmospheres

A dry ideal gas

The negative of the rate of change of the temperature in the vertical is known as the *temperature lapse rate*, or often just the lapse rate, and the lapse rate corresponding to $\partial\theta/\partial z = 0$ is called the *dry adiabatic lapse rate* and denoted Γ_d. Using $\theta = T(p_0/p)^{R/c_p}$ and $\partial p/\partial z = -\rho g$ we find that the lapse rate and the potential temperature lapse rate are related by

$$\frac{\partial T}{\partial z} = \frac{T}{\theta}\frac{\partial\theta}{\partial z} - \frac{g}{c_p}, \tag{2.238}$$

so that the dry adiabatic lapse rate is given by

$$\Gamma_d = \frac{g}{c_p}, \tag{2.239}$$

as in fact we derived in (1.134). (We use the subscript d, for dry, to differentiate it from the moist lapse rate considered below.) The conditions for static stability corresponding to (2.230) are thus:

$$\boxed{\begin{array}{llll} \text{stability}: & \dfrac{\partial\widetilde{\theta}}{\partial z} > 0; & \text{or} & -\dfrac{\partial\widetilde{T}}{\partial z} < \Gamma_d \\ \text{instability}: & \dfrac{\partial\widetilde{\theta}}{\partial z} < 0; & \text{or} & -\dfrac{\partial\widetilde{T}}{\partial z} > \Gamma_d \end{array}}\,, \tag{2.240a,b}$$

where a tilde indicates that the values are those of the environment. The atmosphere is in fact generally stable by this criterion: the observed lapse rate, corresponding to an observed buoyancy frequency of about $10^{-2}\,\mathrm{s}^{-1}$, is often about $7\,\mathrm{K\,km}^{-1}$, whereas a dry adiabatic lapse rate is about $10\,\mathrm{K\,km}^{-1}$. Why the discrepancy? One reason, particularly important in the tropics, is that the atmosphere contains water vapour.

◆ *Effects of water vapour on the lapse rate of an ideal gas*

The amount of water vapour that can be contained in a given volume is an increasing function of temperature (with the presence or otherwise of dry air in that volume being largely irrelevant). Thus, if a parcel of water vapour is cooled, it will eventually become saturated and water vapour will condense into liquid water. A measure of the amount of water vapour in a unit volume is its partial pressure, and the partial pressure of water vapour at saturation, e_s, is given by the Clausius–Clapeyron equation,

$$\frac{\mathrm{d}e_s}{\mathrm{d}T} = \frac{L_c e_s}{R_v T^2}, \tag{2.241}$$

where L_c is the latent heat of condensation or vapourization (per unit mass) and R_v is the gas constant for water vapour. If a parcel rises adiabatically it will cool, and at some height (known as the 'lifting condensation level', a function of its initial temperature and humidity only) the parcel will become saturated and any further ascent will cause the water vapour to condense. The ensuing condensational heating causes the temperature and buoyancy of the parcel to increase; the parcel thus rises further, causing more water vapour to condense, and so on, and the consequence of this is that an environmental profile that is stable if the air is dry may be unstable if saturated. Let us now derive an expression for the lapse rate of a saturated parcel that is ascending adiabatically apart from the affects of condensation.

Let w denote the mass of water vapour per unit mass of dry air, the mixing ratio, and let w_s be the saturation mixing ratio. ($w_s = \alpha e_s/(p - e_s) \approx \alpha_w e_s/p$ where $\alpha_w = 0.62$, the ratio of the mass of a water molecule to one of dry air.) The diabatic heating associated with condensation is then given by

$$Q_{cond} = -L_c \frac{\mathrm{D}w_s}{\mathrm{D}t}, \tag{2.242}$$

so that the thermodynamic equation is

$$c_p \frac{\mathrm{D}\ln\theta}{\mathrm{D}t} = -\frac{L_c}{T}\frac{\mathrm{D}w_s}{\mathrm{D}t}, \tag{2.243}$$

or, in terms of p and and T

$$c_p \frac{\mathrm{D}\ln T}{\mathrm{D}t} - R\frac{\mathrm{D}\ln P}{\mathrm{D}t} = -\frac{L_c}{T}\frac{\mathrm{D}w_s}{\mathrm{D}t}. \tag{2.244}$$

If these material derivatives are due to the parcel ascent then

$$\frac{\mathrm{d}\ln T}{\mathrm{d}z} - \frac{R}{c_p}\frac{\mathrm{d}\ln p}{\mathrm{d}z} = -\frac{L_c}{Tc_p}\frac{\mathrm{d}w_s}{\mathrm{d}z}, \tag{2.245}$$

and using the hydrostatic relationship and the fact that w_s is a function of T and p we obtain

$$\frac{\mathrm{d}T}{\mathrm{d}z} + \frac{g}{c_p} = -\frac{L_c}{c_p}\left[\left(\frac{\partial w_s}{\partial T}\right)_p \frac{\mathrm{d}T}{\mathrm{d}z} - \left(\frac{\partial w_s}{\partial p}\right)_T \rho g\right]. \tag{2.246}$$

Solving for $\mathrm{d}T/\mathrm{d}z$, the lapse rate, Γ_s, of an ascending saturated parcel is given by

$$\Gamma_s = -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}\frac{1 - \rho L_c(\partial w_s/\partial p)_T}{1 + (L_c/c_p)(\partial w_s/\partial T)_p} \approx \frac{g}{c_p}\frac{1 + L_c w_s/(RT)}{1 + L_c^2 w_s/(c_p R T^2)}. \tag{2.247}$$

where the last near equality follows with use of the Clausius–Clapeyron relation. The quantity Γ_s is variously called the *pseudoadiabatic* or *moist adiabatic* or *saturated adiabatic* lapse rate. Because g/c_p is the dry adiabatic lapse rate Γ_d, $\Gamma_s < \Gamma_d$, and values of Γ_s are typically around $6\,\mathrm{K\,km^{-1}}$ in the lower atmosphere; however, $\mathrm{d}w_s/\mathrm{d}T$ is an increasing function of

T so that Γ_s decreases with increasing temperature and can be as low as $3.5\,\mathrm{K\,km^{-1}}$. For a saturated parcel, the stability conditions analogous to (2.240) are

$$\text{stability}: \qquad -\frac{\partial \widetilde{T}}{\partial z} < \Gamma_s, \tag{2.248a}$$

$$\text{instability}: \qquad -\frac{\partial \widetilde{T}}{\partial z} > \Gamma_s. \tag{2.248b}$$

where $\widetilde{T}$ is the environmental temperature. The observed environmental profile in convecting situations is often a combination of the dry adiabatic and moist adiabatic profiles: an unsaturated parcel that is is unstable by the dry criterion will rise and cool following a dry adiabat, Γ_d, until it becomes saturated at the lifting condensation level, above which it will rise following a saturation adiabat, Γ_s. Such convection will proceed until the atmospheric column is stable and, especially in low latitudes, the lapse rate of the atmosphere is largely determined by such convective processes.

◆ *Equivalent potential temperature*

Suppose that all the moisture in a parcel of air condenses, and that all the heat released goes into heating the parcel. The *equivalent potential temperature*, θ_{eq} is the potential temperature that the parcel then achieves. We may obtain an approximate analytic expression for it by noting that the first law of thermodynamics, $\text{đ}Q = T\,\mathrm{d}\eta$, then implies, by definition of potential temperature,

$$-L_c\,\mathrm{d}w = c_p T\,\mathrm{d}\ln\theta, \tag{2.249}$$

where $\mathrm{d}w$ is the change in water vapour mixing ratio, so that a reduction of w via condensation leads to heating. Integrating gives, by definition of equivalent potential temperature,

$$-\int_w^0 \frac{L_c w}{c_p T}\,\mathrm{d}w = \int_\theta^{\theta_{eq}} \mathrm{d}\ln\theta, \tag{2.250}$$

and so, if T and L_c are assumed to be constant,

$$\theta_{eq} = \theta \exp\left(\frac{L_c w}{c_p T}\right). \tag{2.251}$$

The equivalent potential temperature so defined is approximately conserved during condensation, the approximation arising going from (2.250) to (2.251). It is a useful expression for diagnostic purposes, and in constructing theories of convection, but it is not accurate enough to use as a prognostic variable in a putatively realistic numerical model. The 'equivalent temperature' may be defined in terms of the equivalent potential temperature by

$$T_{eq} \equiv \theta_{eq}\left(\frac{p}{p_R}\right)^{\kappa}. \tag{2.252}$$

2.9.4 Gravity waves and convection using the equations of motion

The parcel approach to oscillations and stability, while simple and direct, seems divorced from the fluid-dynamical equations of motion, making it hard to include other effects such as rotation, or to explore the effects of possible differences between the hydrostatic and non-hydrostatic cases. To remedy this, we now use the equations of motion for a stratified Boussinesq fluid to analyze the motion resulting from a small disturbance. We will obtain the dispersion relationship for *internal waves* and further show that, if the lighter fluid lies beneath the heavier fluid then the configuration is unstable. Our treatment here is very brief and introductory, with a fuller treatment given in chapter **??**.

Let us consider a Boussinesq fluid, initially at rest, in which the buoyancy varies linearly with height and the buoyancy frequency, N, is a constant. Linearizing the equations of motion about this basic state gives the linear momentum equations,

$$\frac{\partial u'}{\partial t} = -\frac{\partial \phi'}{\partial x}, \qquad \frac{\partial w'}{\partial t} = -\frac{\partial \phi'}{\partial z} + b', \tag{2.253a,b}$$

the mass continuity and thermodynamic equations,

$$\frac{\partial u'}{\partial x} + \frac{\partial w'}{\partial z} = 0, \qquad \frac{\partial b'}{\partial t} + w' N^2 = 0, \tag{2.254a,b}$$

where for simplicity we assume that the flow is a function only of x and z. A little algebra gives a single equation for w',

$$\left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}\right)\frac{\partial^2}{\partial t^2} + N^2 \frac{\partial^2}{\partial x^2}\right] w' = 0. \tag{2.255}$$

Seeking solutions of the form $w' = \mathrm{Re}\, W \exp[\mathrm{i}(kx + mz - \omega t)]$ (where Re denotes the real part) yields the dispersion relationship for gravity waves:

$$\boxed{\omega^2 = \frac{k^2 N^2}{k^2 + m^2}}. \tag{2.256}$$

The frequency (look ahead to Fig. **??** if you wish to see it plotted) is thus always less than N, approaching N for small horizontal scales, $k \gg m$.

Let us point out two common approximations. First, if we neglect pressure perturbations, as in the parcel argument, then the two equations,

$$\frac{\partial w'}{\partial t} = b', \qquad \frac{\partial b'}{\partial t} + w' N^2 = 0, \tag{2.257}$$

form a closed set, and give $\omega^2 = N^2$. Second, if we make the hydrostatic approximation and omit the $\partial w'/\partial t$ term in (2.253b) then the dispersion relation becomes $\omega^2 = k^2 N^2/m^2$. That is, the frequency grows, artifactually, without bound as the horizontal scale becomes smaller.

If the basic state density increases with height then $N^2 < 0$ and we expect this state to be unstable. Indeed, the disturbance grows exponentially according to $\exp(\sigma t)$ where

$$\sigma = \mathrm{i}\omega = \frac{\pm k\widetilde{N}}{(k^2 + m^2)^{1/2}}, \tag{2.258}$$

where $\widetilde{N}^2 = -N^2$. We have essentially reproduced the result previously obtained by parcel theory, namely that if the basic state density (or more generally potential density) increases with height the flow is unstable. Most convective activity in the ocean and atmosphere is, ultimately, related to an instability of this form, although of course there are many complicating issues — water vapour in the atmosphere, salt in the ocean, the effects of rotation and so forth.

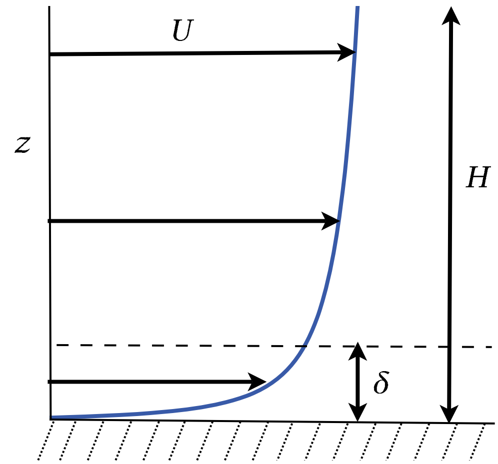

Fig. 2.8 An idealized boundary layer. The values of a field, such as velocity, U, may vary rapidly in a boundary in order to satisfy the boundary conditions at a rigid surface. The parameter δ is a measure of the boundary layer thickness, and H is a typical scale of variation away from the boundary.

2.10 THE EKMAN LAYER

In the final topic of this chapter, we return to geostrophic flow and consider the effects of friction. The fluid fields in the interior of a domain are often set by different physical processes than those occurring at a boundary, and consequently often change rapidly in a thin *boundary layer,* as in Fig. 2.8. Such boundary layers nearly always involve one or both of viscosity and diffusion, because these appear in the terms of highest differential order in the equations of motion, and so are responsible for the number and type of boundary conditions that the equations must satisfy — for example, the presence of molecular viscosity leads to the condition that the tangential flow (as well as the normal flow) must vanish at a rigid surface.

In many boundary layers in non-rotating flow the dominant balance in the momentum equation is between the advective and viscous terms. In some contrast, in large-scale atmospheric and oceanic flow the effects of rotation are large, and this results in a boundary layer, known as the *Ekman layer,* in which the dominant balance is between Coriolis and frictional or stress terms.[12] Now, the direct effects of molecular viscosity and diffusion are nearly always negligible at distances more than a few millimetres away from a solid boundary, but it is inconceivable that the entire boundary layer between the free atmosphere (or free ocean) and the surface is only a few millimetres thick. Rather, in practice a balance occurs between the Coriolis terms and the forces due to the stress generated by small-scale turbulent motion, and this gives rise to a boundary layer that has a typical depth of a few tens to several hundreds of metres. Because the stress arises from the turbulence we cannot with confidence determine its precise form; thus, we should try to determine what general properties Ekman layers may have that are *independent* of the precise form of the friction, and these properties turn out to be integral ones such as the total mass flux in the Ekman layer.

The atmospheric Ekman layer occurs near the ground, and the stress at the ground itself is due to the surface wind (and its vertical variation). In the ocean the main Ekman layer is near the surface, and the stress at ocean surface is largely due to the presence of the overlying wind. There is also a weak Ekman layer at the bottom of the ocean, analogous to the atmospheric Ekman layer. To analyse all these layers, let us assume the following.

- The Ekman layer is Boussinesq. This is a very good assumption for the ocean, and a reasonable one for the atmosphere if the boundary layer is not too deep.

- The Ekman layer has a finite depth that is less than the total depth of the fluid, this depth being given by the level at which the frictional stresses essentially vanish. Within the Ekman layer, frictional terms are important, whereas geostrophic balance holds beyond it.
- The nonlinear and time-dependent terms in the equations of motion are negligible, hydrostatic balance holds in the vertical, and buoyancy is constant, not varying in the horizontal.
- As needed, we shall assume that friction can be parameterized by a viscous term of the form $\rho_0^{-1}\partial\boldsymbol{\tau}/\partial z = A\,\partial^2\boldsymbol{u}/\partial z^2$, where A is constant and $\boldsymbol{\tau}$ is the stress. [In general, stress is a tensor, τ_{ij}, with an associated force given by $F_i = \partial\tau_{ij}/\partial x_j$, summing over the repeated index. It is common in geophysical fluid dynamics that the vertical derivative dominates, and in this case the force is $\boldsymbol{F} = \partial\boldsymbol{\tau}/\partial z$. We still use the word stress for $\boldsymbol{\tau}$, but it now refers to a vector whose derivative in a particular direction (z in this case) is the force on a fluid.] In laboratory settings A may be the molecular viscosity, whereas in the atmosphere and ocean it is a so-called *eddy viscosity*. (In turbulent flows momentum is transferred by the near-random motion of small parcels of fluid and, by analogy with the motion of molecules that produces a molecular viscosity, the associated stress is approximately represented, or parameterized, using a turbulent or eddy viscosity that may be orders of magnitude larger than the molecular one.)

2.10.1 Equations of motion and scaling

Frictional–geostrophic balance in the horizontal momentum equation is:

$$\boldsymbol{f}\times\boldsymbol{u} = -\nabla_z\phi + \frac{\partial\widetilde{\boldsymbol{\tau}}}{\partial z}, \tag{2.259}$$

where $\widetilde{\boldsymbol{\tau}} \equiv \boldsymbol{\tau}/\rho_0$ is the kinematic stress and $\boldsymbol{f} = f\mathbf{k}$, where the Coriolis parameter f is allowed to vary with latitude. If we model the stress with an eddy viscosity, (2.259) becomes

$$\boldsymbol{f}\times\boldsymbol{u} = -\nabla_z\phi + A\frac{\partial^2\boldsymbol{u}}{\partial z^2}. \tag{2.260}$$

The vertical momentum equation is $\partial\phi/\partial z = b$, i.e., hydrostatic balance, and, because buoyancy is constant, we may without loss of generality write this as

$$\frac{\partial\phi}{\partial z} = 0. \tag{2.261}$$

The equation set is completed by the mass continuity equation, $\nabla\cdot\boldsymbol{v} = 0$.

The Ekman number

We non-dimensionalize the equations by setting

$$(u, v) = U(\hat{u}, \hat{v}), \quad (x, y) = L(\hat{x}, \hat{y}), \quad f = f_0\hat{f}, \quad z = H\hat{z}, \quad \phi = \Phi\hat{\phi}, \tag{2.262}$$

where hatted variables are non-dimensional. H is a scaling for the height, and at this stage we will suppose it to be some height scale in the free atmosphere or ocean, not the height of the Ekman layer itself. Geostrophic balance suggests that $\Phi = f_0 UL$. Substituting (2.262) into (2.260) we obtain

$$\hat{\boldsymbol{f}}\times\hat{\boldsymbol{u}} = -\hat{\nabla}\hat{\phi} + Ek\frac{\partial^2\hat{\boldsymbol{u}}}{\partial\hat{z}^2}, \tag{2.263}$$

where the parameter

$$\boxed{Ek \equiv \left(\frac{A}{f_0 H^2}\right)}, \tag{2.264}$$

is the *Ekman number*, and it determines the importance of frictional terms in the horizontal momentum equation. If $Ek \ll 1$ then the friction is small in the flow interior where $\hat{z} = \mathcal{O}(1)$. However, the friction term cannot necessarily be neglected in the boundary layer because it is of the highest differential order in the equation, and so determines the boundary conditions; if Ek is small the vertical scales become small and the second term on the right-hand side of (2.263) remains finite. The case when this term is simply omitted from the equation is therefore a *singular limit,* meaning that it differs from the case with $Ek \to 0$. If $Ek \geq 1$ friction is important everywhere, but it is usually the case that Ek is small for atmospheric and oceanic large-scale flow, and the interior flow is very nearly geostrophic. (In part this is because A itself is only large near a rigid surface where the presence of a shear creates turbulence and a significant eddy viscosity.)

Momentum balance in the Ekman layer

For definiteness, suppose the fluid lies above a rigid surface at $z = 0$. Sufficiently far away from the boundary the velocity field is known, and we suppose this flow to be in geostrophic balance. We then write the velocity field and the pressure field as the sum of the interior geostrophic part, plus a boundary layer correction:

$$\hat{\boldsymbol{u}} = \hat{\boldsymbol{u}}_g + \hat{\boldsymbol{u}}_E, \qquad \hat{\phi} = \hat{\phi}_g + \hat{\phi}_E, \tag{2.265}$$

where the Ekman layer corrections, denoted with a subscript E, are negligible away from the boundary layer. Now, in the fluid interior we have, by hydrostatic balance, $\partial\hat{\phi}_g/\partial\hat{z} = 0$. In the boundary layer we still have $\partial\hat{\phi}_g/\partial\hat{z} = 0$ so that, to satisfy hydrostasy, $\partial\hat{\phi}_E/\partial\hat{z} = 0$. But because $\hat{\phi}_E$ vanishes away from the boundary we have $\hat{\phi}_E = 0$ everywhere. Thus, *there is no boundary layer in the pressure field.* Note that this is a much stronger result than saying that pressure is continuous, which is nearly always true in fluids; rather, it is a special result for Ekman layers.

Using (2.265) with $\hat{\phi}_E = 0$, the dimensional horizontal momentum equation (2.259) becomes, in the Ekman layer,

$$\boldsymbol{f} \times \boldsymbol{u}_E = \frac{\partial \tilde{\boldsymbol{\tau}}}{\partial z}. \tag{2.266}$$

The dominant force balance in the Ekman layer is thus between the Coriolis force and the friction. We can determine the thickness of the Ekman layer if we model the stress with an eddy viscosity so that

$$\boldsymbol{f} \times \boldsymbol{u}_E = A\frac{\partial^2 \boldsymbol{u}_E}{\partial z^2}, \tag{2.267}$$

or, non-dimensionally,

$$\hat{\boldsymbol{f}} \times \hat{\boldsymbol{u}}_E = Ek\frac{\partial^2 \hat{\boldsymbol{u}}_E}{\partial \hat{z}^2}. \tag{2.268}$$

It is evident this equation can only be satisfied if $\hat{z} \neq \mathcal{O}(1)$, implying that H is not a proper scaling for z in the boundary layer. Rather, if the vertical scale in the Ekman layer is $\hat{\delta}$ (meaning $\hat{z} \sim \hat{\delta}$) we must have $\hat{\delta} \sim Ek^{1/2}$. In dimensional terms this means the thickness of the

Ekman layer is

$$\delta = H\hat{\delta} = HEk^{1/2} \tag{2.269}$$

or

$$\boxed{\delta = \left(\frac{A}{f_0}\right)^{1/2}}. \tag{2.270}$$

[This estimate also emerges directly from (2.267).] Note that (2.269) can be written as

$$Ek = \left(\frac{\delta}{H}\right)^2. \tag{2.271}$$

That is, the Ekman number is equal to the square of the ratio of the depth of the Ekman layer to an interior depth scale of the fluid motion. In laboratory flows where A is the molecular viscosity we can thus estimate the Ekman layer thickness, and if we know the eddy viscosity of the ocean or atmosphere we can estimate their respective Ekman layer thicknesses. We can invert this argument and obtain an estimate of A if we know the Ekman layer depth. In the atmosphere, deviations from geostrophic balance are very small in the atmosphere above 1 km, and using this gives $A \approx 10^2\,\mathrm{m^2\,s^{-1}}$. In the ocean Ekman depths are often 50 m or less, and eddy viscosities are about $0.1\,\mathrm{m^2\,s^{-1}}$.

2.10.2 Integral properties of the Ekman layer

What can we deduce about the Ekman layer without specifying the detailed form of the frictional term? Using dimensional notation we recall frictional–geostrophic balance,

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi + \frac{1}{\rho_0}\frac{\partial \boldsymbol{\tau}}{\partial z}, \tag{2.272}$$

where $\boldsymbol{\tau}$ is zero at the edge of the Ekman layer. In the Ekman layer itself we have

$$\boldsymbol{f} \times \boldsymbol{u}_E = \frac{1}{\rho_0}\frac{\partial \boldsymbol{\tau}}{\partial z}. \tag{2.273}$$

Consider either a top or bottom Ekman layer, and integrate over its thickness. From (2.273) we obtain

$$\boldsymbol{f} \times \boldsymbol{M}_E = \boldsymbol{\tau}_T - \boldsymbol{\tau}_B, \tag{2.274}$$

where

$$\boldsymbol{M}_E = \int_{Ek} \rho_0 \boldsymbol{u}_E \,\mathrm{d}z \tag{2.275}$$

is the ageostrophic mass transport in the Ekman layer, and $\boldsymbol{\tau}_T$ and $\boldsymbol{\tau}_B$ are the respective stresses at the top and the bottom of the Ekman layer at hand. The stress at the top (bottom) will be zero in a bottom (top) Ekman layer and therefore, from (2.274),

$$\boxed{\begin{array}{ll} \text{top Ekman layer:} & \boldsymbol{M}_E = -\dfrac{1}{f}\mathbf{k} \times \boldsymbol{\tau}_T \\[2ex] \text{bottom Ekman layer:} & \boldsymbol{M}_E = \dfrac{1}{f}\mathbf{k} \times \boldsymbol{\tau}_B \end{array}}. \tag{2.276a,b}$$

The transport is thus at right angles to the stress at the surface, and proportional to the magnitude of the stress. These properties have a simple physical explanation: integrated over the depth of the Ekman layer the surface stress must be balanced by the Coriolis force, which in turn acts at right angles to the mass transport. A consequence of (2.276) is that the mass transports in adjacent oceanic and atmospheric Ekman layers are equal and opposite, because the stress is continuous across the ocean–atmosphere interface. Equation (2.276a) is particularly useful in the ocean, where the stress at the surface is primarily due to the wind, and is largely independent of the interior oceanic flow. In the atmosphere, the surface stress mainly arises as a result of the interior atmospheric flow, and to calculate it we need to parameterize the stress in terms of the flow.

Finally, we obtain an expression for the vertical velocity induced by an Ekman layer. The mass conservation equation is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0. \tag{2.277}$$

Integrating this over an Ekman layer gives

$$\frac{1}{\rho_0}\nabla\cdot\boldsymbol{M}_{To} = -(w_T - w_B), \tag{2.278}$$

where $\boldsymbol{M}_{To}$ is the total (Ekman plus geostrophic) mass transport in the Ekman layer,

$$\boldsymbol{M}_{To} = \int_{Ek} \rho_0 \boldsymbol{u}\,\mathrm{d}z = \int_{Ek} \rho_0(\boldsymbol{u}_g + \boldsymbol{u}_E)\,\mathrm{d}z \equiv \boldsymbol{M}_g + \boldsymbol{M}_E, \tag{2.279}$$

and w_T and w_B are the vertical velocities at the top and bottom of the Ekman layer; the former (latter) is zero in a top (bottom) Ekman layer. Equations (2.279) and (2.274) give

$$\mathbf{k}\times(\boldsymbol{M}_{To} - \boldsymbol{M}_g) = \frac{1}{f}(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B). \tag{2.280}$$

Taking the curl of this (i.e., cross-differentiating) gives

$$\nabla\cdot(\boldsymbol{M}_{To} - \boldsymbol{M}_g) = \mathrm{curl}_z[(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B)/f], \tag{2.281}$$

where the curl_z operator on a vector $\boldsymbol{A}$ is defined by $\mathrm{curl}_z\boldsymbol{A} \equiv \partial_x A_y - \partial_y A_x$. Using (2.278) we obtain, for top and bottom Ekman layers respectively,

$$\boxed{w_B = \frac{1}{\rho_0}\left(\mathrm{curl}_z\frac{\boldsymbol{\tau}_T}{f} + \nabla\cdot\boldsymbol{M}_g\right), \qquad w_T = \frac{1}{\rho_0}\left(\mathrm{curl}_z\frac{\boldsymbol{\tau}_B}{f} - \nabla\cdot\boldsymbol{M}_g\right)}\,, \tag{2.282a,b}$$

where $\nabla\cdot\boldsymbol{M}_g = -(\beta/f)\boldsymbol{M}_g\cdot\mathbf{j}$ is the divergence of the geostrophic transport in the Ekman layer, and this is often small compared to the other terms in these equations. Thus, friction induces a vertical velocity at the edge of the Ekman layer, proportional to the curl of the stress at the surface, and this is perhaps the most used result in Ekman layer theory. Numerical models sometimes do not have the vertical resolution to explicitly resolve an Ekman layer, and (2.282) provides a means of *parameterizing* the Ekman layer in terms of resolved or known fields. It is particularly useful for the top Ekman layer in the ocean, where the stress can be regarded as a given function of the overlying wind.

2.10.3 Explicit solutions. I: a bottom boundary layer

We now assume that the frictional terms can be parameterized as an eddy viscosity and calculate the explicit form of the solution in the boundary layer. The frictional–geostrophic balance may be written as

$$\boldsymbol{f} \times (\boldsymbol{u} - \boldsymbol{u}_g) = A\frac{\partial^2 \boldsymbol{u}}{\partial z^2}, \tag{2.283a}$$

where

$$f(u_g, v_g) = \left(-\frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial x}\right). \tag{2.283b}$$

We continue to assume there are no horizontal gradients of temperature, so that, via thermal wind, $\partial u_g/\partial z = \partial v_g/\partial z = 0$.

Boundary conditions and solution

Appropriate boundary conditions for a bottom Ekman layer are

$$\text{at } z = 0: \qquad u = 0, \quad v = 0 \qquad \text{(the no slip condition)} \tag{2.284a}$$

$$\text{as } z \to \infty: \qquad u = u_g, \quad v = v_g \qquad \text{(a geostrophic interior).} \tag{2.284b}$$

Let us seek solutions to (2.283a) of the form

$$u = u_g + A_0 \mathrm{e}^{\alpha z}, \qquad v = v_g + B_0 \mathrm{e}^{\alpha z}, \tag{2.285}$$

where A_0 and B_0 are constants. Substituting into (2.283a) gives two homogeneous algebraic equations

$$A_0 f - B_0 A \alpha^2 = 0, \qquad -A_0 A\alpha^2 - B_0 f = 0. \tag{2.286a,b}$$

For non-trivial solutions the solvability condition $\alpha^4 = -f^2/A^2$ must hold, from which we find $\alpha = \pm(1 \pm \mathrm{i})\sqrt{f/2A}$. Using the boundary conditions we then obtain the solution

$$u = u_g - \mathrm{e}^{-z/d}\left[u_g \cos(z/d) + v_g \sin(z/d)\right] \tag{2.287a}$$

$$v = v_g + \mathrm{e}^{-z/d}\left[u_g \sin(z/d) - v_g \cos(z/d)\right], \tag{2.287b}$$

where $d = \sqrt{2A/f}$ is, within a constant factor, the depth of the Ekman layer obtained from scaling considerations. The solution decays exponentially from the surface with this e-folding scale, so that d is a good measure of the Ekman layer thickness. Note that the boundary layer correction depends on the interior flow, since the boundary layer serves to bring the flow to zero at the surface.

To illustrate the solution, suppose that the pressure force is directed in the y-direction (northwards), so that the geostrophic current is eastwards. Then the solution, the now-famous *Ekman spiral*, is plotted in Figs. 2.9 and 2.10. The wind falls to zero at the surface, and its direction just above the surface is northeastwards; that is, it is rotated by 45° to the left of its direction in the free atmosphere. Although this result is independent of the value of the frictional coefficients, it is dependent on the form of the friction chosen. The force balance in the Ekman layer is between the Coriolis force, the stress, and the pressure force. At the surface the Coriolis force is zero, and the balance is entirely between the northward pressure force and the southward stress force.

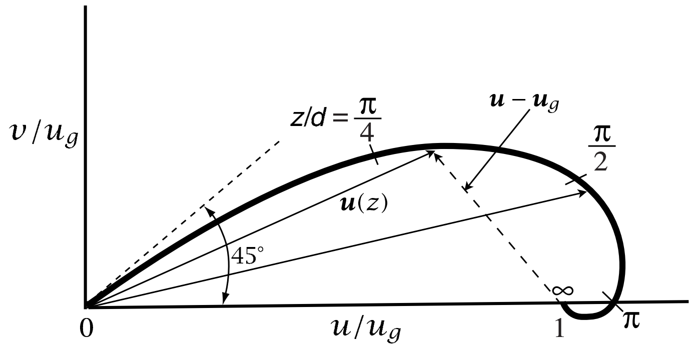

Fig. 2.9 The idealized Ekman layer solution in the lower atmosphere, plotted as a hodograph of the wind components: the arrows show the velocity vectors at a particular heights, and the curve traces out the continuous variation of the velocity. The values on the curve are of the non-dimensional variable z/d, where $d = (2A/f)^{1/2}$, and v_g is chosen to be zero.

Transport, force balance and vertical velocity

The cross-isobaric flow is given by (for $v_g = 0$)

$$V = \int_0^\infty v \, dz = \int_0^\infty u_g e^{-z/d} \sin(z/d) \, dz = \frac{u_g d}{2}. \tag{2.288}$$

For positive f, this is to the left of the geostrophic flow — that is, down the pressure gradient. In the general case ($v_g \neq 0$) we obtain

$$V = \int_0^\infty (v - v_g) \, dz = \frac{d}{2}(u_g - v_g). \tag{2.289}$$

Similarly, the additional zonal transport produced by frictional effects are, for $v_g = 0$,

$$U = \int_0^\infty (u - u_g) \, dz = -\int_0^\infty e^{-z/d} \sin(z/d) \, dz = -\frac{u_g d}{2}, \tag{2.290}$$

and in the general case

$$U = \int_0^\infty (u - u_g) \, dz = -\frac{d}{2}(u_g + v_g). \tag{2.291}$$

Thus, the total transport caused by frictional forces is

$$\boldsymbol{M}_E = \frac{\rho_0 d}{2} \left[-\mathbf{i}(u_g + v_g) + \mathbf{j}(u_g - v_g) \right]. \tag{2.292}$$

The total stress at the bottom surface $z = 0$ induced by frictional forces is

$$\tilde{\boldsymbol{\tau}}_B = A \frac{\partial \boldsymbol{u}}{\partial z}\bigg|_{z=0} = \frac{A}{d} \left[\mathbf{i}(u_g - v_g) + \mathbf{j}(u_g + v_g) \right], \tag{2.293}$$

using the solution (2.287). Thus, using (2.292), (2.293) and $d^2 = 2A/f$, we see that the total frictionally induced transport in the Ekman layer is related to the stress at the surface by $\boldsymbol{M}_E = (\mathbf{k} \times \boldsymbol{\tau}_B)/f$, reprising the result of the previous more general analysis, (2.282). From (2.293), the stress is at an angle of 45° to the left of the velocity at the surface. (However, this result is not generally true for all forms of stress.) These properties are illustrated in Fig. 2.11.

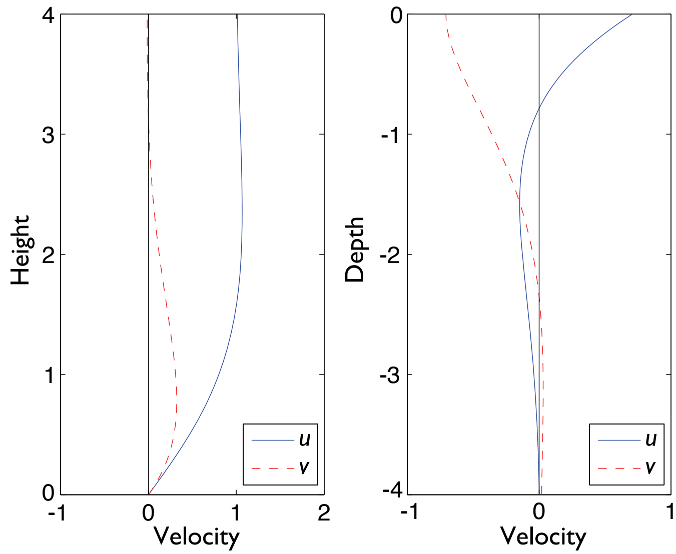

Fig. 2.10 Solutions for a bottom Ekman layer with a given flow in the fluid interior (left), and for a top Ekman layer with a given surface stress (right), both with $d = 1$. On the left we have $u_g = 1$, $v_g = 0$. On the right we have $u_g = v_g = 0$, $\widetilde{\tau}_y = 0$ and $\sqrt{2}\widetilde{\tau}_x/(fd) = 1$.

The vertical velocity at the top of the Ekman layer, w_E, is obtained using (2.292) or (2.293). If f if constant we obtain

$$w_E = -\frac{1}{\rho_0}\nabla \cdot \boldsymbol{M}_E = \frac{1}{f_0}\mathrm{curl}_z\widetilde{\boldsymbol{\tau}}_B = \frac{d}{2}\zeta_g, \tag{2.294}$$

where ζ_g is the vorticity of the geostrophic flow. Thus, the vertical velocity at the top of the Ekman layer, which arises because of the frictionally-induced divergence of the cross-isobaric flow in the Ekman layer, is proportional to the geostrophic vorticity in the free fluid and is proportional to the Ekman layer height $\sqrt{2A/f_0}$.

Another bottom boundary condition

In the analysis above we assumed a *no slip* condition at the surface, namely that the velocity tangential to the surface vanishes. This is formally appropriate if A is a molecular viscosity, but in a turbulent flow, where A is to be interpreted as an eddy viscosity, the flow close to the surface may be far from zero. Then, unless we wish to explicitly calculate the flow in

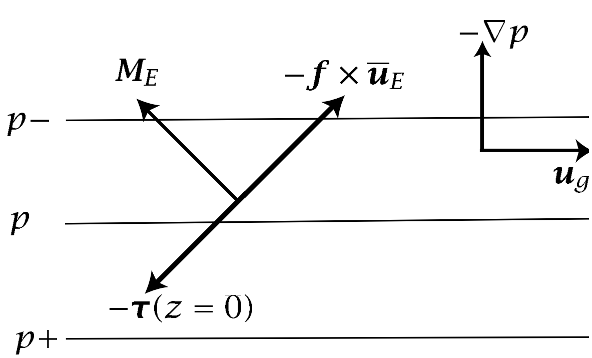

Fig. 2.11 A bottom Ekman layer, generated from an eastward geostrophic flow above it. An overbar denotes a vertical integral over the Ekman layer, so that $-\boldsymbol{f}\times\overline{\boldsymbol{u}}_E$ is the Coriolis force on the vertically integrated Ekman velocity. $\boldsymbol{M}_E$ is the frictionally induced boundary layer transport, and $\boldsymbol{\tau}$ is the stress.

an additional very thin viscous boundary layer the no-slip condition may be inappropriate. An alternative, slightly more general boundary condition is to suppose that the stress at the surface is given by

$$\boldsymbol{\tau} = \rho_0 C \boldsymbol{u}, \tag{2.295}$$

where C is a constant. The surface boundary condition is then

$$A\frac{\partial \boldsymbol{u}}{\partial z} = C\boldsymbol{u}. \tag{2.296}$$

If C is infinite we recover the no-slip condition. If $C = 0$, we have a condition of no stress at the surface, also known as a *free slip* condition. For intermediate values of C the boundary condition is known as a 'mixed condition'. Evaluating the solution in these cases is left as an exercise for the reader (problem 2.27).

2.10.4 Explicit solutions. II: the upper ocean

Boundary conditions and solution

The wind provides a stress on the upper ocean, and the Ekman layer serves to communicate this to the oceanic interior. Appropriate boundary conditions are thus:

$$\text{at } z = 0: \qquad A\frac{\partial u}{\partial z} = \tilde{\tau}^x, \quad A\frac{\partial v}{\partial z} = \tilde{\tau}^y \qquad \text{(a given surface stress)} \tag{2.297a}$$

$$\text{as } z \to -\infty: \qquad u = u_g, \quad v = v_g \qquad \text{(a geostrophic interior)} \tag{2.297b}$$

where $\tilde{\boldsymbol{\tau}}$ is the given (kinematic) wind stress at the surface. Solutions to (2.283a) with (2.297) are found by the same methods as before, and are

$$u = u_g + \frac{\sqrt{2}}{fd}\mathrm{e}^{z/d}\left[\tilde{\tau}^x \cos(z/d - \pi/4) - \tilde{\tau}^y \sin(z/d - \pi/4)\right], \tag{2.298}$$

and

$$v = v_g + \frac{\sqrt{2}}{fd}\mathrm{e}^{z/d}\left[\tilde{\tau}^x \sin(z/d - \pi/4) + \tilde{\tau}^y \cos(z/d - \pi/4)\right]. \tag{2.299}$$

Note that the boundary layer correction depends only on the imposed surface stress, and not the interior flow itself. This is a consequence of the type of boundary conditions chosen, for in the absence of an imposed stress the boundary layer correction is zero — the interior flow already satisfies the gradient boundary condition at the top surface. Similar to the bottom boundary layer, the velocity vectors of the solution trace a diminishing spiral as they descend into the interior (Fig. 2.12, which is drawn for the Southern Hemisphere).

Transport, surface flow and vertical velocity

The transport induced by the surface stress is obtained by integrating (2.298) and (2.299) from the surface to $-\infty$. We explicitly find

$$U = \int_{-\infty}^{0} (u - u_g)\,\mathrm{d}z = \frac{\tilde{\tau}^y}{f}, \qquad V = \int_{-\infty}^{0} (v - v_g)\,\mathrm{d}z = -\frac{\tilde{\tau}^x}{f}, \tag{2.300}$$

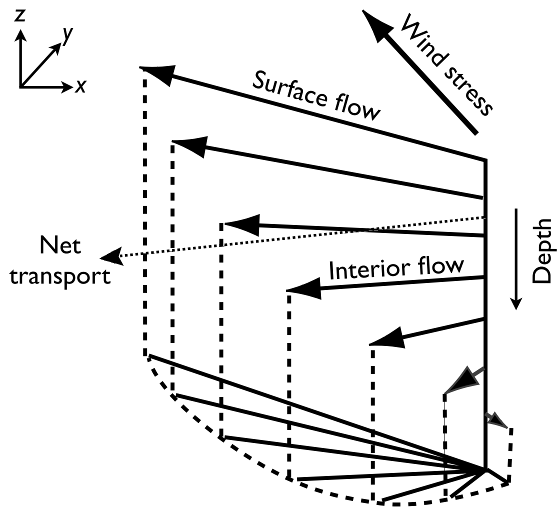

Fig. 2.12 An idealized Ekman spiral in a southern hemisphere ocean, driven by an imposed wind stress. A northern hemisphere spiral would be the reflection of this in the vertical plane. Such a clean spiral is rarely observed in the real ocean. The net transport is at right angles to the wind, independent of the detailed form of the friction. The angle of the surface flow is 45° to the wind only for a Newtonian viscosity.

which indicates that the ageostrophic transport is perpendicular to the wind stress, as noted previously from more general considerations. Suppose that the surface wind is eastward; in this case $\widetilde{\tau}^y = 0$ and the solutions immediately give

$$u(0) - u_g = \widetilde{\tau}^x / fd, \qquad v(0) - v_g = -\widetilde{\tau}^x / fd. \tag{2.301}$$

Therefore the magnitudes of the frictional flow in the x and y directions are equal to each other, and the ageostrophic flow is 45° to the right (for $f > 0$) of the wind. This result depends on the form of the frictional parameterization, but not on the size of the viscosity.

At the edge of the Ekman layer the vertical velocity is given by (2.282), and so is proportional to the curl of the wind stress. (The second term on the right-hand side of (2.282) is the vertical velocity due to the divergence of the geostrophic flow, and is usually much smaller than the first term.) The production of a vertical velocity at the edge of the Ekman layer is one of most important effects of the layer, especially with regard to the large-scale circulation, for it provides an efficient means whereby surface fluxes are communicated to the interior flow (see Fig. 2.13).

2.10.5 Observations of the Ekman layer

Ekman layers — and in particular the Ekman spiral — are generally quite hard to observe, in either the ocean or atmosphere, both because of the presence of phenomena that are not included in the theory, and because of the technical difficulties of actually measuring the vector velocity profile, especially in the ocean. Ekman-layer theory does not take into account the effects of stratification or of inertial and gravity waves (section 2.9.4 and chapter **??**), nor does it account for the effects of convection or buoyancy-driven turbulence. If gravity waves are present, the instantaneous flow will be non-geostrophic and so time-averaging will be required to extract the geostrophic flow. If strong convection is present, the simple eddy-viscosity parameterizations used to derive the Ekman spiral will be rendered invalid, and the spiral Ekman profile cannot be expected to be observed in either atmosphere or ocean.

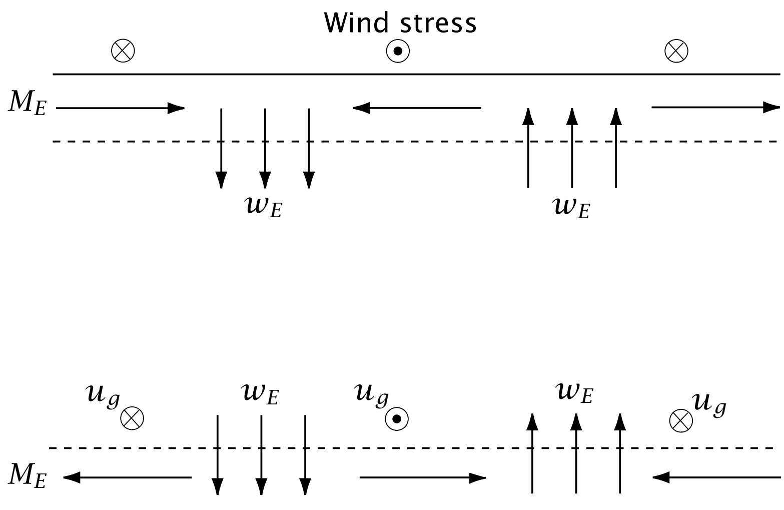

Fig. 2.13 Upper and lower Ekman layers. The upper Ekman layer in the ocean is primarily driven by an imposed wind stress, whereas the lower Ekman layer in the atmosphere or ocean largely results from the interaction of interior geostrophic velocity and a rigid lower surface. The upper part of figure shows the vertical Ekman 'pumping' velocities that result from the given wind stress, and the lower part of the figure shows the Ekman pumping velocities given the interior geostrophic flow.

In the atmosphere, in convectively neutral cases, the Ekman profile can sometimes be qualitatively discerned. In convectively unstable situations the Ekman profile is generally not observed, but the flow is nevertheless cross-isobaric, from high pressure to low, consistent with the theory. (For most purposes, it is in any case the integral properties of the Ekman layer that is most important.) In the ocean, from about 1980 onwards improved instruments have made it possible to observe the vector current with depth, and to average that current and correlate it with the overlying wind, and a number of observations generally consistent with Ekman dynamics have emerged.[13] There are some differences between observations and theory, and these can be ascribed to the effects of stratification (which causes a shallowing and flattening of the spiral), and to the interaction of the Ekman spiral with turbulence (and the inadequacy of the eddy-diffusivity parameterization). In spite of these differences, Ekman layer theory remains a remarkable and enduring foundation of geophysical fluid dynamics.

2.10.6 ♦ Frictional parameterization of the Ekman layer

[Some readers will be reading these sections on Ekman layers after having been introduced to quasi-geostrophic theory; this section is for them. Other readers may return to this section after reading chapter 5, or take (2.302) on faith.]

Suppose that the free atmosphere is described by the quasi-geostrophic vorticity equation,

$$\frac{D\zeta_g}{Dt} = f_0 \frac{\partial w}{\partial z}, \tag{2.302}$$

where ζ_g is the geostrophic relative vorticity. Let us further model the atmosphere as a single homogeneous layer of thickness H lying above an Ekman layer of thickness $d \ll H$. If the vertical velocity is negligible at the top of the layer (at $z = H + d$) the equation of motion becomes

$$\frac{\mathrm{D}\zeta_g}{\mathrm{D}t} = \frac{f_0[w(H+d) - w(d)]}{H} = -\frac{f_0 d}{2H}\zeta_g \tag{2.303}$$

using (2.294). This equation shows that the Ekman layer acts as a *linear drag* on the interior flow, with a drag coefficient r equal to $f_0 d/2H$ and with associated time scale T_{Ek} given by

$$T_{Ek} = \frac{2H}{f_0 d} = \frac{2H}{\sqrt{2f_0 A}}. \tag{2.304}$$

In the oceanic case the corresponding vorticity equation for the interior flow is

$$\frac{\mathrm{D}\zeta_g}{\mathrm{D}t} = \frac{1}{H}\mathrm{curl}_z\,\tau_s, \tag{2.305}$$

where τ_s is the surface stress. The surface stress thus acts as if it were a body force on the interior flow, and neither the Coriolis parameter nor the depth of the Ekman layer explicitly appear in this formula.

The Ekman layer is a very efficient way of communicating surface stresses to the interior. To see this, suppose that eddy mixing were the sole mechanism of transferring stress from the surface to the fluid interior, and there were no Ekman layer. Then the time scale of spindown of the fluid would be given by using

$$\frac{\mathrm{d}\zeta}{\mathrm{d}t} = A\frac{\partial^2\zeta}{\partial z^2}, \tag{2.306}$$

implying a turbulent spin-down time, T_{turb}, of

$$T_{turb} \sim \frac{H^2}{A}, \tag{2.307}$$

where H is the depth over which we require a spin-down. This is much longer than the spin-down of a fluid that has an Ekman layer, for we have

$$\frac{T_{turb}}{T_{Ek}} = \frac{(H^2/A)}{(2H/f_0 d)} = \frac{H}{d} \gg 1, \tag{2.308}$$

using $d = \sqrt{2A/f_0}$. The effects of friction are enhanced because of the presence of a secondary circulation confined to the Ekman layers (as in Fig. 2.13) in which the vertical scales are much smaller than those in the fluid interior and so where viscous effects become significant; these frictional stresses are then communicated to the fluid interior via the induced vertical velocities at the edge of the Ekman layers.

Notes

1 The distinction between Coriolis force and acceleration is not always made in the literature, even after noting that the force is considered as a force per unit mass. For a fluid in geostrophic balance, one might either say that there is a balance between the pressure force and the Coriolis force, with no net acceleration, or that the pressure force produces a Coriolis acceleration. The descriptions are equivalent, because of Newton's second law, but should not be conflated.

The Coriolis forces is named after Gaspard Gustave de Coriolis (1792–1843), who introduced the force in the context of rotating mechanical systems (Coriolis 1832, 1835), but Euler See Persson (1998) for a historical account and interpretation.

2 Phillips (1973). There is also a related discussion in Stommel & Moore (1989).

3 Phillips (1966). See White (2003) for a review. In the early days of numerical modelling these equations were the most primitive — i.e., the least filtered — equations that could practically be integrated numerically. Associated with increasing computer power there is a tendency for comprehensive numerical models to use non-hydrostatic equations of motion that do not make the shallow-fluid or traditional approximations, and it is conceivable that the meaning of the word 'primitive' may evolve to accommodate them.

4 The Boussinesq approximation is named for Boussinesq (1903), although similar approximations were used earlier by Oberbeck (1879, 1888). Spiegel & Veronis (1960) give a physically based derivation for an ideal gas, and Mihaljan (1962) provides a somewhat more general asymptotic derivation. Mahrt (1986) discusses applicability to the atmosphere.

5 As pointed out to me by W. R. Young.

6 Various versions of anelastic equations exist — see Batchelor (1953a), Ogura & Phillips (1962), Gough (1969), Gilman & Glatzmaier (1981), Lipps & Hemler (1982), and Durran (1989), although not all have potential vorticity and energy conservation laws (Bannon 1995, 1996; Scinocca & Shepherd 1992). The system we derive is most similar to that of Ogura & Phillips (1962) and unpublished notes by J. S. A. Green. The connection between the Boussinesq and anelastic equations is discussed by, among others, Lilly (1996) and Ingersoll (2005).

7 A numerical model that includes sound waves must take very small timesteps in order to maintain numerical stability, in particular to satisfy the Courant–Friedrichs–Lewy (CFL) criterion. An alternative is to use an implicit timestepping scheme that effectively lets the numerics do the filtering of the sound waves, and this approach is favoured by many numerical modellers. If we make the hydrostatic approximation then all sound waves except those that propagate horizontally are eliminated, and there is little need, vis-à-vis the numerics, to also make the anelastic approximation.

8 Gill (1980), section 6.17, provides more discussion. Holton (1992) does not differentiate between N and N_* in his section 8.4.

9 The Rossby number is named for C.-G. Rossby (see endnote on page 230), but it was also used by Kibel (1940) and is sometimes called the Kibel or Rossby–Kibel number. The notion of geostrophic balance and so, implicitly, that of a small Rossby number, predates both Rossby and Kibel.

10 After Taylor (1921b) and Proudman (1916). The Taylor–Proudman effect is sometimes called the Taylor–Proudman 'theorem', but it is more usefully thought of as a physical effect, with manifestations even when the conditions for its satisfaction are not precisely met.

11 Foster (1972).

12 After Ekman (1905). The problem was posed to V. W. Ekman (1874-1954), a student of Vilhelm Bjerknes, by Fridtjof Nansen, the polar explorer and statesman, who wanted to understand the motion of pack ice and of his ship, the *Fram*, embedded in the ice.

13 For oceanic observations see Davis *et al.* (1981), Price *et al.* (1987), Rudnick & Weller (1993). For the atmosphere see, e.g., Nicholls (1985).

Further reading

Cushman-Roisin, B., 1994. *An Introduction to Geophysical Fluid Dynamics.*
Provides a compact introduction to a variety of topics in GFD.

Gill, A. E., 1982. *Atmosphere–Ocean Dynamics.*
A rich book, especially strong on equatorial dynamics and gravity wave motion.

Holton, J. R., 1992. *An Introduction to Dynamical Meteorology.*
A deservedly well-known textbook at the upper-division undergraduate/beginning graduate level.

Pedlosky, J., 1987. *Geophysical Fluid Dynamics.*
A primary reference for flow at low Rossby number. Although the book requires some effort, there is a handsome pay-off for those who study it closely.

White (2002) provides a clear and thorough summary of the equations of motion for meteorology, including the non-hydrostatic and primitive equations.

Zdunkowski, W. & Bott, A., 2003. *Dynamics of the Atmosphere: A Course in Theoretical Meteorology.*
Concentrates on the mathematical tools and equations of motion of dynamical meteorology.

Problems

2.1 Show that for an ideal gas in hydrostatic balance, changes in dry static energy ($M = c_pT + gz$) and potential temperature (θ) are related by $\delta M = c_p(T/\theta)\delta\theta$. (The quantity c_pT/θ is known as the 'Exner function', and is denoted Π.)

2.2 For an ideal gas in hydrostatic balance, show that:

(a) The integral of the potential plus internal energy from the surface to the top of the atmosphere [$\int(P + I)\,\mathrm{d}p$] is is equal to its enthalpy;

(b) $\mathrm{d}\sigma/\mathrm{d}z = c_p(T/\theta)\,\mathrm{d}\theta/\mathrm{d}z$, where $\sigma = I + p\alpha + \Phi$ is the dry static energy;

(c) The following expressions for the pressure gradient force are all equal (even without hydrostatic balance):

$$-\frac{1}{\rho}\nabla p = -\theta\,\nabla\Pi = -\frac{c_s^2}{\rho\theta}\nabla(\rho\theta), \tag{P2.1}$$

where $\Pi = c_pT/\theta$ is the Exner function.

(d) Show that item *(a)* also holds for a gas with an arbitrary equation of state.

2.3 Show that, without approximation, the unforced, inviscid momentum equation may be written in the forms

$$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} = T\nabla\eta - \nabla(p\alpha + I) \tag{P2.2}$$

and

$$\frac{\partial\boldsymbol{v}}{\partial t} + \boldsymbol{\omega}\times\boldsymbol{v} = T\nabla\eta - \nabla B \tag{P2.3}$$

where $\boldsymbol{\omega} = \nabla\times\boldsymbol{v}$, η is the specific entropy ($\mathrm{d}\eta = c_p\,\mathrm{d}\ln\theta$) and $B = I + \boldsymbol{v}^2/2 + p\alpha$ where I is the internal energy per unit mass.

Hint: First show that $T\nabla\eta = \nabla I + p\nabla\alpha$, and note also the vector identity $\boldsymbol{v}\times(\nabla\times\boldsymbol{v}) = \frac{1}{2}\nabla(\boldsymbol{v}\cdot\boldsymbol{v}) - (\boldsymbol{v}\cdot\nabla)\boldsymbol{v}$.

2.4 Consider two-dimensional fluid flow in a rotating frame of reference on the f-plane. Linearize the equations about a state of rest.

(a) Ignore the pressure term and determine the general solution to the resulting equations. Show that the speed of fluid parcels is constant. Show that the trajectory of the fluid parcels is a circle with radius $|U|/f$, where $|U|$ is the fluid speed.

(b) What is the period of oscillation of a fluid parcel?

(c) ◆ If parcels travel in straight lines in inertial frames, why is the answer to (b) not the same as the period of rotation of the frame of reference? [To answer this fully you need to understand the dynamics underlying inertial oscillations and inertia circles. See Durran (1993), Egger (1999) and Phillips (2000).]

2.5 A fluid at rest evidently satisfies the hydrostatic relation, which says that the pressure at the surface is given by the weight of the fluid above it. Now consider a *deep* atmosphere on a spherical planet. A unit cross-sectional area at the planet's surface lies beneath a column of fluid whose cross-section increases with height, because the total area of the atmosphere increases with distance away from the centre of the planet. Is the pressure at the surface still given by the hydrostatic relation, or is it greater than this because of the increased mass of fluid in the column? If it is still given by the hydrostatic relation, then the pressure at the surface, integrated over the entire area of the planet, is less than the total weight of the fluid; resolve this paradox. But if the pressure at the surface is greater than that implied by hydrostatic balance, explain how the hydrostatic relation fails.

2.6 In a self-gravitating spherical fluid, like a star, hydrostatic balance may be written

$$\frac{\partial p}{\partial r} = -\frac{GM(r)}{r^2}\rho, \tag{P2.4}$$

where $M(r)$ is the mass interior to a sphere of radius r, and G is a constant. Obtain an expression for the pressure as a function of radius when the fluid (a) has constant density, and (b) is an isothermal ideal gas (if possible). The star is of radius a.

2.7 At what latitude is the angle between the direction of Newtonian gravity (due solely to the mass of the Earth) and that of effective gravity (Newtonian gravity plus centrifugal terms) the largest? At what latitudes, if any, is this angle zero?

2.8 ◆ Write the momentum equations in true spherical coordinates, including the centrifugal and gravitational terms. Show that for reasonable values of the wind, the dominant balance in the meridional component of this equation involves a balance between centrifugal and pressure gradient terms. Can this balance be subtracted out of the equations in a sensible way, so leaving a useful horizontal momentum equation that involves the Coriolis and acceleration terms? If so, obtain a closed set of equations for the flow this way. Discuss the pros and cons of this approach versus the geometric approximation discussed in section 2.2.1.

2.9 For an ideal gas show that the expressions (2.233) and (2.237) are equivalent.

2.10 Consider an ocean at rest with known vertical profiles of potential temperature and salinity, $\theta(z)$ and $S(z)$. Suppose we also know the equation of state in the form $\rho = \rho(\theta, S, p)$. Obtain an expression for the buoyancy frequency. Check your expression by substituting the equation of state for an ideal gas and recovering a known expression for the buoyancy frequency.

2.11 *(a)* The *geopotential height* is the height of a given pressure level. Show that in an atmosphere with a uniform lapse rate (i.e., $\mathrm{d}T/\mathrm{d}z = \Gamma$ = constant) the geopotential height at a pressure p is given by

$$z = \frac{T_0}{\Gamma}\left[1 - \left(\frac{p_0}{p}\right)^{-R\Gamma/g}\right] \tag{P2.5}$$

where T_0 is the temperature at $z = 0$.

(b) In an isothermal atmosphere, obtain an expression for the geopotential height as function of pressure, and show that this is consistent with the expression (P2.5) in the appropriate limit.

2.12 Show that the inviscid, adiabatic, hydrostatic primitive equations for a compressible fluid conserve a form of energy (kinetic plus potential plus internal), and that the kinetic energy has no contribution from the vertical velocity. You may assume Cartesian geometry and a uniform gravitational field in the vertical direction.

2.13 Consider the simple Boussinesq equations, $\mathrm{D}\boldsymbol{v}/\mathrm{D}t = -\nabla\phi + \mathbf{k}b + \nu\nabla^2\boldsymbol{v}$, $\nabla\cdot\boldsymbol{v} = 0$, $\mathrm{D}b/\mathrm{D}t = Q + \kappa\nabla^2 b$. Obtain an energy equation similar to (2.112), but now with the terms on the right-hand side that represent viscous and diabatic effects. Over a closed volume, show that the dissipation of kinetic energy is balanced by a buoyancy source. Also show that, in a statistically steady state, the heating must occur at a lower level than the cooling if a kinetic-energy dissipating circulation is to be maintained.

2.14 ◆ Suppose a fluid is contained in a closed container, with insulating sidewalls, and heated from below and cooled from above. The heating and cooling are adjusted so that there is no net energy flux into the fluid. Let us also suppose that any viscous dissipation of kinetic energy is returned as heating, so the total energy of the fluid is exactly constant. Suppose the fluid starts out at rest and at a uniform temperature, and the heating and cooling are then turned on. A very short time afterwards, the fluid is lighter at the bottom and heavier at the top; that is, its potential energy has increased. Where has this energy come from? Discuss this paradox for both a compressible fluid (e.g., an ideal gas) and for a simple Boussinesq fluid.

2.15 Consider a rapidly rotating (i.e., in near geostrophic balance) Boussinesq fluid on the f-plane.

(a) Show that the pressure divided by the density scales as $\phi \sim fUL$.

(b) Show that the horizontal divergence of the geostrophic wind vanishes. Thus, argue that the scaling $W \sim UH/L$ is an *overestimate* for the magnitude of the vertical velocity. (Optional extra: obtain a scaling estimate for the magnitude of vertical velocity in rapidly rotating flow.)

(c) Using these results, or otherwise, discuss whether hydrostatic balance is more or less likely to hold in a rotating flow than in non-rotating flow.

2.16 Estimate the size of the zonal wind 5 km above the surface in the mid-latitude atmosphere in summer and winter using (approximate) values for the meridional temperature gradient in the atmosphere. Also estimate the shear corresponding to the pole–equator temperature gradient in the ocean.

2.17 Using approximate but realistic values for the observed stratification, what is the buoyancy period for (a) the mid-latitude troposphere, (b) the stratosphere, (c) the oceanic thermocline, (d) the oceanic abyss?

2.18 Consider a dry, hydrostatic, ideal-gas atmosphere whose lapse rate is one of constant potential temperature. What is its vertical extent? That is, at what height does the density vanish? Is this a problem for the anelastic approximation discussed in the text?

2.19 Show that for an ideal gas, the expressions (2.237), (2.232), (2.233) are all equivalent, and express N^2 terms of the temperature lapse rate, $\partial T/\partial z$.

2.20 ◆ Calculate an approximate but reasonably accurate expression for the buoyancy equation for seawater. (From notes by R. de Szoeke)

Solution (i): the buoyancy frequency is given by

$$N^2 = -\frac{g}{\rho}\left(\frac{\partial \rho_{pot}}{\partial z}\right)_{\mathrm{env}} = \frac{g}{\alpha}\left(\frac{\partial \alpha_{pot}}{\partial z}\right)_{\mathrm{env}} = -\frac{g^2}{\alpha^2}\left(\frac{\partial \alpha_{pot}}{\partial p}\right)_{\mathrm{env}} \tag{P2.6}$$

where $\alpha_{pot} = \alpha(\theta, S, p_R)$ is the potential density, and p_R a reference pressure. From (1.155)

$$\alpha_{pot} = \alpha_0\left[1 - \frac{\alpha_0}{c_0^2}p_R + \beta_T(1+\gamma^* p_R)\theta' + \frac{1}{2}\beta_T^*\theta'^2 - \beta_S(S - S_0)\right]. \tag{P2.7}$$

Using this and (P2.6) we obtain the buoyancy frequency,

$$N^2 = -\frac{g^2}{\alpha^2}\alpha_0\left[\beta_T\left(1+\gamma p_R+\frac{\beta_T^*}{\beta_T}\theta\right)\left(\frac{\partial\theta}{\partial p}\right)_{\text{env}} - \beta_S\left(\frac{\partial S}{\partial p}\right)_{\text{env}}\right], \tag{P2.8}$$

although we must substitute local pressure for the reference pressure p_R. (Why?)

Solution (ii): the sound speed is given by

$$c_s^{-2} = -\frac{1}{\alpha^2}\left(\frac{\partial\alpha}{\partial p}\right)_{\theta,S} = \frac{1}{\alpha^2}\left(\frac{\alpha_0^2}{c_0^2} - \gamma\alpha_1\theta\right) \tag{P2.9}$$

and, using (P2.6) and (2.237) the square of the buoyancy frequency may be written

$$N^2 = \frac{g}{\alpha}\left(\frac{\partial\alpha}{\partial z}\right)_{\text{env}} - \frac{g^2}{c_s^2} = -\frac{g^2}{\alpha^2}\left[\left(\frac{\partial\alpha}{\partial p}\right)_{\text{env}} + \frac{\alpha^2}{c_s^2}\right]. \tag{P2.10}$$

Using (1.155), (P2.9) and (P2.10) we recover (P2.8), although now with p explicitly in place of p_R.

2.21 *(a)* Use the chain rule to show that the horizontal gradients of a field in height coordinates and in ξ coordinates are related by

$$\nabla_z \Psi = \nabla_\xi\Psi - (\partial\Psi/\partial\xi)(\partial\xi/\partial z)\nabla_\xi z. \tag{P2.11}$$

(b) Show that w, the vertical velocity in height coordinates, may be expressed in ξ coordinates as

$$w = \mathrm{D}z/\mathrm{D}t = (\partial z/\partial t)_\xi + \boldsymbol{u}\cdot\nabla_\xi z + \dot{\xi}\partial z/\partial\xi. \tag{P2.12}$$

(c) Use the above expressions to verify (2.143), the expression for the material derivative in ξ coordinates.

2.22 Begin with the mass conservation in height coordinates, namely $\mathrm{D}\rho/\mathrm{D}t + \rho\nabla\cdot\boldsymbol{v} = 0$. Transform this into pressure coordinates using the chain rule (or otherwise) and derive the mass conservation equation in the form $\nabla_p\cdot\boldsymbol{u} + \partial\omega/\partial p = 0$.

2.23 ◆ Starting with the primitive equations in pressure coordinates, derive the form of the primitive equations of motion in sigma-pressure coordinates. In particular, show that the prognostic equation for surface pressure is,

$$\frac{\partial p_s}{\partial t} + \nabla\cdot(p_s\boldsymbol{u}) + p_s\frac{\partial\dot{\sigma}}{\partial\sigma} = 0 \tag{P2.13}$$

and that hydrostatic balance may be written $\partial\Phi/\partial\sigma = -RT/\sigma$.

2.24 Starting with the primitive equations in pressure coordinates, derive the form of the primitive equations of motion in log-pressure coordinates in which $Z = -H\ln(p/p_r)$ is the vertical coordinate. Here, H is a reference height (e.g., a scale height RT_r/g where T_r is a typical or an average temperature) and p_r is a reference pressure (e.g., 1000 mb). In particular, show that if the 'vertical velocity' is $W = \mathrm{D}Z/\mathrm{D}t$ then $W = -H\omega/p$ and that

$$\frac{\partial\omega}{\partial p} = -\frac{\partial}{\partial p}\left(\frac{pW}{H}\right) = \frac{\partial W}{\partial Z} - \frac{W}{H}. \tag{P2.14}$$

and obtain the mass conservation equation (2.161). Show that this can be written in the form

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{1}{\rho_s}\frac{\partial}{\partial Z}(\rho_s W) = 0, \tag{P2.15}$$

where $\rho_s = \rho_r\exp(-Z/H)$.

2.25 *(a)* Prove that the argument of the square root in (**??**) is always positive.
Solution: The largest value or the argument occurs when $m = 0$ and $k^2 = 1/(4H^2)$. The argument is then $1 - 4H^2N^2/c_s^2$. But $c_s^2 = \gamma RT_0 = \gamma gH$ and $N^2 = g\kappa/H$ so that $4N^2H^2/c_s^2 = 4\kappa/\gamma \approx 0.8$.
(b) ◆ This argument seems to depend on the parameters in the ideal gas equation of state. Is it more general than this? Is a natural system possible for which the argument is negative, and if so what physical interpretation could one ascribe to the situation?

2.26 Consider a wind stress imposed by a mesoscale cyclonic storm (in the atmosphere) given by

$$\boldsymbol{\tau} = -A\mathrm{e}^{-(r/\lambda)^2}(y\,\mathbf{i} - x\,\mathbf{j}) \tag{P2.16}$$

where $r^2 = x^2 + y^2$, and A and λ are constants. Also assume constant Coriolis gradient $\beta = \partial f/\partial y$ and constant ocean depth H. In the ocean, find (a) the Ekman transport, (b) the vertical velocity $w_E(x, y, z)$ below the Ekman layer, (c) the northward velocity $v(x, y, z)$ below the Ekman layer and (d) indicate how you would find the westward velocity $u(x, y, z)$ below the Ekman layer.

2.27 ◆ In an atmospheric Ekman layer on the f-plane let us write the momentum equation as

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi + \frac{1}{\rho_a}\frac{\partial \boldsymbol{\tau}}{\partial z}, \tag{P2.17}$$

where $\boldsymbol{\tau} = A\rho_a \partial\boldsymbol{u}/\partial z$ and A is a constant eddy viscosity coefficient. An *independent* formula for the stress at the ground is $\boldsymbol{\tau} = C\rho_a\boldsymbol{u}$, where C is a constant. Let us take $\rho_a = 1$, and assume that in the free atmosphere the wind is geostrophic and zonal, with $\boldsymbol{u}_g = U\mathbf{i}$.
(a) Find an expression for the wind vector at the ground. Discuss the limits $C = 0$ and $C = \infty$. Show that when $C = 0$ the frictionally-induced vertical velocity at the top of the Ekman layer is zero.
(b) Find the vertically integrated horizontal mass flux caused by the boundary layer.
(c) When the stress on the atmosphere is $\boldsymbol{\tau}$, the stress on the ocean beneath is also $\boldsymbol{\tau}$. Why? Show how this consistent with Newton's third law.
(d) Determine the direction and strength of the surface current, and the mass flux in the oceanic Ekman layer, in terms of the geostrophic wind in the atmosphere, the oceanic Ekman depth and the ratio ρ_a/ρ_o, where ρ_o is the density of the seawater. Include a figure showing the directions of the various winds and currents. How does the boundary-layer mass flux in the ocean compare to that in the atmosphere? (Assume, as needed, that the stress in the ocean may be parameterized with an eddy viscosity.)

Partial solution for (a): A useful trick in Ekman layer problems is to write the velocity as a complex number, $\hat{u} = u + \mathrm{i}v$ and $\hat{u}_g = u_g + \mathrm{i}v_g$. The Ekman layer equation, (2.283a), may then be written as

$$A\frac{\partial^2 \hat{U}}{\partial z^2} = \mathrm{i}f\hat{U}, \tag{P2.18}$$

where $\hat{U} = \hat{u} - \hat{u}_g$. The solution to this is

$$\hat{u} - \hat{u}_g = [\hat{u}(0) - \hat{u}_g]\exp\left[-\frac{(1+\mathrm{i})z}{d}\right], \tag{P2.19}$$

where $d = \sqrt{2A/f}$ and the boundary condition of finiteness at infinity eliminates the exponentially growing solution. The boundary condition at $z = 0$ is $\partial\hat{u}/\partial z = (C/A)\hat{u}$; applying this gives $[\hat{u}(0) - \hat{u}_g]\exp(\mathrm{i}\pi/4) = -Cd\hat{u}(0)/(\sqrt{2}A)$, from which we obtain $\hat{u}(0)$, and the rest of the solution follows. We may also obtain a solution using the same method that was used to obtain (2.287).

2.28 *The logarithmic boundary layer*

Close to ground rotational effects are unimportant and small-scale turbulence generates a *mixed layer.* In this layer, assume that the stress is constant and that it can be parameterized by an eddy diffusivity the size of which is proportional to the distance from the surface. Show that the velocity then varies logarithmically with height.

Solution: Write the stress as $\tau = \rho_0 u^{*2}$ where the constant u^* is called the friction velocity. Using the eddy diffusivity hypothesis this stress is given by

$$\tau = \rho_0 u^{*2} = \rho_0 A \frac{\partial u}{\partial z} \quad \text{where} \quad A = u^* kz, \tag{P2.20}$$

where k is von Karman's ('universal') constant (approximately equal to 0.4). From (P2.20) we have $\partial u/\partial z = u^*/(kz)$ which integrates to give $u = (u^*/k)\ln(z/z_0)$. The parameter z_0 is known as the roughness length.

Part II

WAVES, INSTABILITIES AND TURBULENCE

And the waves sing because they are moving.
Philip Larkin

CHAPTER SIX

Wave Fundamentals

THIS CHAPTER PROVIDES AN INTRODUCTION BOTH TO WAVE MOTION ITSELF and to what is perhaps the most important kind of wave occurring at large scales in the ocean and atmosphere, namely the Rossby wave.[1] The chapter has three main parts to it. In the first, we provide an introduction to wave kinematics, discussing such basic concepts as phase speed and group velocity. The second part, beginning with section 6.4, is a discussion of the dynamics of Rossby waves, and this part may be considered to be the natural follow-on from the previous chapter. Finally, in section 6.8, we return to group velocity in a somewhat more general way and illustrate the results using Poincaré waves. Wave kinematics is a rather formal topic, yet closely tied to wave dynamics: kinematics without a dynamical example is jejune and dry, yet understanding wave dynamics of any sort is hardly possible without appreciating at least some of the formal structure of waves. Readers should flip pages back and forth through the chapter as needed.

Those readers who already have a knowledge of wave motion and who wish to cut to the chase quickly may wish to skip the first few sections and begin at section 6.4. Other readers may wish to skip the sections on Rossby waves altogether and, after absorbing the sections on the wave theory move on to chapter **??** on gravity waves, returning to Rossby waves (or not) later on. The Rossby wave and gravity wave chapters are largely independent of each other, although they both require that the reader is familiar with the basic ideas of wave analysis such as group velocity and phase speed. Rossby waves and gravity waves can, of course, coexist and we give an introduction to that topic in section 6.7. Close to the equator the two kinds of waves become more intertwined and we deal with the ensuing waves in more depth in chapter **??**. We also extend our discussion of Rossby waves in a global atmospheric context in chapter 16.

6.1 FUNDAMENTALS AND FORMALITIES

6.1.1 Definitions and kinematics

What is a wave? Rather like turbulence, a wave is more easily recognized than defined. Perhaps a little loosely, a wave may be considered to be a propagating disturbance that has a characteristic relationship between its frequency and size; more formally, a wave is a disturbance that satisfies a *dispersion relation*. In order to see what this means, and what a dispersion relation is, suppose that a disturbance, $\psi(\boldsymbol{x}, t)$ (where ψ might be velocity, streamfunction, pressure, etc), satisfies some equation

$$L(\psi) = 0, \tag{6.1}$$

where L is a linear operator, typically a polynomial in time and space derivatives; an example is $L(\psi) = \partial\nabla^2\psi/\partial t + \beta\partial\psi/\partial x$. (Nonlinear waves exist, but the curious reader must look elsewhere to learn about them.[2]) If (6.1) has constant coefficients (if β is constant in this example) then solutions may often be found as a superposition of *plane waves,* each of which satisfy

$$\psi = \mathrm{Re}\,\widetilde{\psi}\mathrm{e}^{\mathrm{i}\theta(x,t)} = \mathrm{Re}\,\widetilde{\psi}\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}. \tag{6.2}$$

where $\widetilde{\psi}$ is a complex constant, θ is the phase, ω is the wave frequency and $\boldsymbol{k}$ is the vector wavenumber (k, l, m) (also written as (k^x, k^y, k^z) or, in subscript notation, k_i). The prefix Re denotes the real part of the expression, but we will drop that notation if there is no ambiguity.

Earlier, we said that waves are characterized by having a particular relationship between the frequency and wavevector known as the *dispersion relation*. This is an equation of the form

$$\omega = \Omega(\boldsymbol{k}) \tag{6.3}$$

where $\Omega(\boldsymbol{k})$ [or $\Omega(k_i)$, and meaning $\Omega(k, l, m)$] is some function determined by the form of L in (6.1) and which thus depends on the particular type of wave — the function is different for sound waves, light waves and the Rossby waves and gravity waves we will encounter in this book (peak ahead to (6.59) and (**??**), and there is more discussion in section 6.1.3). Unless it is necessary to explicitly distinguish the function Ω from the frequency ω, we will often write $\omega = \omega(\boldsymbol{k})$.

If the medium in which the waves are propagating is inhomogeneous then (6.1) will probably not have constant coefficients (for example, β may vary meridionally). Nevertheless, if the medium is varying sufficiently slowly, wave solutions may often still be found with the general form

$$\psi(\boldsymbol{x}, t) = \mathrm{Re}\,a(\boldsymbol{x}, t)\mathrm{e}^{\mathrm{i}\theta(\boldsymbol{x},t)}, \tag{6.4}$$

where $a(\boldsymbol{x}, t)$ varies slowly compared to the variation of the phase, θ. The frequency and wavenumber are then *defined* by

$$\boldsymbol{k} \equiv \nabla\theta, \qquad \omega \equiv -\frac{\partial\theta}{\partial t}. \tag{6.5}$$

The example of (6.2) is clearly just a special case of this. Eq. (6.5) implies the formal relation between $\boldsymbol{k}$ and ω:

$$\frac{\partial\boldsymbol{k}}{\partial t} + \nabla\omega = 0. \tag{6.6}$$

6.1.2 Wave propagation and phase speed

An almost universal property of waves is that they propagate through space with some velocity (which in special cases might be zero). Waves in fluids may carry energy and momentum but not normally, at least to a first approximation, fluid parcels themselves. Further, it turns out that the speed at which properties like energy are transported (the group speed) may be different from the speed at which the wave crests themselves move (the phase speed). Let's try to understand this statement, beginning with the phase speed.

Phase speed

Let us consider the propagation of monochromatic plane waves, for that is all that is needed to introduce the phase speed. Given (6.2) a wave will propagate in the direction of $\boldsymbol{k}$ (Fig. 6.1). At a given instant and location we can align our coordinate axis along this direction, and we write $\boldsymbol{k}\cdot\boldsymbol{x} = Kx^*$, where x^* increases in the direction of $\boldsymbol{k}$ and $K^2 = |\boldsymbol{k}|^2$ is the magnitude of the wavenumber. With this, we can write (6.2) as

$$\psi = \operatorname{Re}\widetilde{\psi}\mathrm{e}^{\mathrm{i}(Kx^*-\omega t)} = \operatorname{Re}\widetilde{\psi}\mathrm{e}^{\mathrm{i}K(x^*-ct)}, \tag{6.7}$$

where $c = \omega/K$. From this equation it is evident that the phase of the wave propagates at the speed c in the direction of $\boldsymbol{k}$, and we define the *phase speed* by

$$c_p \equiv \frac{\omega}{K}. \tag{6.8}$$

The wavelength of the wave, λ, is the distance between two wavecrests — that is, the distance between two locations along the line of travel whose phase differs by 2π — and evidently this is given by

$$\lambda = \frac{2\pi}{K}. \tag{6.9}$$

In (for simplicity) a two-dimensional wave, and referring to Fig. 6.1(a), the wavelength and wave vectors in the x- and y-directions are given by,

$$\lambda^x = \frac{\lambda}{\cos\phi}, \quad \lambda^y = \frac{\lambda}{\sin\phi}, \qquad k^x = K\cos\phi, \quad k^y = K\sin\phi. \tag{6.10}$$

In general, lines of constant phase intersect both the coordinate axes and propagate along them. The speed of propagation along these axes is given by

$$c_p^x = c_p\frac{l^x}{l} = \frac{c_p}{\cos\phi} = c_p\frac{K}{k^x} = \frac{\omega}{k^x}, \qquad c_p^y = c_p\frac{l^y}{l} = \frac{c_p}{\sin\phi} = c_p\frac{K}{k^y} = \frac{\omega}{k^y}, \tag{6.11}$$

using (6.8) and (6.10). The speed of phase propagation along any one of the axis is in general *larger* than the phase speed in the primary direction of the wave. The phase speeds are clearly *not* components of a vector: for example, $c_p^x \neq c_p\cos\phi$. Analogously, the wavevector $\boldsymbol{k}$ is a true vector, whereas the wavelength λ is not.

To summarize, the phase speed and its components are given by

$$\boxed{c_p = \frac{\omega}{K}, \qquad c_p^x = \frac{\omega}{k^x}, \quad c_p^y = \frac{\omega}{k^y}}. \tag{6.12}$$

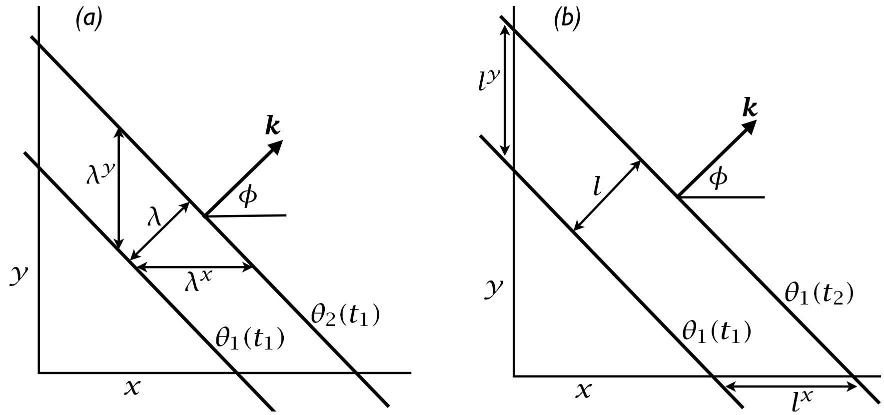

Fig. 6.1 The propagation of a two-dimensional wave. (a) Two lines of constant phase (e.g., two wavecrests) at a time t_1. The wave is propagating in the direction $\boldsymbol{k}$ with wavelength λ. (b) The same line of constant phase at two successive times. The phase speed is the speed of advancement of the wavecrest in the direction of travel, and so $c_p = l/(t_2 - t_1)$. The phase speed in the x-direction is the speed of propagation of the wavecrest along the x-axis, and $c_p^x = l^x/(t_2 - t_1) = c_p/\cos\phi$.

Phase velocity

Although it is not particularly useful, there is a way of defining a phase speed so that is a true vector, and which might then be called phase velocity. We define the phase velocity to be the phase speed in the direction in which the wave crests are propagating; that is

$$\boldsymbol{c}_p \equiv \frac{\omega}{K}\frac{\boldsymbol{k}}{|K|}, \tag{6.13}$$

where $\boldsymbol{k}/|K|$ is the unit vector in the direction of wave-crest propagation. The components of the phase velocity in the the x- and y-directions are then given by

$$c_p^x = \boldsymbol{c}_p \cos\phi, \qquad c_p^y = \boldsymbol{c}_p \sin\phi. \tag{6.14}$$

Defined this way, the quantity given by (6.13) is indeed a true vector velocity. However, the components in the x- and y-directions are manifestly not the speed at which wave crests propagate in those directions. It is therefore a misnomer to call these quantities phase speeds, although it is helpful to ascribe a direction to the phase speed and so the quantity given by (6.13) can be useful.

6.1.3 The dispersion relation

The above description is mostly kinematic and a little abstract, applying to almost any disturbance that has a wavevector and a frequency. The particular *dynamics* of a wave are determined by the relationship between the wavevector and the frequency; that is, by the *dispersion relation*. Once the dispersion relation is known a great many of the properties of the wave follow in a more-or-less straightforward manner, as we will see. Picking up from (6.3), the dispersion relation is a functional relationship between the frequency and the wavevector of the general form

$$\omega = \Omega(\boldsymbol{k}). \tag{6.15}$$

Perhaps the simplest example of a linear operator that gives rise to waves is the one-dimensional equation

$$\frac{\partial \psi}{\partial t} + c\frac{\partial \psi}{\partial x} = 0. \tag{6.16}$$

Substituting a trial solution of the form $\psi = \mathrm{Re}\, A\mathrm{e}^{\mathrm{i}(kx-\omega t)}$, where Re denotes the real part, we obtain $(-\mathrm{i}\omega + c\mathrm{i}k)A = 0$, giving the dispersion relation

$$\omega = ck. \tag{6.17}$$

The phase speed of this wave is $c_p = \omega/k = c$. A few other examples of governing equations, dispersion relations and phase speeds are:

$$\frac{\partial \psi}{\partial t} + \boldsymbol{c}\cdot\nabla\psi = 0, \quad \omega = \boldsymbol{c}\cdot\boldsymbol{k}, \quad c_p = |\boldsymbol{c}|\cos\theta, \quad c_p^x = \frac{\boldsymbol{c}\cdot\boldsymbol{k}}{k}, \quad c_p^y = \frac{\boldsymbol{c}\cdot\boldsymbol{k}}{l} \tag{6.18a}$$

$$\frac{\partial^2 \psi}{\partial t^2} - c^2\nabla^2\psi = 0, \quad \omega^2 = c^2K^2, \quad c_p = \pm c, \quad c_p^x = \pm\frac{cK}{k}, \quad c_p^y = \pm\frac{cK}{l}, \tag{6.18b}$$

$$\frac{\partial}{\partial t}\nabla^2\psi + \beta\frac{\partial \psi}{\partial x} = 0, \quad \omega = \frac{-\beta k}{K^2}, \quad c_p = \frac{\omega}{K}, \quad c_p^x = -\frac{\beta}{K^2}, \quad c_p^y = -\frac{\beta k/l}{K^2}. \tag{6.18c}$$

where $K^2 = k^2 + l^2$ and θ is the angle between $\boldsymbol{c}$ and $\boldsymbol{k}$, and the examples are all two-dimensional, with variation in x and y only.

A wave is said to be *nondispersive* or *dispersionless* if the phase speed is independent of the wavelength. This condition is clearly satisfied for the simple example (6.16) but is manifestly not satisfied for (6.18c), and these waves (Rossby waves, in fact) are *dispersive*. Waves of different wavelengths then travel at different speeds so that a group of waves will spread out — disperse — even if the medium is homogeneous. When a wave is dispersive there is another characteristic speed at which the waves propagate, known as the group velocity, and we come to this in the next section.

Most media are, of course, inhomogeneous, but if the medium varies sufficiently slowly in space and time — and in particular if the variations are slow compared to the wavelength and period — we may still have a *local* dispersion relation between frequency and wavevector,

$$\omega = \Omega(\boldsymbol{k}; \boldsymbol{x}, t). \tag{6.19}$$

Although Ω is a function of $\boldsymbol{k}, \boldsymbol{x}$ and t the semi-colon in (6.19) is used to suggest that x and t are slowly varying parameters of a somewhat different nature than $\boldsymbol{k}$. We'll resume our discussion of this topic in section 6.3, but before that we must introduce the group velocity.

6.2 GROUP VELOCITY

Information and energy clearly cannot travel at the phase speed, for as the direction of propagation of the phase line tends to a direction parallel to the y-axis, the phase speed in the x-direction tends to infinity! Rather, it turns out that most quantities of interest, including energy, propagate at the *group velocity*, a quantity of enormous importance in wave theory.[3]

Wave Fundamentals

- A wave is a propagating disturbance that has a characteristic relationship between its frequency and size, known as the dispersion relation. Waves typically arise as solutions to a linear problem of the form
$$L(\psi) = 0, \tag{WF.1}$$
where L is (commonly) a linear operator in space and time. Two examples are
$$\frac{\partial^2 \psi}{\partial t^2} - c^2 \nabla^2 \psi = 0 \qquad \text{and} \qquad \frac{\partial}{\partial t}\nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = 0. \tag{WF.2}$$
The first example is so common in all areas of physics it is sometimes called 'the' wave equation. The second example gives rise to Rossby waves.

- Solutions to the governing equation are often sought in the form of plane waves that have the form
$$\psi = \mathrm{Re}\, A \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}, \tag{WF.3}$$
where A is the wave amplitude, $\boldsymbol{k} = (k, l, m)$ is the wavevector, and ω is the frequency.

- The dispersion relation connects the frequency and wavevector through an equation of the form $\omega = \Omega(\boldsymbol{k})$ where Ω is some function. The relation is normally derived by substituting a trial solution like (WF.3) into the governing equation (WF.1). For the examples of (WF.2) we obtain $\omega = c^2K^2$ and $\omega = -\beta k/K^2$ where $K^2 = k^2 + l^2 + m^2$ or, in two dimensions, $K^2 = k^2 + l^2$.

- The phase speed is the speed at which the wave crests move. In the direction of propagation and in the x, y and z directions the phase speed is given by, respectively,
$$c_p = \frac{\omega}{K}, \qquad c_p^x = \frac{\omega}{k}, \quad c_p^y = \frac{\omega}{l}, \quad c_p^z = \frac{\omega}{m}. \tag{WF.4}$$
where $K = 2\pi/\lambda$ where λ is the wavelength. The wave crests have both a speed (c_p) and a direction of propagation (the direction of $\boldsymbol{k}$), like a vector, but the components defined in (WF.4) are not the components of that vector.

- The group velocity is the velocity at which a wave packet or wave group moves. It is a vector and is given by
$$\boldsymbol{c}_g = \frac{\partial \omega}{\partial \boldsymbol{k}} \quad \text{with components} \quad c_g^x = \frac{\partial \omega}{\partial k}, \quad c_g^y = \frac{\partial \omega}{\partial l}, \quad c_g^z = \frac{\partial \omega}{\partial m}. \tag{WF.5}$$
Most physical quantities of interest are transported at the group velocity.

- If the coefficients of the wave equation are not constant (for example if the medium is inhomogeneous) then, if the coefficients are only slowly varying, approximate solutions may sometimes be found in the form
$$\psi = \mathrm{Re}\, A(\boldsymbol{x}, t) \mathrm{e}^{\mathrm{i}\theta(\boldsymbol{x}, t)}, \tag{WF.6}$$
where the amplitude A is also slowly varying and the local wavenumber and frequency are related to the phase, θ, by $\boldsymbol{k} = \nabla\theta$ and $\omega = -\partial\theta/\partial t$. The dispersion relation is then a *local* one of the form $\omega = \Omega(\boldsymbol{k}; x, t)$.

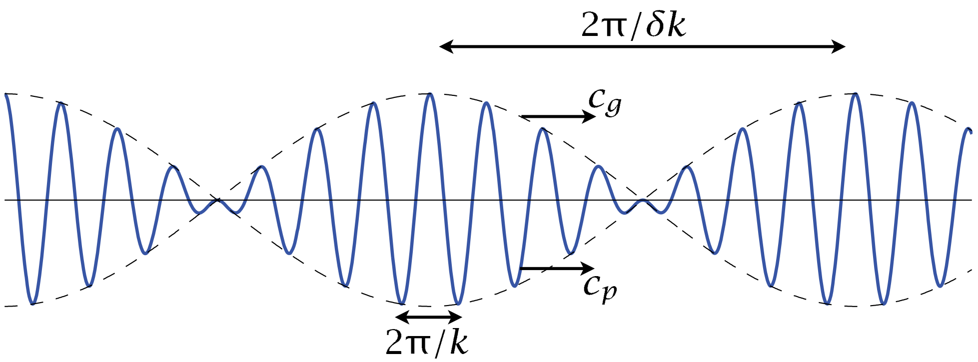

Fig. 6.2 Superposition of two sinusoidal waves with wavenumbers k and $k + \delta k$, producing a wave (solid line) that is modulated by a slowly varying wave envelope or wave packet (dashed line). The envelope moves at the group velocity, $c_g = \partial\omega/\partial k$ and the phase of the wave moves at the group speed $c_p = \omega/k$.

Roughly speaking, group velocity is the velocity at which a packet or a group of waves will travel, whereas the individual wave crests travel at the phase speed. To introduce the idea we will consider the superposition of plane waves, noting that a monochromatic plane wave already fills space uniformly so that there can be no propagation of energy from place to place. We will restrict attention to waves propagating in one direction, but the argument may be extended to two or three dimensions.

6.2.1 Superposition of two waves

Consider the linear superposition of two waves. Limiting attention to the one-dimensional case for simplicity, consider a disturbance represented by

$$\psi = \mathrm{Re}\,\widetilde{\psi}(\mathrm{e}^{\mathrm{i}(k_1 x - \omega_1 t)} + \mathrm{e}^{\mathrm{i}(k_2 x - \omega_2 t)}). \tag{6.20}$$

Let us further suppose that the two waves have similar wavenumbers and frequency, and, in particular, that $k_1 = k + \Delta k$ and $k_2 = k - \Delta k$, and $\omega_1 = \omega + \Delta\omega$ and $\omega_2 = \omega - \Delta\omega$. With this, (6.20) becomes

$$\begin{aligned}\psi &= \mathrm{Re}\,\widetilde{\psi}\mathrm{e}^{\mathrm{i}(kx-\omega t)}[\mathrm{e}^{\mathrm{i}(\Delta k\, x - \Delta\omega\, t)} + \mathrm{e}^{-\mathrm{i}(\Delta k\, x - \Delta\omega\, t)}] \\ &= 2\,\mathrm{Re}\,\widetilde{\psi}\mathrm{e}^{\mathrm{i}(kx-\omega t)}\cos(\Delta k\, x - \Delta\omega\, t).\end{aligned} \tag{6.21}$$

The resulting disturbance, illustrated in Fig. 6.2 has two aspects: a rapidly varying component, with wavenumber k and frequency ω, and a more slowly varying envelope, with wavenumber Δk and frequency $\Delta\omega$. The envelope modulates the fast oscillation, and moves with velocity $\Delta\omega/\Delta k$; in the limit $\Delta k \to 0$ and $\Delta\omega \to 0$ this is the *group velocity*, $c_g = \partial\omega/\partial k$. Group velocity is equal to the phase speed, ω/k, only when the frequency is a linear function of wavenumber. The energy in the disturbance must move at the group velocity — note that the node of the envelope moves at the speed of the envelope and no energy can cross the node. These concepts generalize to more than one dimension, and if the wavenumber is the three-dimensional vector $\boldsymbol{k} = (k, l, m)$ then the three-dimensional envelope propagates at the group velocity given by

$$\boxed{\boldsymbol{c}_g = \frac{\partial \omega}{\partial \mathbf{k}} \equiv \left(\frac{\partial \omega}{\partial k}, \frac{\partial \omega}{\partial l}, \frac{\partial \omega}{\partial m}\right)}. \tag{6.22}$$

The group velocity is also written as $\boldsymbol{c}_g = \nabla_{\boldsymbol{k}} \omega$ or, in subscript notation, $c_{gi} = \partial \Omega / \partial k_i$, with the subscript i denoting the component of a vector.

6.2.2 ♦ Superposition of many waves

Now consider a generalization of the above arguments to the case in which many waves are excited. In a homogeneous medium, nearly all wave patterns can be represented as a superposition of an infinite number of plane waves; mathematically the problem is solved by evaluating a Fourier integral. For mathematical simplicity we'll continue to treat only the one-dimensional case but the three dimensional generalization is possible.

A superposition of plane waves, each satisfying some dispersion relation, can be represented by the Fourier integral

$$\psi(x, t) = \int_{-\infty}^{\infty} \widetilde{A}(k) \mathrm{e}^{\mathrm{i}(kx - \omega t)} \, \mathrm{d}k. \tag{6.23a}$$

The function $\widetilde{A}(k)$ is given by the initial conditions:

$$\widetilde{A}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \psi(x, 0) \mathrm{e}^{-\mathrm{i}kx} \, \mathrm{d}x. \tag{6.23b}$$

As an aside, note that if the waves are dispersionless and $\omega = ck$ where c is a constant, then

$$\psi(x, t) = \int_{-\infty}^{+\infty} \widetilde{A}(k) \mathrm{e}^{\mathrm{i}k(x - ct)} \, \mathrm{d}k = \psi(x - ct, 0), \tag{6.24}$$

by comparison with (6.23a) at $t = 0$. That is, the initial condition simply translates at a speed c, with no change in structure.

Although the above procedure is quite general it doesn't get us very far: it doesn't provide us with any physical intuition, and the integrals themselves may be hard to evaluate. A physically more revealing case is to consider the case for which the disturbance is a *wave packet* — essentially a nearly plane wave or superposition of waves but confined to a finite region of space. We will consider a case with the initial condition

$$\psi(x, 0) = a(x, 0) \mathrm{e}^{\mathrm{i}k_0 x} \tag{6.25}$$

where $a(x, t)$, rather like the envelope in Fig. 6.3, modulates the amplitude of the wave on a scale much longer than that of the wavelength $2\pi / k_0$, and more slowly than the wave period. That is,

$$\frac{1}{a} \frac{\partial a}{\partial x} \ll k_0, \qquad \frac{1}{a} \frac{\partial a}{\partial t} \ll k_0 c, \tag{6.26a,b}$$

and the disturbance is essentially a slowly modulated plane wave. We suppose that $a(x, 0)$ is peaked around some value x_0 and is very small if $|x - x_0| \gg k_0^{-1}$; that is, $a(x, 0)$ is small

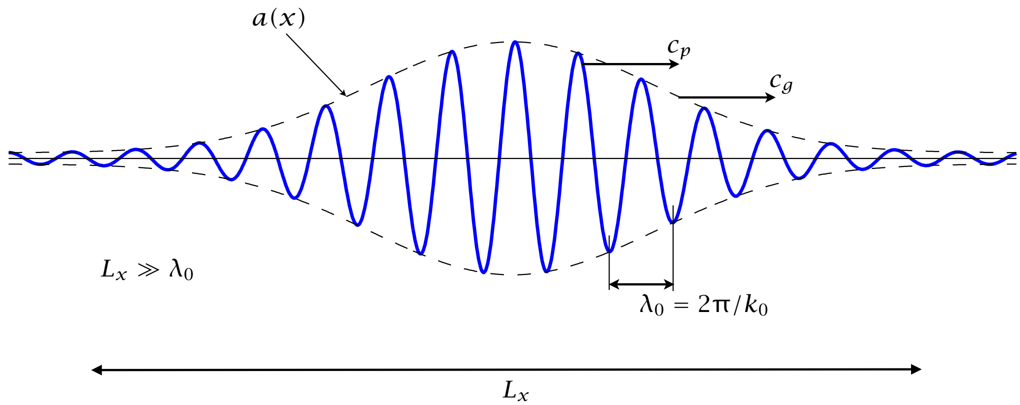

Fig. 6.3 A generic wave packet. The envelope, $a(x)$, has a scale L_x that is much larger than the wavelength, λ_0, of the wave embedded within in. The envelope moves at the group velocity, c_g, and the phase of the waves at the phase speed, c_p.

if we are sufficiently many wavelengths of the plane wave away from the peak, as is the case in Fig. 6.3. We would like to know how such a packet evolves.

We can express the envelope as a Fourier integral by first noting that that we can write the initial conditions as a Fourier integral,

$$\psi(x,0) = \int_{-\infty}^{\infty} \widetilde{A}(k)\mathrm{e}^{\mathrm{i}kx}\,\mathrm{d}k \qquad \text{where} \qquad \widetilde{A}(k) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \psi(x,0)\mathrm{e}^{-ikx}\,\mathrm{d}x, \tag{6.27a,b}$$

so that, using (6.25),

$$\widetilde{A}(k) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} a(x,0)\mathrm{e}^{\mathrm{i}(k_0-k)x}\,\mathrm{d}x \qquad \text{and} \qquad a(x,0) = \int_{-\infty}^{\infty} \widetilde{A}(k)\mathrm{e}^{\mathrm{i}(k-k_0)x}\,\mathrm{d}k. \tag{6.28a,b}$$

We still haven't made much progress beyond (6.23). To do so, we note first that $a(x)$ is confined in space, so that to a good approximation the limits of the integral in (6.28a) can be made finite, $\pm L$ say, provided $L \gg k_0^{-1}$. We then note that when $(k_0 - k)x$ is large the integrand in (6.28a) oscillates rapidly; successive intervals in x therefore cancel each other and make a small net contribution to the integral. Thus, the integral is dominated by values of k near k_0, and $\widetilde{A}(k)$ is peaked near k_0. (Note that the finite spatial extent of $a(x,0)$ is crucial for this argument.)

We can now evaluate how the wave packet evolves. Beginning with (6.23a) we have

$$\psi(x,t) = \int_{-\infty}^{\infty} \widetilde{A}(k)\exp\{\mathrm{i}(kx - \omega(k)t)\}\,\mathrm{d}k \tag{6.29a}$$

$$\approx \int \widetilde{A}(k)\exp\left\{\mathrm{i}[k_0x - \omega(k_0)t] + \mathrm{i}(k-k_0)x - \mathrm{i}(k-k_0)\left.\frac{\partial\omega}{\partial k}\right|_{k=k_0} t\right\}\mathrm{d}k \tag{6.29b}$$

having expanded $\omega(k)$ in a Taylor series about k and kept only the first two terms, noting that the wavenumber band is effectively limited. We therefore have

$$\psi(x,t) = \exp\{\mathrm{i}[k_0x - \omega(k_0)t]\}\int \widetilde{A}(k)\exp\left\{\mathrm{i}(k-k_0)\left[x - \left.\frac{\partial\omega}{\partial k}\right|_{k=k_0} t\right]\right\}\mathrm{d}k \tag{6.30a}$$

$$= \exp\{\mathrm{i}[k_0 x - \omega(k_0)t]\}\, a\,(x - c_g t) \tag{6.30b}$$

where $c_g = \partial\omega/\partial k$ evaluated at $k = k_0$. That is to say, the envelope $a(x,t)$ moves at the group velocity, keeping its initial shape.

The group velocity has a meaning beyond that implied by the derivation above: there is no need to restrict attention to narrow band processes, and it turns out to be a quite general property of waves that energy (and certain other quadratic properties) propagate at the group velocity. This is to be expected, at least in the presence of coherent wave packets, because there is no energy outside of the wave envelope so the energy must propagate with the envelope. Let's now examine this more closely.

6.2.3 ♦ The method of stationary phase

We will now relax the assumption that wavenumbers are confined to a narrow band but (since there is no free lunch) we confine ourselves to seeking solutions at large t; that is, we will be seeking a description of waves far from their source. Consider a disturbance of the general form

$$\psi(x,t) = \int_{-\infty}^{\infty} \widetilde{A}(k)\mathrm{e}^{\mathrm{i}[kx-\omega(k)t]}\,\mathrm{d}k = \int_{-\infty}^{\infty} \widetilde{A}(k)\mathrm{e}^{\mathrm{i}\Theta(k;x,t)t}\,\mathrm{d}k \tag{6.31}$$

where $\Theta(k;x,t) \equiv kx/t - \omega(k)$. (Here we regard Θ as a function of k with parameters x and t; we will sometimes just write $\Theta(k)$ with $\Theta'(k) = \partial\Theta/\partial k$.) Now, a standard result in mathematics (known as the 'Riemann–Lebesgue lemma') states that

$$I = \lim_{t\to\infty} \int_{-\infty}^{\infty} f(k)\mathrm{e}^{\mathrm{i}kt}\,\mathrm{d}k = 0 \tag{6.32}$$

provided that $f(k)$ is integrable and $\int_{-\infty}^{\infty} f(k)\,\mathrm{d}k$ is finite. Intuitively, as t increases the oscillations in the integral increase and become much faster than any variation in $f(k)$; successive oscillations thus cancel and the integral becomes very small (Fig. 6.4).

Looking at (6.31), with $\widetilde{A}$ playing the role of $f(k)$, the integral will be small if Θ is everywhere varying with k. However, if there is a region where Θ does not vary with k — that is, if there is a region where the phase is stationary and $\partial\Theta/\partial k = 0$ — then there will be a contribution to the integral from that region. Thus, for large t, an observer will predominantly see waves for which $\Theta'(k) = 0$ and so, using the definition of Θ, for which

$$\frac{x}{t} = \frac{\partial\omega}{\partial k}. \tag{6.33}$$

In other words, at some space-time location (x,t) the waves that dominate are those whose group velocity $\partial\omega/\partial k$ is x/t. In the example plotted in Fig. 6.4, $\omega = -\beta/k$ so that the wavenumber that dominates, k_0 say, is given by solving $\beta/k_0^2 = x/t$, which for $x/t = 1$ and $\beta = 400$ gives $k_0 = 20$.

We may actually approximately calculate the contribution to $\psi(x,t)$ from waves moving with the group velocity. Let us expand $\Theta(k)$ around the point, k_0, where $\Theta'(k_0) = 0$. We obtain

$$\psi(x,t) = \int_{-\infty}^{\infty} \widetilde{A}(k)\exp\left\{\mathrm{i}t\left[\Theta(k_0) + (k-k_0)\Theta'(k_0) + \tfrac{1}{2}(k-k_0)^2\Theta''(k_0)\ldots\right]\right\}\mathrm{d}k \tag{6.34}$$

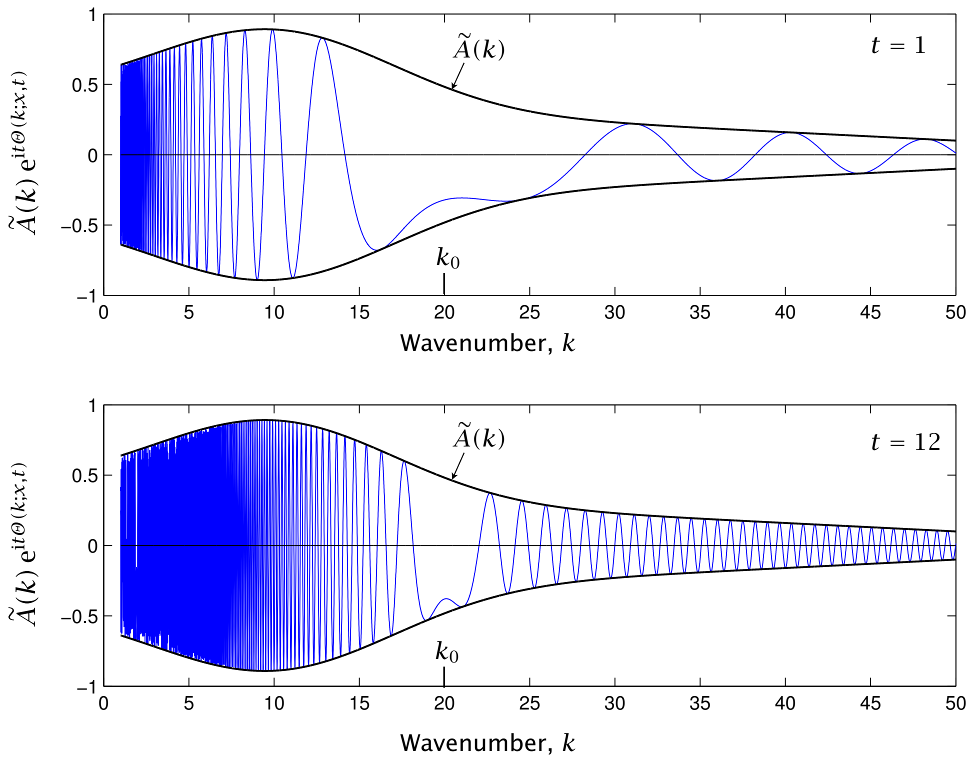

Fig. 6.4 The integrand of (6.31), namely the function that when integrated over wavenumber gives the wave amplitude at a particular x and t. The example shown is for a Rossby wave with $\omega = -\beta/k$, with $\beta = 400$ and $x/t = 1$, and hence $k_0 = 20$, for two times $t = 1$ and $t = 12$. (The amplitude of the envelope, $\tilde{A}(k)$, diminishes at high wavenumber but is otherwise arbitrary.) At the later time the oscillations are much more rapid in k, so that the contribution is more peaked from wavenumbers near to k_0.

The higher order terms are small because $k - k_0$ is presumed small (for if it is large the integral vanishes), and the term involving $\Theta'(k_0)$ is zero. The integral becomes

$$\psi(x,t) = \tilde{A}(k_0)e^{i\Theta(k_0)} \int_{-\infty}^{\infty} \exp\left\{it\tfrac{1}{2}(k-k_0)^2\Theta''(k_0)\right\} dk. \tag{6.35}$$

We therefore have to evaluate a Gaussian, and because $\int_{-\infty}^{\infty} e^{-cx^2}\,dx = \sqrt{\pi/c}$ we obtain

$$\psi(x,t) \approx \tilde{A}(k_0)e^{i\Theta(k_0)}\left[-2\pi/(it\theta''(k_0))\right]^{1/2} = \tilde{A}(k_0)e^{i(k_0x-\omega(k_0)t)}\left[2i\pi/(t\theta''(k_0))\right]^{1/2}. \tag{6.36}$$

The solution is therefore a plane wave, with wavenumber k_0 and frequency $\omega(k_0)$, slowly modulated by an envelope determined by the form of $\Theta(k_0; x, t)$, where k_0 is the wavenumber such that $x/t = c_g = \partial\omega/\partial k|_{k=k_0}$

6.3 RAY THEORY

Most waves propagate in a media that is inhomogeneous. In the Earth's atmosphere and ocean the stratification varies with altitude and the Coriolis parameter varies with latitude. In these cases it can be hard to obtain the solution of a wave problem by Fourier methods, even approximately. Nonetheless, the ideas of signals propagating at the group velocity is a very robust one, and it turns out that we can often obtain much of the information we want — and in particular the trajectory of a wave — using an approximate recipe known as *ray theory,* using the word theory a little generously.[4]

In an inhomogeneous medium let us suppose that the solution to a particular wave problem is of the form

$$\psi(\boldsymbol{x}, t) = a(\boldsymbol{x}, t)\mathrm{e}^{\mathrm{i}\theta(\boldsymbol{x},t)}, \tag{6.37}$$

where a is the wave amplitude and θ the phase, and a varies slowly in a sense we will make more precise shortly. The local wavenumber and frequency are defined by,

$$k_i \equiv \frac{\partial \theta}{\partial x_i}, \qquad \omega \equiv -\frac{\partial \theta}{\partial t}. \tag{6.38}$$

where the first expression is equivalent to $\boldsymbol{k} \equiv \nabla\theta$ and so $\nabla \times \boldsymbol{k} = 0$. We suppose that the amplitude a varies slowly over a wavelength and a period; that is $|\Delta a|/|a|$ is small over the length $1/k$ and the period $1/\omega$ or

$$\frac{|\partial a/\partial x|}{a} \ll |k|, \qquad \frac{|\partial a/\partial t|}{a} \ll \omega, \tag{6.39}$$

and similarly in the other directions. We will assume that the wavenumber and frequency as defined by (6.38) are the same as those that would arise if the medium were homogeneous and a were a constant. Thus, we may obtain a local dispersion relation from the governing equation by keeping the spatially (and possibly temporally) varying parameters fixed and obtain

$$\omega = \Omega(k_i; x_i, t), \tag{6.40}$$

and then allow x_i and t to vary, albeit slowly.

Let us now consider how the wavevector and frequency might change with position and time. It follows from their definitions above that the wavenumber and frequency are related by

$$\frac{\partial k_i}{\partial t} + \frac{\partial \omega}{\partial x_i} = 0, \tag{6.41}$$

where we use a subscript notation for vectors and repeated indices are summed. Using (6.41) and (6.40) gives

$$\frac{\partial k_i}{\partial t} + \frac{\partial \Omega}{\partial x_i} + \frac{\partial \Omega}{\partial k_j}\frac{\partial k_j}{\partial x_i} = 0 \qquad \text{or} \qquad \frac{\partial k_i}{\partial t} + \frac{\partial \Omega}{\partial x_i} + \frac{\partial \Omega}{\partial k_j}\frac{\partial k_i}{\partial x_j} = 0, \tag{6.42a,b}$$

using $\partial k_j/\partial x_i = \partial k_i/\partial x_j$ as wave vector has no curl. Equation (6.42b) may be written as

$$\frac{\partial \boldsymbol{k}}{\partial t} + \boldsymbol{c}_g \cdot \nabla \boldsymbol{k} = -\nabla\Omega \tag{6.43}$$

where

$$\boldsymbol{c}_g = \frac{\partial \Omega}{\partial \boldsymbol{k}} = \left(\frac{\partial \Omega}{\partial k}, \frac{\partial \Omega}{\partial l}, \frac{\partial \Omega}{\partial m}\right) \tag{6.44}$$

is, once more, the group velocity. The left-hand side of (6.43) is similar to an advective derivative, but with the velocity being a group velocity not a fluid velocity. Evidently, if the dispersion relation for frequency is not an explicit function of space *the wavevector is propagated at the group velocity*.

The frequency is, in general, a function of space, wavenumber and time, and from the dispersion relation, (6.40), its variation is governed by

$$\frac{\partial \omega}{\partial t} = \frac{\partial \Omega}{\partial t} + \frac{\partial \Omega}{\partial k_i}\frac{\partial k_i}{\partial t} = \frac{\partial \Omega}{\partial t} - \frac{\partial \Omega}{\partial k_i}\frac{\partial \omega}{\partial x_i} \tag{6.45}$$

using (6.41). Thus, using the definition of group velocity, we may write (6.45) as

$$\frac{\partial \omega}{\partial t} + \boldsymbol{c}_g \cdot \nabla \omega = \frac{\partial \Omega}{\partial t}. \tag{6.46}$$

As with (6.43) the left-hand side is like an advective derivative, but with the velocity being a group velocity. Thus, if the dispersion relation is not a function of time, the frequency also propagates at the group velocity.

Motivated by (6.43) and (6.46) we define a *ray* as the trajectory traced by the group velocity, and we see that if the function Ω is not an explicit function of space or time, then *both the wavevector and the frequency are constant along a ray.*

6.3.1 Ray theory in practice

What use is ray theory? The idea is that we may use (6.43) and (6.46) to track a group of waves from one location to another without solving the full wave equations of motion. Indeed, it turns out that we can sometimes solve problems using ordinary differential equations (ODEs) rather than partial differential equations (PDEs).

Suppose that the initial conditions consist of a group of waves at a position x_0, for which the amplitude and wavenumber vary only slowly with position. We also suppose that we know the dispersion relation for the waves at hand; that is, we know the functional form of $\Omega(k; x, t)$. Now, the total derivate following the group velocity is given by

$$\frac{\mathrm{d}}{\mathrm{d}t} = \frac{\partial}{\partial t} + \boldsymbol{c}_g \cdot \nabla, \tag{6.47}$$

so that (6.43) and (6.46) may be written as

$$\frac{\mathrm{d}\boldsymbol{k}}{\mathrm{d}t} = -\nabla \Omega, \tag{6.48a}$$

$$\frac{\mathrm{d}\omega}{\mathrm{d}t} = -\frac{\partial \Omega}{\partial t}. \tag{6.48b}$$

These are ordinary differential equations for wavevector and frequency, solvable provided we know the right-hand sides; that is, provided we know the space and time location at which the dispersion relation [i.e., $\Omega(k; x, t)$] is to be evaluated. But the location *is* known because it is moving with the group velocity and so

$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = \boldsymbol{c}_g. \tag{6.48c}$$

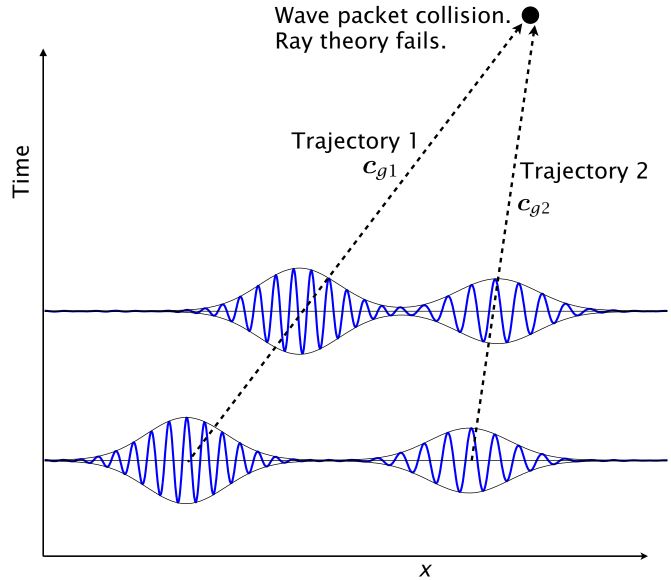

Fig. 6.5 Schema of the trajectory of two wavepackets, each with a different wavelength and moving with a different group velocity, as might be calculated using ray theory. If the wave packets collide ray theory must fail. Ray theory gives only the trajectory of the wave packet, not the detailed structure of the waves within a packet.

where $\boldsymbol{c}_g = \partial\Omega/\partial\boldsymbol{k}|_{\boldsymbol{x},t}$ (i.e., $c_{gi} = \partial\Omega/\partial k_i|_{\boldsymbol{x},t}$). The set (6.48) is a triplet of ordinary differential equations for the wavevector, frequency and position of a wave group. The equations may be solved, albeit sometimes numerically, to give the trajectory of a wave packet or collection of wave packets as schematically illustrated in Fig. 6.5. Of course, if the medium or the wavepacket amplitude is not slowly varying ray theory will fail, and this will perforce happen if two wave packets collide.

The evolution of the amplitude of the wave packet is not given by ray theory. However, the evolution of a quantity related to the amplitude of a wave packet — specifically, the wave activity — may be calculated if the group velocity is known. It may be shown that the wave activity, A, satisfies $\partial A/\partial t + \nabla\cdot(\boldsymbol{c}_g A) = 0$; that is, the flux of wave activity is along a ray, but we leave further discussion to chapter 10. Another way to calculate the evolution of a wave and its amplitude in a varying medium is to use 'WKB theory' — see the appendix to chapter **??**, with examples in section **??** and chapters 16 and 17. Before all that we turn our attention to a specific form of wave — Rossby waves — but the reader whose interest is more in the general properties of waves may skip forward to section 6.8.

6.4 ROSSBY WAVES

We now shift gears and consider in some detail a particular wave, namely the Rossby wave in a quasi-geostrophic system. Rossby waves are perhaps the most important large-scale wave in the atmosphere and ocean (although gravity waves, discussed in the next chapter, are arguably as important in some contexts).[5]

6.4.1 The linear equation of motion

For most of the rest of this chapter we will be concerned with the quasi-geostrophic equations of motion for which (as discussed in chapter 5) the inviscid, adiabatic potential vorticity equation is

$$\frac{\partial q}{\partial t} + \boldsymbol{u} \cdot \nabla q = 0, \tag{6.49}$$

where $q(x, y, z, t)$ is the potential vorticity and $\boldsymbol{u}(x, y, z, t)$ is the horizontal velocity. The velocity is related to a streamfunction by $u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$ and the potential vorticity is some function of the streamfunction, which might differ from system to system. Two examples, one applying to a continuously stratified system and the second to a single layer system, are

$$q = f + \zeta + \frac{\partial}{\partial z}\left(S(z)\frac{\partial\psi}{\partial z}\right), \qquad q = \zeta + f - k_d^2\psi. \tag{6.50a,b}$$

where $S(z) = f_0^2/N^2$, $\zeta = \nabla^2\psi$ is the relative vorticity and $k_d = 1/L_d$ is the inverse radius of deformation for a shallow water system. (Note that definitions of k_d and L_d can vary, typically by factors of 2, π, etc.) Boundary conditions may be needed to form a complete system.

We now *linearize* (6.49); that is, we suppose that the flow consists of a time-independent component (the 'basic state') plus a perturbation, with the perturbation being small compared with the mean flow. The basic state must satisfy the time-independent equation of motion, and it is common and useful to linearize about a zonal flow, $\overline{u}(y, z)$. The basic state is then purely a function of y and so we write

$$q = \overline{q}(y, z) + q'(x, y, t), \qquad \psi = \overline{\psi}(y, z) + \psi'(x, y, z, t) \tag{6.51}$$

with a similar notation for the other variables. Note that $\overline{u} = -\partial\overline{\psi}/\partial y$ and $\overline{v} = 0$. Substituting into (6.49) gives, without approximation,

$$\frac{\partial q'}{\partial t} + \overline{\boldsymbol{u}} \cdot \nabla\overline{q} + \overline{\boldsymbol{u}} \cdot \nabla q' + \boldsymbol{u}' \cdot \nabla\overline{q} + \boldsymbol{u}' \cdot \nabla q' = 0. \tag{6.52}$$

The primed quantities are presumptively small so we neglect terms involving their products. Further, we are assuming that we are linearizing about a state that is a solution of the equations of motion, so that $\overline{\boldsymbol{u}} \cdot \nabla\overline{q} = 0$. Finally, since $\overline{v} = 0$ and $\partial\overline{q}/\partial x = 0$ we obtain

$$\boxed{\frac{\partial q'}{\partial t} + \overline{u}\frac{\partial q'}{\partial x} + v'\frac{\partial\overline{q}}{\partial y} = 0}\,. \tag{6.53}$$

This equation or one very similar appears very commonly in studies of Rossby waves. To proceed, let us consider the simple example of waves in a single layer.

6.4.2 Waves in a single layer

Consider a system obeying (6.49) and (6.50b). The equation could be written in spherical coordinates with $f = 2\Omega\sin\vartheta$, but the dynamics are more easily illustrated on Cartesian β-plane for which $f = f_0 + \beta y$, and since f_0 is a constant it does not appear in our subsequent derivations.

Infinite deformation radius

If the scale of motion is much less than the deformation scale then we make the approximation that $k_d = 0$ and the equation of motion may be written as

$$\frac{\partial \zeta}{\partial t} + \boldsymbol{u} \cdot \nabla \zeta + \beta v = 0 \tag{6.54}$$

We linearize about a constant zonal flow, $\overline{u}$, by writing

$$\psi = \overline{\psi}(y) + \psi'(x, y, t), \tag{6.55}$$

where $\overline{\psi} = -\overline{u}y$. Substituting(6.55) into (6.54) and neglecting the nonlinear terms involving products of ψ' to give

$$\frac{\partial}{\partial t}\nabla^2\psi' + \overline{u}\frac{\partial \nabla^2\psi'}{\partial x} + \beta\frac{\partial \psi'}{\partial x} = 0. \tag{6.56}$$

This equation is just a single-layer version of (6.53), with $\partial\overline{q}/\partial y = \beta$, $q' = \nabla^2\psi'$ and $v' = \partial\psi'/\partial x$.

The coefficients in (6.56) are not functions of y or z; this is not a requirement for wave motion to exist but it does enable solutions to be found more easily. Let us seek solutions in the form of a plane wave, namaely

$$\psi' = \mathrm{Re}\, \widetilde{\psi} \mathrm{e}^{\mathrm{i}(kx + ly - \omega t)}, \tag{6.57}$$

where $\widetilde{\psi}$ is a complex constant and Re indicates the real part of the function (and this will sometimes be omitted if no ambiguity is so-caused). Solutions of this form are valid in a domain with doubly-periodic boundary conditions; solutions in a channel can be obtained using a meridional variation of $\sin ly$, with no essential changes to the dynamics. The amplitude of the oscillation is given by $\widetilde{\psi}$ and the phase by $kx + ly - \omega t$, where k and l are the x- and y-wavenumbers and ω is the frequency of the oscillation.

Substituting (6.57) into (6.56) yields

$$[(-\omega + Uk)(-K^2) + \beta k]\widetilde{\psi} = 0, \tag{6.58}$$

where $K^2 = k^2 + l^2$. For non-trivial solutions this implies

$$\boxed{\omega = Uk - \frac{\beta k}{K^2}}. \tag{6.59}$$

This is the *dispersion relation* for barotropic Rossby waves, and evidently the velocity U Doppler shifts the frequency. The components of the phase speed and group velocity are given by, respectively,

$$c_p^x \equiv \frac{\omega}{k} = \overline{u} - \frac{\beta}{K^2}, \qquad c_p^y \equiv \frac{\omega}{l} = \overline{u}\frac{k}{l} - \frac{\beta k}{K^2 l}, \tag{6.60a,b}$$

and

$$c_g^x \equiv \frac{\partial \omega}{\partial k} = \overline{u} + \frac{\beta(k^2 - l^2)}{(k^2 + l^2)^2}, \qquad c_g^y \equiv \frac{\partial \omega}{\partial l} = \frac{2\beta k l}{(k^2 + l^2)^2}. \tag{6.61a,b}$$

The phase speed in the absence of a mean flow is *westwards,* with waves of longer wavelengths travelling more quickly, and the eastward current speed required to hold the waves of a particular wavenumber stationary (i.e., $c_p^x = 0$) is $U = \beta/K^2$. The background flow $\overline{u}$ evidently just provides a uniform shift to the phase speed, and could be transformed away by a change of coordinate.

Finite deformation radius

For a finite deformation radius the basic state $\Psi = -Uy$ is still a solution of the original equations of motion, but the potential vorticity corresponding to this state is $q = Uyk_d^2 + \beta y$ and its gradient is $\nabla q = (\beta + Uk_d^2)\mathbf{j}$. The linearized equation of motion is thus

$$\left(\frac{\partial}{\partial t} + \overline{u}\frac{\partial}{\partial x}\right)(\nabla^2\psi' - \psi' k_d^2) + (\beta + \overline{u}k_d^2)\frac{\partial \psi'}{\partial x} = 0. \tag{6.62}$$

Substituting $\psi' = \widetilde{\psi}e^{i(kx+ly-\omega t)}$ we obtain the dispersion relation,

$$\boxed{\omega = \frac{k(UK^2 - \beta)}{K^2 + k_d^2} = Uk - k\frac{\beta + Uk_d^2}{K^2 + k_d^2}}. \tag{6.63}$$

The corresponding components of phase speed and group velocity are

$$c_p^x = \overline{u} - \frac{\beta + \overline{u}k_d^2}{K^2 + k_d^2} = \frac{\overline{u}K^2 - \beta}{K^2 + k_d^2}, \qquad c_p^y = \overline{u}\frac{k}{l} - \frac{k}{l}\left(\frac{\overline{u}K^2 - \beta}{K^2 + k_d^2}\right) \tag{6.64a,b}$$

and

$$c_g^x = \overline{u} + \frac{(\beta + \overline{u}k_d^2)(k^2 - l^2 - k_d^2)}{(k^2 + l^2 + k_d^2)^2}, \qquad c_g^y = \frac{2kl(\beta + \overline{u}k_d^2)}{(k^2 + l^2 + k_d^2)^2}. \tag{6.65a,b}$$

The uniform velocity field now no longer provides just a simple Doppler shift of the frequency, nor a uniform addition to the phase speed. From (6.64a) the waves are stationary when $K^2 = \beta/\overline{u} \equiv K_s^2$; that is, the current speed required to hold waves of a particular wavenumber stationary is $\overline{u} = \beta/K^2$. However, this is *not* simply the magnitude of the phase speed of waves of that wavenumber in the absence of a current — this is given by

$$c_p^x = \frac{-\beta}{K_s^2 + k_d^2} = \frac{-\overline{u}}{1 + k_d^2/K_s^2}. \tag{6.66}$$

Why is there a difference? It is because the current does not just provide a uniform translation, but, if k_d is non-zero, it also modifies the basic potential vorticity gradient. The basic state height field η_0 is sloping; that is $\eta_0 = -(f_0/g)\overline{u}y$, and the ambient potential vorticity field increases with y and $q = (\beta + Uk_d^2)y$. Thus, the basic state defines a preferred frame of reference, and the problem is not Galilean invariant.[6] We also note that, from (6.64b), the group velocity is negative (westward) if the x-wavenumber is sufficiently small, compared to the y-wavenumber or the deformation wavenumber. That is, said a little loosely, *long waves move information westward and short waves move information eastward,* and this is a common property of Rossby waves. The x-component of the phase speed, on the other hand, is always westward relative to the mean flow.

6.4.3 The mechanism of Rossby waves

The fundamental mechanism underlying Rossby waves is easily understood. Consider a material line of stationary fluid parcels along a line of constant latitude, and suppose that some disturbance causes their displacement to the line marked $\eta(t = 0)$ in Fig. 6.6. In the displacement, the potential vorticity of the fluid parcels is conserved, and in the simplest case of

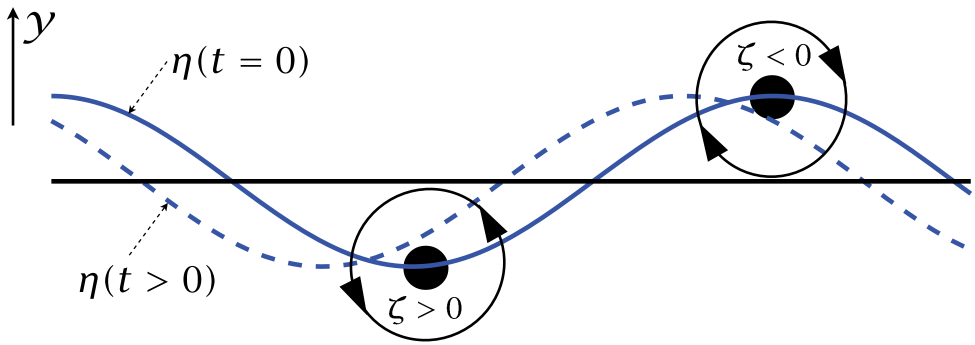

Fig. 6.6 The mechanism of a two-dimensional (x–y) Rossby wave. An initial disturbance displaces a material line at constant latitude (the straight horizontal line) to the solid line marked $\eta(t=0)$. Conservation of potential vorticity, $\beta y+\zeta$, leads to the production of relative vorticity, as shown for two parcels. The associated velocity field (arrows on the circles) then advects the fluid parcels, and the material line evolves into the dashed line. The phase of the wave has propagated westwards.

barotropic flow on the β-plane the potential vorticity is the absolute vorticity, $\beta y+\zeta$. Thus, in either hemisphere, a northward displacement leads to the production of negative relative vorticity and a southward displacement leads to the production of positive relative vorticity. The relative vorticity gives rise to a velocity field which, in turn, advects the parcels in material line in the manner shown, and the wave propagates westwards.

In more complicated situations, such as flow in two layers, considered below, or in a continuously stratified fluid, the mechanism is essentially the same. A displaced fluid parcel carries with it its potential vorticity and, in the presence of a potential vorticity gradient in the basic state, a potential vorticity anomaly is produced. The potential vorticity anomaly produces a velocity field (an example of potential vorticity inversion) which further displaces the fluid parcels, leading to the formation of a Rossby wave. The vital ingredient is a basic state potential vorticity gradient, such as that provided by the change of the Coriolis parameter with latitude.

6.4.4 Rossby waves in two layers

Now consider the dynamics of the two-layer model, linearized about a state of rest. The two, coupled, linear equations describing the motion in each layer are

$$\frac{\partial}{\partial t}\left[\nabla^2\psi_1'+F_1(\psi_2'-\psi_1')\right]+\beta\frac{\partial\psi_1'}{\partial x}=0, \tag{6.67a}$$

$$\frac{\partial}{\partial t}\left[\nabla^2\psi_2'+F_2(\psi_1'-\psi_2')\right]+\beta\frac{\partial\psi_2'}{\partial x}=0, \tag{6.67b}$$

where $F_1=f_0^2/g'H_1$ and $F_2=f_0^2/g'H_2$. By inspection (6.67) may be transformed into two uncoupled equations: the first is obtained by multiplying (6.67a) by F_2 and (6.67b) by F_1 and adding, and the second is the difference of (6.67a) and (6.67b). Then, defining

$$\overline{\psi}=\frac{F_1\psi_2'+F_2\psi_1'}{F_1+F_2},\qquad \tau=\frac{1}{2}(\psi_1'-\psi_2'), \tag{6.68a,b}$$

(think 'τ for temperature'), (6.67) become

$$\frac{\partial}{\partial t}\nabla^2\overline{\psi} + \beta\frac{\partial\overline{\psi}}{\partial x} = 0, \tag{6.69a}$$

$$\frac{\partial}{\partial t}\left[(\nabla^2 - k_d^2)\tau\right] + \beta\frac{\partial \tau}{\partial x} = 0, \tag{6.69b}$$

where now $k_d = (F_1 + F_2)^{1/2}$. The internal radius of deformation for this problem is the inverse of this, namely

$$L_d = k_d^{-1} = \frac{1}{f_0}\left(\frac{g' H_1 H_2}{H_1 + H_2}\right)^{1/2}. \tag{6.70}$$

The variables $\overline{\psi}$ and τ are the *normal modes* for the two-layer model, as they oscillate independently of each other. [For the continuous equations the analogous modes are the eigenfunctions of $\partial_z[(f_0^2/N^2)\partial_z\phi] = \lambda^2\phi$.] The equation for $\overline{\psi}$, *the barotropic mode,* is identical to that of the single-layer, rigid-lid model, namely (6.56) with $U = 0$, and its dispersion relation is just

$$\omega = -\frac{\beta k}{K^2}. \tag{6.71}$$

The barotropic mode corresponds to synchronous, depth-independent, motion in the two layers, with no undulations in the dividing interface.

The displacement of the interface is given by $2f_0\tau/g'$ and so proportional to the amplitude of τ, the *baroclinic mode.* The dispersion relation for the baroclinic mode is

$$\omega = -\frac{\beta k}{K^2 + k_d^2}. \tag{6.72}$$

The mass transport associated with this mode is identically zero, since from (6.68) we have

$$\psi_1 = \overline{\psi} + \frac{2F_1\tau}{F_1 + F_2}, \qquad \psi_2 = \overline{\psi} - \frac{2F_2\tau}{F_1 + F_2}, \tag{6.73a,b}$$

and this implies

$$H_1\psi_1 + H_2\psi_2 = (H_1 + H_2)\overline{\psi}. \tag{6.74}$$

The left-hand side is proportional to the total mass transport, which is evidently associated with the barotropic mode.

The dispersion relation and associated group and phase velocities are plotted in Fig. 6.7. The x-component of the phase speed, ω/k, is negative (westwards) for both baroclinic and barotropic Rossby waves. The group velocity of the barotropic waves is always positive (eastwards), but the group velocity of long baroclinic waves may be negative (westwards). For very short waves, $k^2 \gg k_d^2$, the baroclinic and barotropic velocities coincide and their phase and group velocities are equal and opposite. With a deformation radius of 50 km, typical for the mid-latitude ocean, then a non-dimensional frequency of unity in the figure corresponds to a dimensional frequency of $5 \times 10^{-7}\,\mathrm{s}^{-1}$ or a period of about 100 days. In an atmosphere with a deformation radius of 1000 km a non-dimensional frequency of unity corresponds to $1 \times 10^{-5}\,\mathrm{s}^{-1}$ or a period of about 7 days. Non-dimensional velocities of unity correspond to respective dimensional velocities of about $0.25\,\mathrm{m\,s^{-1}}$ (ocean) and $10\,\mathrm{m\,s^{-1}}$ (atmosphere).

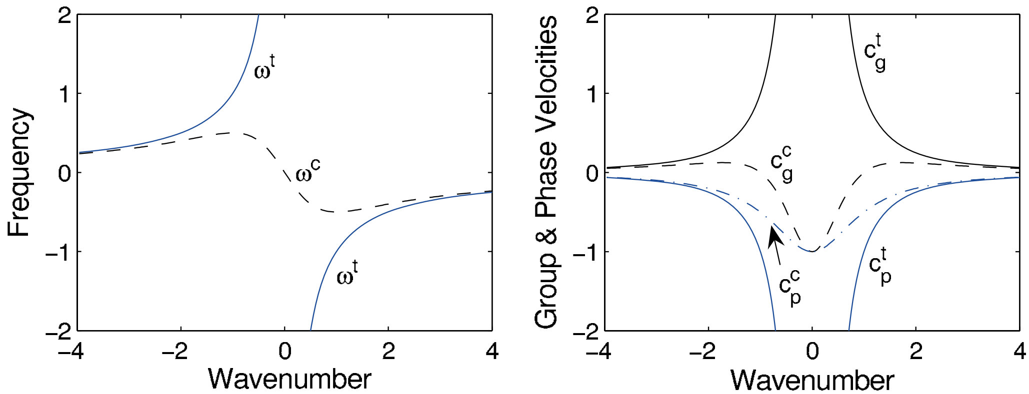

Fig. 6.7 Left: the dispersion relation for barotropic (ω^t, solid line) and baroclinic (ω^c, dashed line) Rossby waves in the two-layer model, calculated using (6.71) and (6.72) with $k^y = 0$, plotted for both positive and negative zonal wavenumbers and frequencies. The wavenumber is non-dimensionalized by k_d, and the frequency is non-dimensionalized by β/k_d. Right: the corresponding zonal group and phase velocities, $c_g = \partial\omega/\partial k^x$ and $c_p = \omega/k^x$, with superscript 't' or 'c' for the barotropic or baroclinic mode, respectively. The velocities are non-dimensionalized by β/k_d^2.

The deformation radius only affects the baroclinic mode. For scales much smaller than the deformation radius, $K^2 \gg k_d^2$, we see from (6.69b) that the baroclinic mode obeys the same equation as the barotropic mode so that

$$\frac{\partial}{\partial t}\nabla^2\tau + \beta\frac{\partial\tau}{\partial x} = 0. \tag{6.75}$$

Using this and (6.69a) implies that

$$\frac{\partial}{\partial t}\nabla^2\psi_i + \beta\frac{\partial\psi_i}{\partial x} = 0, \qquad i = 1, 2. \tag{6.76}$$

That is to say, the two layers themselves are uncoupled from each other. At the other extreme, for very long baroclinic waves the relative vorticity is unimportant.

6.5 ROSSBY WAVES IN STRATIFIED QUASI-GEOSTROPHIC FLOW

6.5.1 Setting up the problem

Let us now consider the dynamics of linear waves in stratified quasi-geostrophic flow on a β-plane, with a resting basic state. (In chapter 16 we explore the role of Rossby waves in a more realistic setting.) The interior flow is governed by the potential vorticity equation, (5.118), and linearizing this about a state of rest gives

$$\frac{\partial}{\partial t}\left[\nabla^2\psi' + \frac{1}{\tilde\rho(z)}\frac{\partial}{\partial z}\left(\tilde\rho(z)F(z)\frac{\partial\psi'}{\partial z}\right)\right] + \beta\frac{\partial\psi'}{\partial x} = 0, \tag{6.77}$$

where $\tilde\rho$ is the density profile of the basic state and $F(z) = f_0^2/N^2$. (F is the square of the inverse Prandtl ratio, N/f_0.) In the Boussinesq approximation $\tilde\rho = \rho_0$, i.e., a constant. The vertical boundary conditions are determined by the thermodynamic equation, (5.120). If the boundaries are flat, rigid, slippery surfaces then $w = 0$ at the boundaries and if there is no surface buoyancy gradient the linearized thermodynamic equation is

$$\frac{\partial}{\partial t}\left(\frac{\partial \psi'}{\partial z}\right) = 0. \tag{6.78}$$

We apply this at the ground and, with somewhat less justification, at the tropopause — the higher static stability of the stratosphere inhibits vertical motion. If the ground is not flat or if friction provides a vertical velocity by way of an Ekman layer, the boundary condition must be correspondingly modified, but we will stay with the simplest case here and apply (6.78) at $z = 0$ and $z = H$.

6.5.2 Wave motion

As in the single-layer case, we seek solutions of the form

$$\psi' = \mathrm{Re}\,\widetilde{\psi}(z)\mathrm{e}^{\mathrm{i}(kx+ly-\omega t)}, \tag{6.79}$$

where $\widetilde{\psi}(z)$ will determine the vertical structure of the waves. The case of a sphere is more complicated but introduces no truly new physical phenomena.

Substituting (6.79) into (6.77) gives

$$\omega\left[-K^2\widetilde{\psi}(z) + \frac{1}{\widetilde{\rho}}\frac{\partial}{\partial z}\left(\widetilde{\rho}F(z)\frac{\partial\widetilde{\psi}}{\partial z}\right)\right] - \beta k\widetilde{\psi}(z) = 0. \tag{6.80}$$

Now, if $\widetilde{\psi}$ satisfies

$$\frac{1}{\widetilde{\rho}}\frac{\partial}{\partial z}\left(\widetilde{\rho}F(z)\frac{\partial\widetilde{\psi}}{\partial z}\right) = -\Gamma\widetilde{\psi}, \tag{6.81}$$

where Γ is a constant, then the equation of motion becomes

$$-\,\omega\left[K^2 + \Gamma\right]\widetilde{\psi} - \beta k\widetilde{\psi} = 0, \tag{6.82}$$

and the dispersion relation follows, namely

$$\boxed{\omega = -\frac{\beta k}{K^2 + \Gamma}}. \tag{6.83}$$

Equation (6.81) constitutes an eigenvalue problem for the vertical structure; the boundary conditions, derived from (6.78), are $\partial\widetilde{\psi}/\partial z = 0$ at $z = 0$ and $z = H$. The resulting eigenvalues, Γ are proportional to the inverse of the squares of the deformation radii for the problem and the eigenfunctions are the vertical structure functions.

A simple example

Consider the case in which $F(z)$ and $\widetilde{\rho}$ are constant, and in which the domain is confined between two rigid surfaces at $z = 0$ and $z = H$. Then the eigenvalue problem for the vertical structure is

$$F\frac{\partial^2\widetilde{\psi}}{\partial z^2} = -\Gamma\widetilde{\psi} \tag{6.84a}$$

with boundary conditions of

$$\frac{\partial \widetilde{\psi}}{\partial z} = 0, \qquad \text{at } z = 0, H. \tag{6.84b}$$

There is a sequence of solutions to this, namely

$$\widetilde{\psi}_n(z) = \cos(n\pi z/H), \qquad n = 1, 2 \dots \tag{6.85}$$

with corresponding eigenvalues

$$\Gamma_n = n^2 \frac{F\pi^2}{H^2} = (n\pi)^2 \left(\frac{f_0}{NH}\right)^2, \qquad n = 1, 2 \dots. \tag{6.86}$$

Equation (6.86) may be used to define the deformation radii for this problem, namely

$$L_n \equiv \frac{1}{\sqrt{\Gamma_n}} = \frac{NH}{n\pi f_0}. \tag{6.87}$$

The first deformation radius is the same as the expression obtained by dimensional analysis, namely NH/f, except for a factor of π. (Definitions of the deformation radii both with and without the factor of π are common in the literature, and neither is obviously more correct. In the latter case, the first deformation radius in a problem with uniform stratification is given by NH/f, equal to $\pi/\sqrt{\Gamma_1}$.) In addition to these baroclinic modes, the case with $n = 0$, that is with $\widetilde{\psi} = 1$, is also a solution of (6.84) for any $F(z)$.

Using (6.83) and (6.86) the dispersion relation becomes

$$\boxed{\omega = -\frac{\beta k}{K^2 + (n\pi)^2 (f_0/NH)^2}, \qquad n = 0, 1, 2 \dots} \tag{6.88}$$

and, of course, the horizontal wavenumbers k and l are also quantized in a finite domain. The dynamics of the barotropic mode are independent of height and independent of the stratification of the basic state, and so these Rossby waves are *identical* with the Rossby waves in a homogeneous fluid contained between two flat rigid surfaces. The structure of the baroclinic modes, which in general depends on the structure of the stratification, becomes increasingly complex as the vertical wavenumber n increases. This increasing complexity naturally leads to a certain delicacy, making it rare that they can be unambiguously identified in nature. The eigenproblem for a realistic atmospheric profile is further complicated because of the lack of a rigid lid at the top of the atmosphere.[7]

6.6 ENERGY PROPAGATION AND REFLECTION OF ROSSBY WAVES

We now consider how energy is fluxed in Rossby waves. To keep matters reasonably simple from an algebraic point of view we will consider waves in a single layer and without a mean flow, but we will allow for a finite radius of deformation. To remind ourselves, the dynamics are governed by the evolution of potential vorticity and the linearized evolution equation is

$$\frac{\partial}{\partial t}\left(\nabla^2 - k_d^2\right)\psi + \beta \frac{\partial \psi}{\partial x} = 0. \tag{6.89}$$

Essentials of Rossby Waves

- Rossby waves owe their existence to a gradient of potential vorticity in the fluid. If a fluid parcel is displaced, it conserves its potential vorticity and so its relative vorticity will in general change. The relative vorticity creates a velocity field that displaces neighbouring parcels, whose relative vorticity changes and so on.

- A common source of a potential vorticity gradient is differential rotation, or the β-effect. In the presence of non-zero β the ambient potential vorticity increases northward and the phase of the Rossby waves propagates westward. In general, Rossby waves propagate pseudo-westwards, meaning to the left of the direction of the potential vorticity gradient.

- A common equation of motion for Rossby waves is

$$\frac{\partial q'}{\partial t} + \overline{u}\frac{\partial q'}{\partial x} + v'\frac{\partial \overline{q}}{\partial y} = 0, \qquad \text{(RW.1)}$$

with an overbar denoting the basic state and a prime a perturbation. In the case of a single layer of fluid with no mean flow this equation becomes

$$\frac{\partial}{\partial t}(\nabla^2 + k_d^2)\psi' + \beta\frac{\partial \psi'}{\partial x} = 0 \qquad \text{(RW.2)}$$

with dispersion relation

$$\omega = \frac{-\beta k}{k^2 + l^2 + k_d^2}. \qquad \text{(RW.3)}$$

- The phase speed in the zonal direction ($c_p^x = \omega/k$) is always negative, or westward, and is larger for large waves. For (RW.2) components of the group velocity are given by

$$c_g^x = \frac{\beta(k^2 - l^2 - k_d^2)}{(k^2 + l^2 + k_d^2)^2}, \qquad c_g^y = \frac{2\beta k l}{(k^2 + l^2 + k_d^2)^2}. \qquad \text{(RW.4)}$$

The group velocity is westward if the zonal wavenumber is sufficiently small, and eastward if the zonal wavenumber is sufficiently large.

- Rossby waves exist in stratified fluids, and have a similar dispersion relation to (RW.3) with an appropriate vertical wavenumber appearing in place of the inverse deformation radius, k_d.

- The reflection of such Rossby waves at a wall is specular, meaning that the group velocity of the reflected wave makes the same angle with the wall as the group velocity of the incident wave. The energy flux of the reflected wave is equal and opposite to that of the incoming wave in the direction normal to the wall.

The dispersion relation follows in the usual way and is

$$\omega = \frac{-k\beta}{K^2 + k_d^2}. \tag{6.90}$$

which is a simplification of (6.63), and the group velocities are

$$c_g^x = \frac{\beta(k^2 - l^2 - k_d^2)}{(K^2 + k_d^2)^2}, \qquad c_g^y = \frac{2\beta k l}{(K^2 + k_d^2)^2}, \tag{6.91a,b}$$

which are simplifications of (6.65), and as usual $K^2 = k^2 + l^2$.

To obtain an energy equation multiply (6.89) by $-\psi$ to obtain, after a couple of lines of algebra,

$$\frac{1}{2}\frac{\partial}{\partial t}\left((\nabla\psi)^2 + k_d^2\psi^2\right) - \nabla\cdot\left(\psi\nabla\frac{\partial\psi}{\partial t} + \mathbf{i}\beta\psi^2\right) = 0. \tag{6.92}$$

where $\mathbf{i}$ is the unit vector in the x direction. The first group of terms are the energy itself, or more strictly the energy density. (An energy density is an energy per unit mass or per unit volume, depending on the context.) The term $(\nabla\psi)^2/2 = (u^2 + v^2)/2$ is the kinetic energy and $k_d^2\psi^2/2$ is the potential energy, proportional to the displacement of the free surface, squared. The second term is the energy flux, so that we may write

$$\frac{\partial E}{\partial t} + \nabla\cdot\boldsymbol{F} = 0. \tag{6.93}$$

where $E = (\nabla\psi)^2/2 + k_d^2\psi^2$ and $\boldsymbol{F} = -\left(\psi\nabla\partial\psi/\partial t + \mathbf{i}\beta\psi^2\right)$. We haven't yet used the fact that the disturbance has a dispersion relation, and if we do so we may expect, following the derivations of section 6.2, that the energy moves at the group velocity. Let us now demonstrate this explicitly.

We assume solution of the form

$$\psi = A(x)\cos(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t) = A(x)\cos(kx + ly - \omega t) \tag{6.94}$$

where $A(x)$ is assumed to vary slowly compared to the nearly plane wave. (Note that $\boldsymbol{k}$ is the wave vector, to be distinguished from $\mathbf{k}$, the unit vector in the z-direction.) The kinetic energy in a wave is given by

$$KE = \frac{A^2}{2}\left(\psi_x^2 + \psi_y^2\right) \tag{6.95}$$

so that, averaged over a wave period,

$$\overline{KE} = \frac{A^2}{2}(k^2 + l^2)\frac{\omega}{2\pi}\int_0^{2\pi/\omega}\sin^2(\mathbf{k}\cdot\boldsymbol{x} - \omega t)\,\mathrm{d}t. \tag{6.96}$$

The time-averaging produces a factor of one half, and applying a similar procedure1 to the potential energy we obtain

$$\overline{KE} = \frac{A^2}{4}(k^2 + l^2), \qquad \overline{PE} = \frac{A^2}{4}k_d^2, \tag{6.97}$$

so that the average total energy is

$$\overline{E} = \frac{A^2}{4}(K^2 + k_d^2), \tag{6.98}$$

where $K^2 = k^2 + l^2$.

The flux, $\boldsymbol{F}$, is given by

$$\boldsymbol{F} = -\left(\psi\nabla\frac{\partial\psi}{\partial t} + \mathbf{i}\beta\psi^2\right) = -A^2\cos^2(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)\left(\boldsymbol{k}\omega - \mathbf{i}\frac{\beta}{2}\right), \tag{6.99}$$

so that evidently the energy flux has a component in the direction of the wavevector, $\boldsymbol{k}$, and a component in the x-direction. Averaging over a wave period straightforwardly gives us additional factors of one half:

$$\overline{\boldsymbol{F}} = -\frac{A^2}{2}\left(\boldsymbol{k}\omega + \mathbf{i}\frac{\beta}{2}\right). \tag{6.100}$$

We now use the dispersion relation $\omega = -\beta k/(K^2 + k_d^2)$ to eliminate the frequency, giving

$$\overline{\boldsymbol{F}} = \frac{A^2\beta}{2}\left(\boldsymbol{k}\frac{k}{K^2 + k_d^2} - \mathbf{i}\frac{1}{2}\right), \tag{6.101}$$

and writing this in component form we obtain

$$\overline{\boldsymbol{F}} = \mathbf{i}\frac{A^2\beta}{4}\left(\frac{k^2 - l^2 - k_d^2}{K^2 + k_d^2}\right) + \mathbf{j}\frac{2kl}{K^2 + k_d^2}. \tag{6.102}$$

Comparison of (6.102) with (6.91) and (6.98) reveals that

$$\overline{\boldsymbol{F}} = \boldsymbol{c}_g\overline{E} \tag{6.103}$$

so that the energy propagation equation, (6.93), when averaged over a wave, becomes

$$\boxed{\frac{\partial\overline{E}}{\partial t} + \nabla\cdot\boldsymbol{c}_g\overline{E} = 0}. \tag{6.104}$$

It is interesting that the variation of A plays no role in the above manipulations, so that the derivation appears to go through if the amplitude $A(\boldsymbol{x}, t)$ is in fact a constant and the wave is a single plane wave. This seems hard to reconcile with our previous discussion, in which we noted that the group velocity was the velocity of a wave *packet* involving a superposition of plane waves. Indeed, the derivative of the frequency with respect to wavenumber means little if there is only one wavenumber. In fact there is nothing wrong with the above derivation if A is a constant and only a single plane wave is present. The resolution of the paradox arises by noting that a plane wave fills all of space and time; in this case there is no convergence of the energy flux and the energy propagation equation is trivially true.

6.6.1 ♦ Rossby wave reflection

We now consider how Rossby waves might be reflected from a solid boundary. The topic has an obvious oceanographic relevance, for the reflection of Rossby waves turns out to one way of interpreting why intense oceanic boundary currents form on the western sides of ocean basins, not the east. There is also an atmospheric relevance, for meridionally propagating Rossby waves may effectively be reflected as they approach a 'turning latitude' where the meridional wavenumber goes to zero, as considered in chapter 16. As a preliminary, let us give a useful graphic interpretation of Rossby wave propagation.[8]

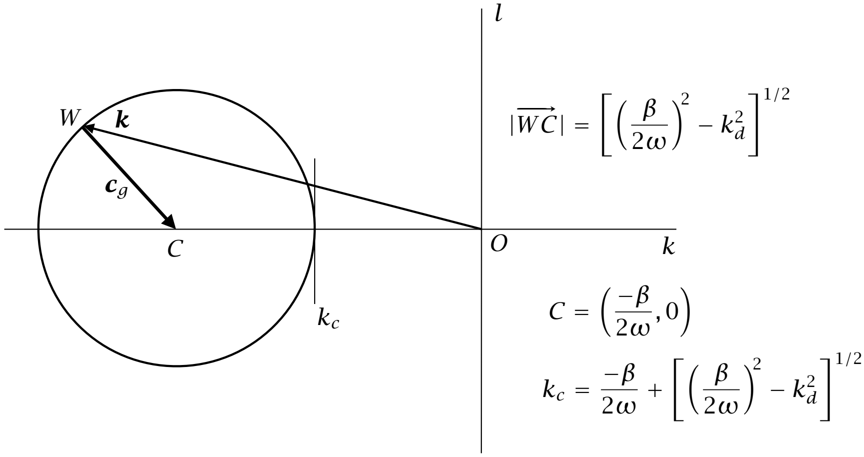

Fig. 6.8 The energy propagation diagram for Rossby waves. The wavevectors of a given frequency all lie in a circle of radius $[(\beta/2\omega)^2 - k_d^2]^{1/2}$, centered at the point C. The closest distance of the circle to the origin is k_c, and if the deformation radius is infinite k_c the circle touches the origin. For a given wavenumber $\boldsymbol{k}$, the group velocity is along the line directed from W to C.

The energy propagation diagram

The dispersion relation for Rossby waves, $\omega = -\beta k/(k^2 + l^2 + k_d^2)$, may be rewritten as

$$(k + \beta/2\omega)^2 + l^2 = (\beta/2\omega)^2 - k_d^2. \tag{6.105}$$

This equation is the parametric representation of a circle, meaning that the wavevector (k, l) must lie on a circle centered at the point $(-\beta/2\omega, 0)$ and with radius $[(\beta/2\omega)^2 - k_d^2]^{1/2}$, as illustrated in Fig. 6.8. If the deformation radius is zero the circle touches the origin, and if it is nonzero the distance of the closest point to the circle, k_c say, is given by $k_c = -\beta/2\omega + [(\beta/2\omega)^2 - k_d^2]^{1/2}$. For low frequencies, specifically if $\omega \ll \beta/2k$, then $k_c \approx -\omega k_d^2/\beta$. The radius of the circle is a positive real number only when $\omega < \beta/2k_d$. This is the maximum frequency possible, and it occurs when $l = 0$ and $k = k_d$ and when $c_g^x = c_g^y = 0$.

It turns out that the group velocity, and hence the energy flux, can be visualized graphically from Fig. 6.8. By direct manipulation of the expressions for group velocity and frequency we find that

$$c_g^x = \frac{2\omega}{\left(K^2 + k_d^2\right)^2}\left(k + \frac{\beta}{2\omega}\right), \tag{6.106a}$$

$$c_g^y = \frac{2\omega}{\left(K^2 + k_d^2\right)^2}\, l. \tag{6.106b}$$

(To check this, it is easiest to begin with the right-hand sides and use the dispersion relation for ω.) Now, since the center of the circle of wavevectors is at the position $(-\beta/2\omega, 0)$, and referring to Fig. 6.8, we have that

$$\boldsymbol{c}_g = \frac{2\omega}{(K^2 + k_d^2)^2}\boldsymbol{R} \tag{6.107}$$

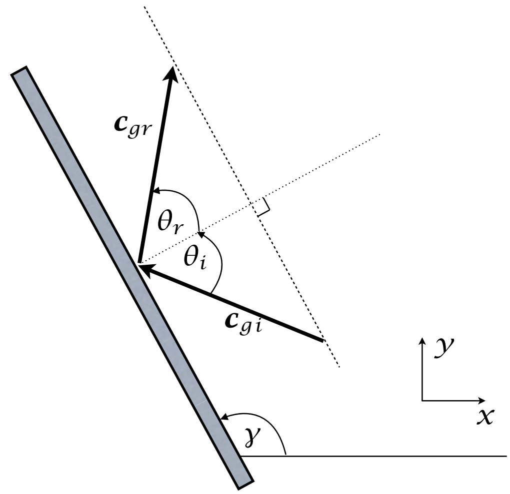

Fig. 6.9 The reflection of a Rossby wave at a western wall, in physical space. A Rossby wave with a westward group velocity impinges at an angle θ_i to a wall, inducing a reflected wave moving eastward at an angle θ_r. The reflection is specular, with $\theta_r = \theta_i$, and energy conserving, with $|\boldsymbol{c}_{gr}| = |\boldsymbol{c}_{gi}|$ — see text and Fig. 6.10.

where $\boldsymbol{R} = \overrightarrow{WC}$ is the vector directed from W to C, that is from the end of the wavevector itself to the center of the circle around which all the wavevectors lie.

Eq. (6.107) and Fig. 6.8 allow for a useful visualization of the energy and phase. The phase propagates in the direction of the wave vector, and for Rossby waves this is always westward. The group velocity is in the direction of the wave vector to the center of the circle, and this can be either eastward (if $k^2 > l^2 + k_d^2$) or westward ($k^2 < l^2 + k_d^2$). Interestingly, the velocity vector is normal to the wave vector. To see this, consider a purely westward propagating wave for which $l = 0$. Then $v = \partial\psi/\partial x = \mathrm{i}k\widetilde{\psi}$ and $u = -\partial\psi/\partial y = -\mathrm{i}l\widetilde{\psi} = 0$. We now see how some of these properties can help us understand the reflection of Rossby waves.

[Do we need a gray box summarizing some of the properties of reflection? xxx]

Reflection at a wall

Consider Rossby waves incident on wall making an angle γ with the x-axis, and suppose that somehow these waves are reflected back into the fluid interior. This is a reasonable expectation, for the wall cannot normally simply absorb all the wave energy. We first note a couple of general properties about reflection, namely that the incident and reflected wave will have the same wavenumber component along the wall and their frequencies must be the same To see these properties, consider the case in which the wall is oriented meridionally, along the y-axis with $\gamma = 90°$. There is no loss of generality in this choice, because we may simply choose coordinates so that y is parallel to the wall and the β-effect, which differentiates x from y, does not enter the argument. The incident and reflected waves are

$$\psi_i(x,y,t) = A_i \mathrm{e}^{\mathrm{i}(k_i x + l_i y - \omega_i t)}, \qquad \psi_r(x,y,t) = A_r \mathrm{e}^{\mathrm{i}(k_r x + l_r y - \omega_r t)}, \tag{6.108}$$

with subscripts i and r denoting incident and reflected. At the wall, which we take to be at $x = 0$, the normal velocity $u = -\partial\psi/\partial y$ must be zero so that

$$A_i l_i \mathrm{e}^{\mathrm{i}(l_i y - \omega_i t)} + A_r l_r \mathrm{e}^{\mathrm{i}(l_r y - \omega_r t)} = 0. \tag{6.109}$$

For this equation to hold for all y and all time then we must have

$$l_r = l_i, \qquad \omega_r = \omega_i. \tag{6.110}$$

This result is independent of the detailed dynamics of the waves, requiring only that the velocity is determined from a streamfunction. When we consider Rossby-wave dynamics specifically, the x- and y-coordinates are not arbitrary and so the wall cannot be taken to be aligned with the y-axis; rather, the result means that the *projection* of the incident wavevector, $\boldsymbol{k}_i$ on the wall must equal the *projection* of the reflected wavevector, $\boldsymbol{k}_r$. The magnitude of the wavevector (the wavenumber) is not in general conserved by reflection. Finally, given these results and using (6.109) we see that the incident and reflected amplitudes are related by

$$A_r = -A_i. \tag{6.111}$$

Now let's delve a little deeper into the wave-reflection properties.

Generally, when we consider a wave to be incident on a wall, we are supposing that the *group velocity* is directed toward the wall. Suppose that a wave of given frequency, ω, and wavevector, $\boldsymbol{k}_i$, and with westward group velocity is incident on a predominantly western wall, as in Fig. 6.9. (Similar reasoning, *mutatis mutandis,* can be applied to a wave incident on an eastern wall.) Let us suppose that incident wave, $\boldsymbol{k}_i$ lies at the point I on the wavenumber circle, and the group velocity is found by drawing a line from I to the center of the circle, C (so $\boldsymbol{c}_{gi} \propto \overrightarrow{IC}$), and in this case the vector is directed westward.

The projection of the $\boldsymbol{k}_i$ must be equal to the projection of the reflected wave vector, $\boldsymbol{k}_r$, and both wavevectors must lie in the same wavenumber circle, centered at $-\beta/2\omega$, because the frequencies of the two waves are the same. We may then graphically determines the wavevector of the reflected wave using the construction of Fig. 6.10. Given the wavevector, the group velocity of the reflected wave follows by drawing a line from the wavevector to the center of the circle (the line $\overrightarrow{RC}$). We see from the figure that the reflected group velocity is directed eastward and that it forms the same angle to the wall as does the incident wave; that is, the reflection is *specular*. Since the amplitude of the incoming and reflected wave are the same, the components of the energy flux perpendicular to the wall are equal and opposite. Furthermore, we can see from the figure that the wavenumber of the reflected wave has a larger magnitude than that of the incident wave. For waves reflecting of an easter boundary, the reverse is true. Put simply, at a western boundary incident long waves are reflected as short waves, whereas at an eastern boundary incident short waves are reflected as long waves.

Quantitatively solving for the wavenumbers of the reflected wave is a little tedious in the case when the wall is at angle, but easy enough if the wall is a meridional, along the y-axis. We know the frequency, ω, and the y-wavenumber, l, so that the x-wavenumber is may be deduced from the dispersion relation

$$\omega = \frac{-\beta k_i}{k_i^2 + l^2 + k_d^2} = \frac{-\beta k_r}{k_r^2 + l^2 + k_d^2}. \tag{6.112}$$

We obtain

$$k_i = \frac{-\beta}{2\omega} + \sqrt{\left(\frac{\beta}{2\omega}\right)^2 - \left(l^2 + k_d^2\right)}, \qquad k_r = \frac{-\beta}{2\omega} - \sqrt{\left(\frac{\beta}{2\omega}\right)^2 - \left(l^2 + k_d^2\right)}. \tag{6.113a,b}$$

The signs of the square-root terms are chosen for reflection at a western boundary, for which, as we noted, the reflected wave has a larger (absolute) wavenumber than the incident wave. For reflection at an eastern boundary, we simply reverse the signs.

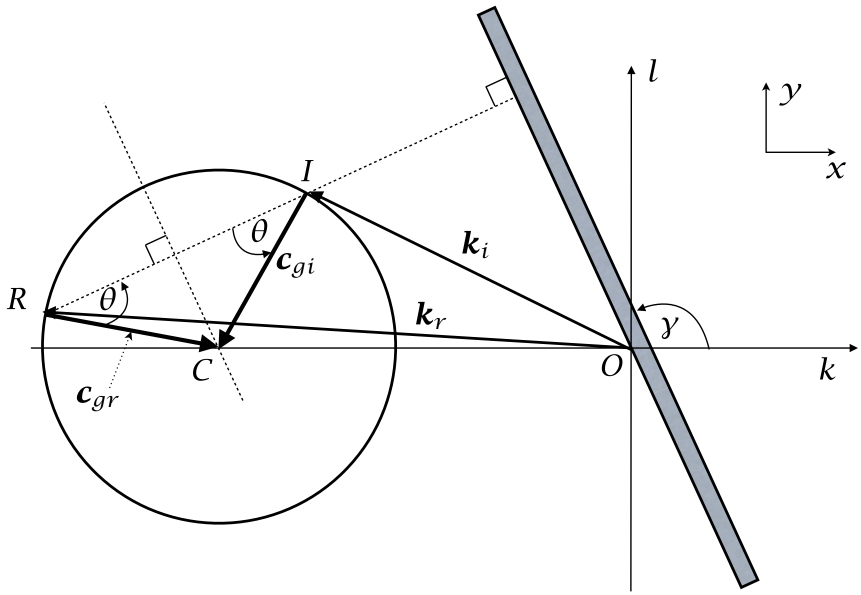

Fig. 6.10 Graphical representation of the reflection of a Rossby wave at a western wall, in spectral space. The incident wave has wavevector $\boldsymbol{k}_i$, ending at point I. Construct the wavevector circle through point I with radius $\sqrt{(\beta/2\omega)^2 - k_d^2}$ and center $C = (-\beta/2\omega, 0)$; the group velocity vector then lies along $\overrightarrow{IC}$ and is directed westward. The reflected wave has a wavevector $\boldsymbol{k}_r$ such that its projection on the wall is equal to that of $\boldsymbol{k}_i$, and this fixes the point R. The group velocity of the reflected wave then lies along $\overrightarrow{RC}$, and it can be seen that $\boldsymbol{c}_{gr}$ makes the same angle to the wall as does $\boldsymbol{c}_{gi}$, except that it is directed eastward. The reflection is therefore both specular and is such that the energy flux directed away from the wall is equal to the energy flux directed toward the wall.

Oceanographic relevance

The behaviour of Rossby waves at lateral boundaries is not surprisingly of some oceanographic importance, there being two particularly important examples. One of them concerns the equatorial ocean, and the other the formation of western boundary currents, common in midlatitudes. We only touch on these topics here, deferring a more extensive treatments to later chapters.

Suppose that Rossby waves are generated in the middle of the ocean, for example by the wind or possibly by some fluid dynamical instability in the ocean. Shorter waves will tend to propagate eastward, and be reflected back at the eastern boundary as long waves, and long waves will tend to propagate westward, being reflected back as short waves. The reflection at the western boundary is believed to be particularly important in the dynamics of *El Niño*. although the situation is further complicated because the reflection may also generate eastward moving equatorial Kelvin waves, which we discuss more in the next chapter.

In mid-latitudes the reflection at a western boundary generates Rossby waves that have a short *zonal* length scale (the meridional scale is the same as the incident wave if the wall is meridional), which means that their *meridional* velocity is large. Now, if the zonal wavenumber is much larger than both the meridional wavenumber l and the inverse deformation radius k_d then, using either (6.61) or (6.65) the group velocity in the x-direction is given by $c_g^x = \overline{u} + \beta/k^2$, where $\overline{u}$ is the zonal mean flow. If the mean flow is westward, so that U is negative, then very short waves will be unable to escape from the boundary; specifically, if $k > \sqrt{\beta / -U}$ then the waves will be trapped in a western boundary layer. [More here ?? xxx]

6.7 ROSSBY-GRAVITY WAVES: AN INTRODUCTION

We now consider Rossby waves and shallow water gravity waves together. To keep the treatment tractable we will consider the simplest possible case, namely a single layer of shallow water on the beta plane in which the Coriolis parameter, f, is held constant except where it is differentiated, an approximation similar to that made when deriving the quasi-geostrophic equations.[9] A (perforce more complex) treatment of the analogous problem on the equatorial beta-plane, in which we allow both f and β to vary fully with latitude, in is given in chapter **??**.

Our equations of motion are the shallow water equations in Cartesian coordinates in a rotating frame of reference, namely

$$\frac{\partial u}{\partial t} - fv = -\frac{\partial \phi}{\partial x}, \qquad \frac{\partial v}{\partial t} + fu = -\frac{\partial \phi}{\partial y}, \tag{6.114a,b}$$

$$\frac{\partial \phi}{\partial t} + c^2\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0 \tag{6.114c}$$

where, in terms of possibly more familiar shallow water variables, $\phi = g'\eta$ and $c^2 = g'H$, where ϕ is the kinematic pressure, η is the free surface height, H is the reference depth of the fluid and g' is the reduced gravity.

After some manipulation (described more fully in section **??**) we obtain, without additional approximation, a single equation for v:

$$\frac{1}{c^2}\frac{\partial^3 v}{\partial t^3} + \frac{f^2}{c^2}\frac{\partial v}{\partial t} - \frac{\partial}{\partial t}\nabla^2 v - \beta\frac{\partial v}{\partial x} = 0. \tag{6.115}$$

In this equation the Coriolis parameter is given by the β-plane expression $f = f_0 + \beta y$; thus, the equation has a non-constant coefficient, entailing considerable algebraic difficulties. We will address some of these difficulties in chapter **??**, but for now we take a simpler approach: we assume that f is constant except where differentiated, an approximation that is reasonable in mid-latitudes provided we are concerned with sufficiently small variations in latitude. Equation (6.115) then has constant coefficients and we may look for plane wave solutions of the form $v = \widetilde{v}\exp\left[\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)\right]$, whence

$$\frac{\omega^2 - f_0^2}{c^2} - (k^2 + l^2) - \frac{\beta k}{\omega} = 0. \tag{6.116a}$$

or, written differently,

$$\left(k + \frac{\beta}{2\omega}\right)^2 + l^2 = \left(\frac{\beta}{2\omega}\right)^2 + \frac{\omega^2 - f_0^2}{c^2}. \tag{6.116b}$$

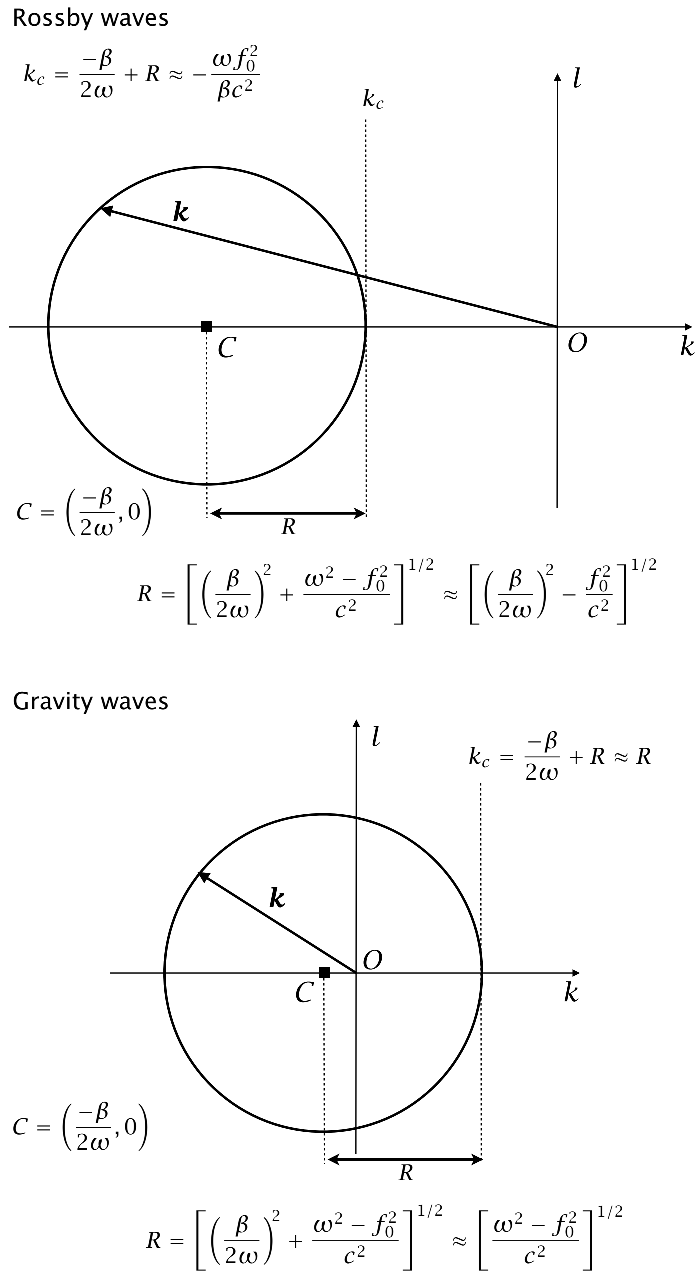

Fig. 6.11 Wave propagation diagrams for Rossby-gravity waves, obtained using (6.116). The top figure shows the diagram in the low frequency, Rossby wave limit, and the bottom figure shows the high frequency, gravity wave limit. In each case the the locus of wavenumbers for a given frequency is a circle centered at $C = (-\beta/2\omega, 0)$ with a radius R given by (6.117), but the approximate expressions differ significantly at high and low frequency.

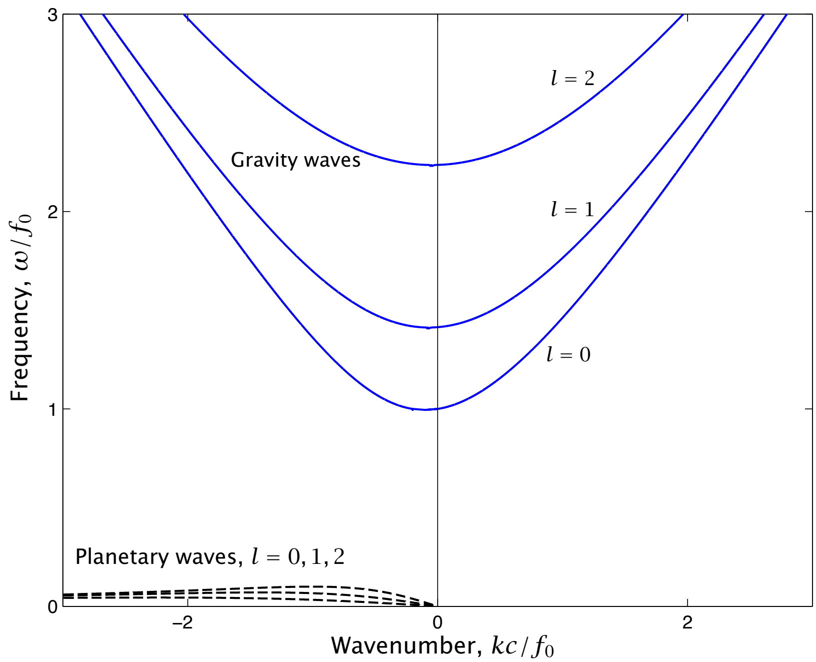

Fig. 6.12 Dispersion relation for Rossby-gravity waves, obtained from (6.129) with $\widehat{\beta} = 0.2$ for three values of l. There a frequency gap between the Rossby or planetary waves and the gravity waves. For the stratified mid-latitude atmosphere or ocean the frequency gap is in fact much larger.

This equation may be compared to (6.105): noting that $k_d^2 = f_0^2/g'H = f_0^2/c^2$, the two equations are identical except for the appearance of a term involving frequency on last term on the right-hand side of (6.116b). The wave propagation diagram is illustrated in Fig. 6.11. The wave vectors a a given frequency all lie on a circle centered at $(-\beta/2\omega, 0)$ and with radius r given by

$$R = \left[\left(\frac{\beta}{2\omega}\right)^2 + \frac{\omega^2 - f_0^2}{c^2}\right]^{1/2}, \tag{6.117}$$

and the radius must be positive in order for the waves to exist. In the low frequency case the diagram is essentially the same as that shown in Fig. 6.8, but is quantitatively significantly different in the high frequency case. These limiting cases are discussed further in section 6.7.1 below.

To plot the full dispersion relation it is useful to nondimensionalize using the following scales for time (T), distance (L) and velocity (U)

$$T = f_0^{-1}, \qquad L = L_d = k_d^{-1} = c/f_0, \qquad U = L/T = c, \tag{6.118a,b}$$

so that, denoting nondimensional quantities with a hat,

$$\omega = \widehat{\omega} f_0, \qquad (k, l) = (\hat{k}, \hat{l})k_d, \qquad \beta = \widehat{\beta}\frac{f_0^2}{c} = \widehat{\beta}\frac{f_0}{L_d} = \widehat{\beta} f_0 k_d. \tag{6.119}$$

The dispersion relation (6.116) may then be written as

$$\widehat{\omega}^2 - 1 - (\hat{k}^2 + \hat{l}^2) - \widehat{\beta}\frac{\hat{k}}{\widehat{\omega}} = 0 \tag{6.120}$$

This is a cubic equation in ω, as might be expected given the governing equations (6.114). We may expect that two of the roots correspond to gravity waves and the third to Rossby waves. The only parameter in the dispersion relation is $\widehat{\beta} = \beta c/f_0^2 = \beta L_d/f_0$. In the atmosphere a representative value for L_d is 1000 km, whence $\widehat{\beta} = 0.1$. In the ocean $L_d \sim 100\,\mathrm{km}$, whence $\widehat{\beta} = 0.01$. If we allow ourselves to consider 'external' Rossby waves (which are of some oceanographic relevance) then $c = \sqrt{gH} = 200\,\mathrm{m\,s^{-1}}$ and $L_d = 2000\,\mathrm{km}$, whence $\widehat{\beta} = 0.2$.

To actually obtain a solution we regard the equation as a quadratic in k and solve in terms of the frequency, giving

$$\hat{k} = -\frac{\widehat{\beta}}{2\widehat{\omega}} \pm \frac{1}{2}\left[\frac{\widehat{\beta}^2}{\widehat{\omega}^2} + 4(\widehat{\omega}^2 - \hat{l}^2 - 1)\right]^{1/2}. \tag{6.121}$$

The solutions are plotted in Fig. 6.12, with $\widehat{\beta} = 0.2$, and we see that the waves fall into two groups, labelled gravity waves and planetary waves in the figure. The gap between the two groups of waves is in fact still larger if a smaller (and generally more relevant) value of $\widehat{\beta}$ is used. To interpret all this let us consider some limiting cases.

6.7.1 Special cases and properties of the waves

We now consider a few special cases of the dispersion relation.

(i) Constant Coriolis parameter

If $\beta = 0$ then the dispersion relation becomes

$$\omega\left[\omega^2 - f_0^2 - (k^2 + l^2)c^2\right] = 0, \tag{6.122}$$

with the roots

$$\omega = 0, \qquad \omega^2 = f_0^2 + c^2(k^2 + l^2). \tag{6.123a,b}$$

The root $\omega = 0$ corresponds to geostrophic motion (and, since $\beta = 0$, Rossby waves are absent), with the other root corresponding to Poincaré waves, considered in chapter 3. Note that $\omega^2 > f_0^2$.

(ii) High frequency waves

If we take the limit of $\omega \gg f_0$ then (6.116a) gives

$$\frac{\omega^2}{c^2} - (k^2 + l^2) - \frac{\beta k}{\omega} = 0. \tag{6.124}$$

To be physically realistic we should also now eliminate the β term, because if $\omega \gg f_0$ then, from geometric considerations on a sphere, $k^2 \gg \beta k/\omega$. Thus, the dispersion relation is simply $\omega^2 = c^2(k^2 + l^2)$. These waves are just gravity waves uninfluenced by rotation, and are a special case of Poincaré waves.

Rossby and Gravity Waves

- Generically speaking, Rossby-gravity waves are waves that arise under the combined effects of a potential vorticity gradient and stratification. Sometimes the definition is restricted to a wave on a single branch of the dispersion curve connecting Rossby and Gravity waves. In mid-latitudes Rossby waves and gravity waves are well separated with distinct physical mechanisms

- The simplest setting in which such waves occur is in the linearized shallow water equations which may be written as a single equation for v, namely

$$\frac{1}{c^2}\frac{\partial^3 v}{\partial t^3} + \frac{f^2}{c^2}\frac{\partial v}{\partial t} - \frac{\partial}{\partial t}\nabla^2 v - \beta\frac{\partial v}{\partial x} = 0. \tag{RG.1}$$

- If we take both f and β to be constants then the equation above admits of plane-wave solutions with dispersion relation

$$\omega^2 - \frac{\beta k c^2}{\omega} = f_0^2 + c^2(k^2 + l^2). \tag{RG.2}$$

- In Earth's atmosphere and ocean it is common, especially in mid-latitudes, for there to be a frequency separation between two classes of solution. To a good approximation, high frequency waves satisfy

$$\omega^2 = f_0^2 + c^2(k^2 + l^2). \tag{RG.3}$$

  These are gravity waves and which in this context, because of the presence of rotation, are known as Poincaré waves. The low frequency waves satisfy

$$\omega = \frac{-\beta k c^2}{f_0^2 + c^2(k^2 + l^2)} = \frac{-\beta k}{k_d^2 + k^2 + l^2}, \tag{RG.4}$$

  where $k_d^2 = f_0^2/c^2$, and these are called Rossby waves or planetary waves.

- Rossby-gravity waves also exist in the stratified equations. Solutions may be found be decomposing the vertical structure into a series of orthogonal modes, and a sequence of shallow water equations for each mode results, with a different c for each mode. Solutions may also be found if f is allowed to vary in (RG.1), at the price of some algebraic complexity, as discussed in chapter **??**.

(iii) Low frequency waves

Consider the limit of $\omega \ll f_0$. The dispersion relation reduces to

$$\omega = \frac{-\beta k}{k^2 + l^2 + k_d^2}. \tag{6.125}$$

This is just the dispersion relation for quasi-geostrophic Rossby waves as previously obtained — see (6.63) or (6.90). In this limit, the requirement that the radius of the circle be positive becomes

$$\omega^2 < \frac{\beta^2}{4k_d^2}. \tag{6.126}$$

That is to say, the Rossby waves have a maximum frequency, and directly from (6.125) this occurs when $k = k_d$ and $l = 0$.

The frequency gap

The maximum frequency of Rossby waves is usually much less than the frequency of the Poincaré waves: the lowest frequency of the Poincare waves is f_0 and the highest frequency of the Rossby waves is $\beta/2k_d$. Thus,

$$\frac{\text{Low gravity wave frequency}}{\text{High Rossby wave frequency}} = \frac{f_0}{\beta/2k_d} = \frac{f_0^2}{2\beta c}. \tag{6.127}$$

If $f_0 = 10^{-4}\,\mathrm{s^{-1}}$, $\beta = 10^{-11}\,\mathrm{m^{-1}\,s^{-1}}$ and $k_d = 1/100\,\mathrm{km^{-1}}$ (a representative oceanic baroclinic deformation radius) then $f_0/(\beta/2k_d) = 200$. If $L_d = 1000\,\mathrm{km}$ (an atmospheric baroclinic radius) then the ratio is 20. If we use a barotropic deformation radius of $L_d = 2000\,\mathrm{km}$ then the ratio is 10. Evidently, for most midlatitude applications there is a large gap between the Rossby wave frequency and the gravity wave frequency. Because of this frequency gap, to a good approximation Fig. 6.12 may be obtained by separately plotting (6.123b) for the gravity waves, and (6.125) for the Rossby or planetary waves. The differences between these and the exact results become smaller as $\widehat{\beta}$ gets smaller, and for the Rossby waves are in fact less than the thickness of the line on the plot shown.

Finally, we remark that a 'Rossby-gravity wave' is sometimes defined to be the wave on a single branch of the dispersion curve that connects Rossby waves and gravity waves, depending on the value of the wavenumber. The equatorial beta plane does support such a wave — the 'Yanai wave' derived in chapter **??** and shown in Fig. **??**. However, in the mid-latitude system above there is no such wave; rather, there are Rossby waves and gravity waves, separated by a frequency gap.

6.7.2 Planetary geostrophic Rossby waves

A good approximation for the large-scale ocean circulation involves ignoring the time-derivatives and nonlinear terms in the momentum equation, allowing evolution only to occur in the thermodynamic equation. This is the planetary-geostrophic approximation, introduced in section 5.2 202, and it is interesting to see to what extent that system supports Rossby waves.[10] It is easiest just to begin with the linear shallow water equations themselves, and omitting time derivatives in the momentum equation gives

$$-fv = -\frac{\partial \phi}{\partial x}, \qquad fu = -\frac{\partial \phi}{\partial y}, \tag{6.128a,b}$$

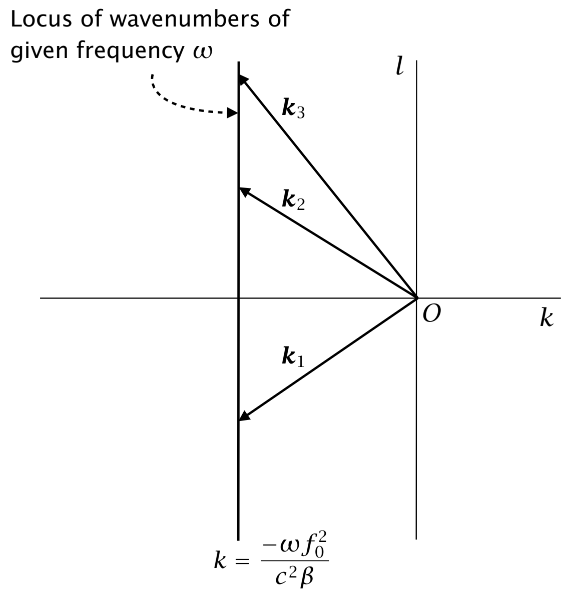

Fig. 6.13 The locus of points on planetary-geostrophic Rossby waves. Waves of a given frequency all have the same x-wavenumber, given by xxx

$$\frac{\partial \phi}{\partial t} + c^2\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0. \tag{6.128c}$$

From these equations we straightforwardly obtain

$$\frac{\partial \phi}{\partial t} - \frac{c^2\beta}{f^2}\frac{\partial \phi}{\partial x} = 0. \tag{6.129}$$

Again we will treat both f and β as constants so that we may look for solutions in the form $\phi = \widetilde{\phi}\exp[\mathrm{i}(\boldsymbol{k}\cdot x - \omega t)]$. The ensuing dispersion relation is

$$\omega = -\frac{c^2\beta}{f_0^2}k = -\frac{\beta k}{k_d^2} \tag{6.130}$$

which is a limiting case of (**??**) with $k^2, l^2 \ll k_d^2$. The waves are a form of Rossby waves with phase and group speeds given by

$$c_p = -\frac{c^2\beta}{f_0^2}, \qquad c_g^x = -\frac{c^2\beta}{f_0^2}. \tag{6.131}$$

That is, the waves are non-dispersive and propagate westward. Eq. (6.129) has the general solution $\phi = G(x + \beta c^2/f^2 t)$, where G is any function, so an initial disturbance will just propagate westward at a speed given by (6.131), without any change in form.

Note finally that the locus of wavenumbers in k–l space is no longer a circle, as it is for the usual Rossby waves. Rather, since the frequency does not depend on the y-wavenumber, the locus is a straight line, parallel to the y-axis, as in Fig. 6.13. Waves of a given frequency all have the same x-wavenumber, given by $k = -\omega f_0^2/(c^2\beta) = -\omega k_d^2/\beta$, as shown in Fig. 6.13.

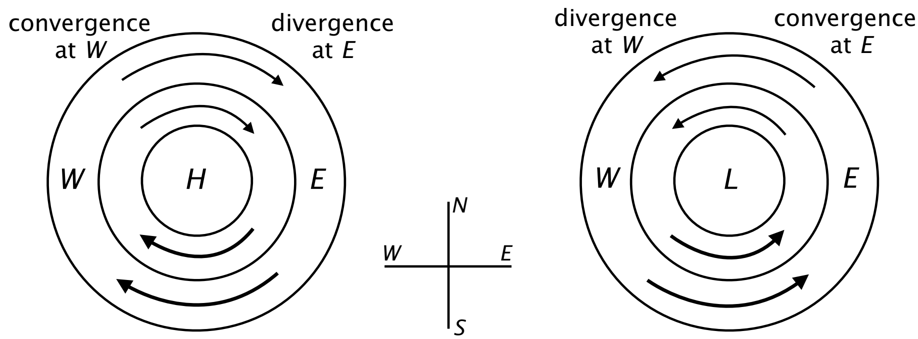

Fig. 6.14 The westward propagation of planetary-geostrophic Rossby waves. The circular lines are isobars centered around high and low pressure centres. Because of the variation of the Coriolis force, the mass flux between two isobars is greater to the south of a pressure center than it is to the north. Hence, in the left-hand sketch there is convergence to the west of the high pressure and the pattern propagates westward. Similarly, if the pressure centre is a low, as in the right-hand sketch, there is divergence to the west of the pressure centre and the pattern still propagates westward.

Physical mechanism

Because the waves *are* a form of Rossby wave their physical mechanism is related to that discussed in section 6.4.3, but with an important difference: relative vorticity is no longer important, but the flow divergence is. Thus, consider flow round a region of high pressure, as illustrated in Fig. 6.14. If the pressure is circularly symmetric as shown, the flow to the south of H in the left-hand sketch, and to the south of L in the right-hand sketch, is larger than that to the north. Hence, in the left sketch the flow converges at W and diverges at E, and the flow pattern moves westward. In the flow depicted in the right sketch the low pressure propagates westward in a similar fashion.

6.8 ◆ THE GROUP VELOCITY PROPERTY

We now return to a more general discussion of group velocity. Our goal is to show that the group velocity arises in fairly general ways, not just from methods stemming from Fourier analysis or from ray theory. In a purely logical sense this discussion follows most naturally from the end of the section on ray theory (section 6.3), but for most humans it is helpful to have had a concrete introduction to at least one nontrivial form of waves before considering more abstract material. We first give a simple and direct derivation of group velocity that valid in the simple but important special case of a homogeneous medium.[11] Then, in section 6.8.2, we give a rather general derivation of the *group velocity property,* namely that conserved quantities that are quadratic in the wave amplitude — that is,s *wave activities* — are transported at the group velocity.

6.8.1 Group velocity in homogeneous media

Consider waves propagating in a homogeneous medium in which the wave equation is a polynomial of the general form

$$L(\psi) = \Lambda\left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x}\right)\psi(x,t) = 0. \tag{6.132}$$

where Λ is a polynomial operator in the space and time derivatives. For algebraic simplicity we restrict attention to waves in one dimension, and a simple example is $\Lambda = \partial(\partial_{xx})/\partial t + \beta\partial/\partial x$ so that $L(\psi) = \partial(\partial_{xx}\psi)/\partial t + \beta\partial\psi/\partial x$. We will seek a solution of the form [c.f., (6.4)]

$$\psi(x,t) = A(x,t)\mathrm{e}^{i\theta(x,t)}, \tag{6.133}$$

where θ is the phase of the disturbance and $A(x,t)$ is the slowly varying amplitude, so that the solution has the form of a wave packet. The phase is such that $k = \partial\theta/\partial x$ and $\omega = -\partial\theta/\partial t$, and the slowly varying nature of the envelope $A(x,t)$ is formalized by demanding that

$$\frac{1}{A}\frac{\partial A}{\partial x} \ll k, \qquad \frac{1}{A}\frac{\partial A}{\partial t} \ll \omega, \tag{6.134}$$

The space and time derivatives of ψ are then given by

$$\frac{\partial\psi}{\partial x} = \left(\frac{\partial A}{\partial x} + \mathrm{i}A\frac{\partial\theta}{\partial x}\right)\mathrm{e}^{i\theta} = \left(\frac{\partial A}{\partial x} + \mathrm{i}Ak\right)\mathrm{e}^{i\theta}, \tag{6.135a}$$

$$\frac{\partial\psi}{\partial t} = \left(\frac{\partial A}{\partial t} + \mathrm{i}A\frac{\partial\theta}{\partial t}\right)\mathrm{e}^{i\theta} = \left(\frac{\partial A}{\partial t} - \mathrm{i}A\omega\right)\mathrm{e}^{i\theta}, \tag{6.135b}$$

so that the wave equation becomes

$$\Lambda\psi = \Lambda\left(\frac{\partial}{\partial t} - \mathrm{i}\omega, \frac{\partial}{\partial x} + \mathrm{i}k\right)A = 0. \tag{6.136}$$

Noting that the space and time derivative of A are small compared to k and ω we expand the polynomial in a Taylor series about (ω, k) to obtain

$$\Lambda(-\mathrm{i}\omega, \mathrm{i}k)A + \frac{\partial\Lambda}{\partial(-\mathrm{i}\omega)}\frac{\partial A}{\partial t} + \frac{\partial\Lambda}{\partial(\mathrm{i}k)}\frac{\partial A}{\partial x} = 0. \tag{6.137}$$

The first term is nothing but the linear dispersion relation; that is $\Lambda(-\mathrm{i}\omega, \mathrm{i}k)A = 0$ is the dispersion relation for plane waves. Taking this to be satisfied, (6.137) gives

$$\frac{\partial A}{\partial t} - \frac{\partial\Lambda/\partial k}{\partial\Lambda/\partial\omega}\frac{\partial A}{\partial x} = \frac{\partial A}{\partial t} + \frac{\partial\omega}{\partial k}\frac{\partial A}{\partial x} = 0. \tag{6.138}$$

That is, the envelope moves at the group velocity $\partial\omega/\partial k$.

6.8.2 ♦ Group velocity property: a general derivation

In our discussion of Rossby waves in section 6.6 in (6.104) we showed that the energy of the waves is conserved in the sense that

$$\frac{\partial E}{\partial t} + \nabla\cdot\boldsymbol{F} = 0, \tag{6.139}$$

where E is the energy density of the waves and $\boldsymbol{F}$ is its flux. In (6.104) we further showed that, when averaged over a wavelength and a period, the average flux, $\overline{\boldsymbol{F}}$, was related to the average energy, $\overline{E}$, by $\overline{\boldsymbol{F}} = \boldsymbol{c}_g\overline{E}$. This property is called the group velocity property and it is a very general property, not restricted to Rossby waves or even to energy. In the previous section we gave a more general derivation valid in homogeneous media. In fact, the property is still more general and it holds for almost any conserved quantity that is quadratic in the wave amplitude, and we now demonstrate this in a rather general way.[12] A quantity that is quadratic and conserved is known as a *wave activity*. (The corresponding local quantity, such as the wave activity per unit volume, might strictly we called the *wave activity density*.) The group velocity property is useful because if we can determine $\boldsymbol{c}_g$ then we know straightaway how wave activities propagate. Energy itself can be a wave activity but is not always. In a growing baroclinic wave energy is drawn from the background state; however, we will see in chapter 10 that even in a growing baroclinic disturbance it is possible to define a conserved wave activity.

◆ *The formal procedure*

The derivation, which is rather formal, will hold for waves and wave activities that satisfy the following three assumptions.

(i) The wave activity, A, and flux, $\boldsymbol{F}$, obey the general conservation relation

$$\frac{\partial A}{\partial t} + \nabla\cdot\boldsymbol{F} = 0. \tag{6.140}$$

(ii) Both the wave activity and the flux are quadratic functions of the wave amplitude.

(iii) The waves themselves are of the general form

$$\psi = \widetilde{\psi}\mathrm{e}^{\mathrm{i}\theta(\boldsymbol{x},t)} + \text{c.c.}, \qquad \theta = \boldsymbol{k}\cdot\boldsymbol{x} - \omega t, \qquad \omega = \omega(\boldsymbol{k}), \tag{6.141a,b,c}$$

where (6.141c) is the dispersion relation, and ψ is any wave field. We will carry out the derivation in case in which $\widetilde{\psi}$ is a constant, but the derivation may be extended to the case in which it varies slowly over a wavelength.

Given assumption *(ii)*, the wave activity must have the general form

$$A = b + a\mathrm{e}^{2\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)} + a^*\mathrm{e}^{-2\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}, \tag{6.142a}$$

where the asterisk, *, denotes complex conjugacy, and b is a real constant and a is a complex constant. For example, suppose that $A = \psi^2$ and $\psi = c\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)} + c^*\mathrm{e}^{-\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}$, then we find that (6.142a) is satisfied with $a = c^2$ and $b = 2cc^*$. Similarly, the flux has the general form

$$\boldsymbol{F} = \boldsymbol{g} + \boldsymbol{f}\mathrm{e}^{2\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)} + \boldsymbol{f}^*\mathrm{e}^{-2\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}. \tag{6.142b}$$

where $\boldsymbol{g}$ is a real constant vector and $\boldsymbol{f}$ is a complex constant vector. The mean activity and mean flux are obtained by averaging over a cycle; the oscillating terms vanish on integration and therefore the wave activity and flux are given by

$$\overline{A} = b, \qquad \overline{\boldsymbol{F}} = \boldsymbol{g}, \tag{6.143}$$

where the overbar denotes the mean.

Now formally consider a wave with a slightly different phase, $\theta + \mathrm{i}\,\delta\theta$, where $\delta\theta$ is small compared with θ. Thus, we formally replace $\boldsymbol{k}$ by $\boldsymbol{k} + \mathrm{i}\,\boldsymbol{\delta k}$ and ω by $\omega + \mathrm{i}\,\delta\omega$ where, to satisfy the dispersion relation, we have

$$\omega + \mathrm{i}\,\delta\omega = \omega(\boldsymbol{k} + \mathrm{i}\,\boldsymbol{\delta k}) \approx \omega(\boldsymbol{k}) + \mathrm{i}\,\boldsymbol{\delta k} \cdot \frac{\partial \omega}{\partial \boldsymbol{k}}, \tag{6.144}$$

and therefore

$$\delta\omega = \boldsymbol{\delta k} \cdot \frac{\partial \omega}{\partial \boldsymbol{k}} = \boldsymbol{\delta k} \cdot \boldsymbol{c}_g, \tag{6.145}$$

where $\boldsymbol{c}_g \equiv \partial\omega/\partial\boldsymbol{k}$ is the group velocity.

The new wave has the general form

$$\psi' = (\widetilde{\psi} + \delta\widetilde{\psi})\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}\mathrm{e}^{-\boldsymbol{\delta k}\cdot\boldsymbol{x} + \delta\omega t} + \text{c.c.}, \tag{6.146}$$

and, analogously to (6.142), the associated wave activity and flux have the forms:

$$A' = \left[b + \delta b + (a + \delta a)\mathrm{e}^{2\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)} + (a^* + \delta a^*)\mathrm{e}^{-2\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}\right]\mathrm{e}^{-2\boldsymbol{\delta k}\cdot\boldsymbol{x} + 2\delta\omega t} \tag{6.147a}$$

$$\boldsymbol{F}' = \left[\boldsymbol{g} + \delta\boldsymbol{g} + (\boldsymbol{f} + \delta\boldsymbol{f})\mathrm{e}^{2\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)} + (\boldsymbol{f}^* + \delta\boldsymbol{f}^*)\mathrm{e}^{-2\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}\right]\mathrm{e}^{-2\boldsymbol{\delta k}\cdot\boldsymbol{x} + 2\delta\omega t}, \tag{6.147b}$$

where the δ quantities are small. If we now demand that A' and $\boldsymbol{F}'$ satisfy assumption (i), then substituting (6.147) into (6.140) gives, after a little algebra,

$$(\boldsymbol{g} + \delta\boldsymbol{g}) \cdot \boldsymbol{\delta k} = (b + \delta b)\delta\omega \tag{6.148}$$

and therefore at first order in δ quantities, $\boldsymbol{g} \cdot \boldsymbol{\delta k} = b\delta\omega$. Using (6.145) and (6.143) we obtain

$$\boldsymbol{c}_g = \frac{\boldsymbol{g}}{b} = \frac{\overline{\boldsymbol{F}}}{\overline{A}}, \tag{6.149}$$

and using this the conservation law, (6.140), becomes

$$\frac{\partial \overline{A}}{\partial t} + \nabla \cdot (\boldsymbol{c}_g \overline{A}) = 0. \tag{6.150}$$

Thus, for waves satisfying our three assumptions, the flux velocity — that is, the propagation velocity of the wave activity — is equal to the group velocity.

6.9 ENERGY PROPAGATION OF POINCARÉ WAVES

In the final section of this chapter we discuss the energetics of Poincaré waves, and show explicitly that the energy propagation occurs at the group velocity. (Poincaré waves were first introduced in section 3.7.2 and the reader may wish to review that section before continuing.) We begin with the one-dimensional problem as this shows the essential aspects and the algebra is a little simpler.

6.9.1 Energetics in one dimension

The one-dimensional (i.e., no variations in the y-direction), inviscid linear shallow-water equations on the f-plane, linearized about a state of rest, are

$$\frac{\partial u}{\partial t} - f_0 v = -g\frac{\partial h}{\partial x}, \qquad \frac{\partial v}{\partial t} + f_0 u = 0, \qquad \frac{\partial \eta}{\partial t} = -H\frac{\partial u}{\partial x}. \tag{6.151a,b,c}$$

To obtain the dispersion relation we differentiate the first equation with respect to t and substitute from the second and third to obtain

$$\frac{\partial^2 u}{\partial t^2} - Hg\frac{\partial u}{\partial x} + f_0^2 u = 0, \tag{6.152}$$

whence, assuming solutions of the form $u = \mathrm{Re}\,\tilde{u}e^{i(kx-\omega t)}$, we obtain the dispersion relation

$$\omega^2 = f^2 + Hgk^2. \tag{6.153}$$

This is immediately recognizable as a special case of the two-dimensional dispersion relation. An interesting property of this equation is obtained by differentiating with respect to k, giving $2\omega\partial\omega/\partial k = 2kHg$ or

$$c_g = \frac{Hg}{c_p}, \tag{6.154}$$

where $c_g = \partial\omega/\partial k$ and $c_p = \omega/k$ are the group and phase velocities, respectively. Using (6.153) and (6.154) the ratio of the group and phase velocities is found to be

$$\frac{c_g}{c_p} = \frac{L_d^2 k^2}{1 + L_d^2 k^2}, \tag{6.155}$$

where $L_d = \sqrt{gH}/f$ is the deformation radius. This ratio is always less than unity, tending to zero in the long-wave limit ($kL_d \ll 1$) and to unity for short waves ($kL_d \gg 1$).

The energy equations are obtained by multiplying the three equations of (**??**) by u, v and η respectively, and adding, to give

$$\frac{\partial E}{\partial t} + \frac{\partial F}{\partial x} = 0, \tag{6.156a}$$

where

$$E = \frac{1}{2}(Hu^2 + Hv^2 + g\eta^2), \qquad F = gHu\eta, \tag{6.156b}$$

are the energy density and the energy flux, respectively. Note that in the linear approximation the energy is transported only by the pressure term, whereas in the full nonlinear equations there is also an advective transport.

The group velocity property

To specialize to the case of propagating *waves* we need to average over a wavelength and use the phase relationships between u, v and η implied by the equations of motion. Writing $u = \mathrm{Re}\,\tilde{u}e^{i(kx-\omega t)}$, and similarly for v and η, we have

$$\tilde{v} = -\mathrm{i}f\frac{\tilde{\eta}}{Hk}, \qquad \tilde{u} = \omega\frac{\tilde{\eta}}{Hk}. \tag{6.157a,b}$$

The kinetic energy, averaged over a wavelength, is then

$$KE = \frac{1}{2}H(\overline{u^2} + \overline{v^2}) = \frac{1}{4}(\omega^2 + f^2)\frac{\tilde{\eta}^2}{Hk^2} = \frac{1}{4}\frac{\omega^2 + f^2}{\omega^2 - f^2}g\tilde{\eta}^2 \tag{6.158}$$

using (6.157) and the dispersion relation, with the extra factor of one half arising from the averaging over a wavelength. Similarly, the potential energy of the wave is

$$PE = \frac{1}{2}g\overline{\eta^2} = \frac{1}{4}g\tilde{\eta}^2 \tag{6.159}$$

Thus, the ratio of kinetic to potential energy is just

$$\frac{KE}{PE} = \frac{\omega^2 + f^2}{\omega^2 - f^2} = 1 + \frac{2}{k^2 L_d^2} \tag{6.160}$$

using the dispersion relation, and where $l_d = \sqrt{gH/f}$ is the deformation radius. Thus, the kinetic energy is always *greater* than the potential energy (there is no equipartition in this problem), with the ratio approaching unity for small scales (large k).

The total energy (kinetic plus potential) is then

$$KE + PE = \frac{1}{4}\left(\frac{\omega^2 + f^2}{\omega^2 - f^2} + 1\right)g\tilde{\eta}^2 = \frac{1}{2}\frac{\omega^2}{k^2 H}\tilde{\eta}^2 = \frac{1}{2}\frac{c_p^2}{H}\tilde{\eta}^2, \tag{6.161}$$

again using the dispersion relation. The energy flux, F, averaged over a wavelength, is

$$F = gH\overline{u\eta} = \frac{1}{2}\frac{g\omega}{k}\tilde{\eta}^2 = \frac{1}{2}gc_p\tilde{\eta}^2. \tag{6.162}$$

From (6.161) an (6.162) the flux and the energy are evidently related by

$$\boxed{F = \frac{Hg}{c_p}E = c_g E}, \tag{6.163}$$

using (6.154). That is, the energy flux is equal to the group velocity times the energy itself. Note that in this problem there is no flux in the y direction, because v and η are exactly out of phase from (6.157a).

6.9.2 ◆ Energetics in two dimensions

The derivations of the preceding section carry through, *mutatis mutandis,* in the full two-dimensional case. We will give only the key results and allow the reader to fill in the algebra. As derived in section 3.7.2 the dispersion relation is

$$\omega^2 = f_0^2 + gH(k^2 + l^2). \tag{6.164}$$

The relation between the components of the group velocity and the phase speed is very similar to the the one-dimensional case, and in particular we have

$$c_g^x = \frac{\partial\omega}{\partial k} = gH\frac{k}{\omega} = \frac{gH}{c_p^x}, \qquad c_g^y = \frac{\partial\omega}{\partial l} = gH\frac{l}{\omega} = \frac{gH}{c_p^y}. \tag{6.165}$$

The magnitude of the group velocity is $c_g \equiv |\boldsymbol{c}_g| = (c_g^{x\,2} + c_g^{y\,2})^{1/2}$. The magnitude of the phase speed, in the direction of travel of the wave crests, is $c_p = \omega/(k^2 + l^2)^{1/2}$ (note that in general this is *smaller* than the phase speed in either the x or y directions, ω/k or ω/l). Thus, we have

$$c_g^2 = (gH)^2 \frac{k^2 + l^2}{\omega^2} = \frac{(gH)^2}{c_p^2}, \qquad \boldsymbol{c}_g = \left(\frac{gH}{c_p K}\right) \boldsymbol{k}, \tag{6.166}$$

which is analogous to (6.154). The ratio of the magnitudes of the group and phase velocities is, analogously to (6.155),

$$\frac{c_g}{c_p} = \frac{gH}{c_p^2} = \frac{L_d^2 K^2}{1 + L_d^2 K^2}, \tag{6.167}$$

where $K^2 = k^2 + l^2$. As in the one-dimensional case the group velocity is large for short waves, in which rotation plays no role, and small for long waves.

The energy equation is found to be

$$\frac{\partial E}{\partial t} + \nabla \cdot \boldsymbol{F} = 0 \tag{6.168a}$$

with

$$E = \frac{1}{2}(Hu^2 + Hv^2 + g\eta^2), \qquad F = gH(u\mathbf{i} + v\mathbf{j})\eta. \tag{6.168b}$$

From the equations of motion the phase relations between the fields are found to be

$$\tilde{v} = \frac{\omega l - \mathrm{i}kf}{HK^2}\tilde{\eta}, \qquad \tilde{u} = \frac{\omega k - \mathrm{i}lf}{HK^2}\tilde{\eta}, \tag{6.169}$$

so that the kinetic energy is given by, similar to (6.158),

$$KE = \frac{1}{2}H(\overline{u^2} + \overline{v^2}) = \frac{1}{4}(\omega^2 + f^2)\frac{\tilde{\eta}^2}{Hk^2} = \frac{1}{4}\frac{\omega^2 + f^2}{\omega^2 - f^2}g\tilde{\eta}^2, \tag{6.170}$$

and the potential energy by

$$PE = \frac{1}{2}g\overline{\eta^2} = \frac{1}{4}g\tilde{\eta}^2 \tag{6.171}$$

The ratio of the kinetic and potential energies is given by

$$\frac{KE}{PE} = \frac{\omega^2 + f^2}{\omega^2 - f^2} = 1 + \frac{2}{K^2 L_d^2} \tag{6.172}$$

The total (kinetic plus potential) energy is given by

$$E = KE + PE = \frac{1}{4}\left(\frac{\omega^2 + f^2}{\omega^2 - f^2} + 1\right)g\tilde{\eta}^2 = \frac{1}{2}\frac{\omega^2}{K^2 H}\tilde{\eta}^2 = \frac{1}{2}\frac{c_p^2}{H}\tilde{\eta}^2, \tag{6.173}$$

The energy flux, $\boldsymbol{F}$, averaged over a wavelength, is

$$\boldsymbol{F} = gH\overline{\boldsymbol{u}\eta} = \frac{1}{2}\frac{g\omega}{k^2 + l^2}\tilde{\eta}^2\mathbf{k} = \frac{1}{2}\frac{g\omega}{K^2}\tilde{\eta}^2\boldsymbol{k}, \tag{6.174}$$

using (6.169) and where $\boldsymbol{k} = k\mathbf{i} + l\mathbf{j}$ is the wavevector of the wave.

From (6.173) and (6.174), and using (6.166), the flux and the energy are found to be related by

$$\boxed{\boldsymbol{F} = \boldsymbol{c}_g E}. \tag{6.175}$$

That is, the energy flux is equal to the group velocity times the energy itself.

Notes

1 A useful introduction to wave motion, from which this chapter has benefited, can be found in unpublished lecture notes by Chapman *et al.* (1989). Other useful material can be found in the 'further reading' section below.

2 For example *Linear and Nonlinear Waves* by G. B. Whitham.

3 For a review of group velocity, see Lighthill (1965).

4 More detailed treatments of ray theory and related matters are given by Whitham (1974), Lighthill (1978) and LeBlond & Mysak (1980).

5 Rossby waves were probably first discovered in a theoretical context by **??**. He considered the linear shallow water equations on a sphere, expanding the solution in powers of the sine of latitude, so obtaining both long gravity waves and Rossby waves. However, the discovery did not garner very much attention in the meteorological or oceanographic community until it was reprised by Rossby (1939). Rossby used the beta-plane approximation in Cartesian coordinates, and the simplicity of the presentation along with the meteorological context lead to the work attracting significant notice.

6 This non-Doppler effect also arises quite generally, even in models in height coordinates. See White (1977) and problem 5.5.

7 See Chapman & Lindzen (1970).

8 Following Longuet-Higgins (1964).

9 To read more about this problem, see Paldor *et al.* (2007) and Heifetz & Caballero (2014).

10 The ensuing waves seem to have been first noted by **?**.

11 Following Pedlosky (2003).

12 The form of this derivation was originally given by Hayes (1977) in the context of wave energy. See Vanneste & Shepherd (1998) for generalizations.

Further reading

Majda, A. J., 2003. *Introduction to PDEs and Waves for the Atmosphere and Ocean.*
Provides a compact, somewhat mathematical introduction to various equation sets and their properties, including quasi-geostrophy.

Problems

6.1 Consider the flat-bottomed shallow water potential vorticity equation in the form

$$\frac{\mathrm{D}}{\mathrm{D}t}\frac{\zeta + f}{h} = 0 \tag{P6.1}$$

(a) Suppose that deviations of the height field are small compared to the mean height field, and that the Rossby number is small (so $|\zeta| \ll f$). Further consider flow on a β-plane such that $f = f_0 + \beta y$ where $|\beta y| \ll f_0$. Show that the evolution equation becomes

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\zeta + \beta y - \frac{f_0 \eta}{H}\right) = 0 \tag{P6.2}$$

where $h = H + \eta$ and $|\eta| \ll H$. Using geostrophic balance in the form $f_0 u = -g\partial\eta/\partial y$, $f_0 v = g\partial\eta/\partial x$, obtain an expression for ζ in terms of η.

(b) Linearize (P6.2) about a state of rest, and show that the resulting system supports two-dimensional Rossby waves that are similar to those of the usual two-dimensional barotropic

system. Discuss the limits in which the wavelength is much shorter or much longer than the deformation radius.

(c) Linearize (P6.2) about a *geostrophically balanced state* that is translating uniformly eastwards. Note that this means that:

$$u = U + u' \qquad \eta = \eta(y) + \eta',$$

where $\eta(y)$ is in geostrophic balance with U. Obtain an expression for the form of $\eta(y)$.

(d) Obtain the dispersion relation for Rossby waves in this system. Show that their speed is different from that obtained by adding a constant U to the speed of Rossby waves in part (b), and discuss why this should be so. (That is, why is the problem not *Galilean invariant?*)

6.2 Obtain solutions to the two-layer Rossby wave problem by seeking solutions of the form

$$\psi_1 = \text{Re}\,\widetilde{\psi}_1 e^{i(k_x x + k_y y - \omega t)}, \qquad \psi_2 = \text{Re}\,\widetilde{\psi}_2 e^{i(k_x x + k_y y - \omega t)}. \tag{P6.3}$$

Substitute (P6.3) directly into (6.67) to obtain the dispersion relation, and show that the ensuing two roots correspond to the baroclinic and barotropic modes.

6.3 ◆ (Not difficult, but messy.) Obtain the vertical normal modes and the dispersion relationship of the two-layer quasi-geostrophic problem with a free surface, for which the equations of motion linearized about a state of rest are

$$\frac{\partial}{\partial t}\left[\nabla^2\psi_1 + F_1(\psi_2 - \psi_1)\right] + \beta\frac{\partial \psi_1}{\partial x} = 0 \tag{P6.4a}$$

$$\frac{\partial}{\partial t}\left[\nabla^2\psi_2 + F_2(\psi_1 - \psi_2) - F_{ext}\psi_1\right] + \beta\frac{\partial \psi_2}{\partial x} = 0, \tag{P6.4b}$$

where $F_{ext} = f_0/(gH_2)$.

6.4 Given the baroclinic dispersion relation, $\omega = -\beta k^x/(k^{x2} + k_d^2)$, for what value of k_x is the x-component of the group velocity the largest (i.e., the most positive), and what is the corresponding value of the group velocity?

6.5 ◆ Show that the non-Doppler effect arises using geometric height as the vertical coordinate, using the modified quasi-geostrophic set of White (1977). In particular, obtain the dispersion relation for stratified quasi-geostrophic flow with a resting basic state. Then obtain the dispersion relation for the equations linearized about a uniformly translating state, paying attention to the lower boundary condition, and note the conditions under which the waves are stationary. Discuss.

6.6 *(a)* Obtain the dispersion relationship for Rossby waves in the single-layer quasi-geostrophic potential vorticity equation with linear drag.

(b) Obtain the dispersion relation for Rossby waves in the linearized two-layer potential vorticity equation with linear drag in the lowest layer.

(c) ◆ Obtain the dispersion relation for Rossby waves in the continuously stratified quasi-geostrophic equations, with the effects of linear drag appearing in the thermodynamic equation for the lower boundary condition. That is, the boundary condition at $z = 0$ is $\partial_t(\partial_z\psi) + N^2 w = 0$ where $w = \alpha\zeta$ with α being a constant. You may make the Boussinesq approximation and assume N^2 is constant if you like.

Part III

LARGE-SCALE ATMOSPHERIC CIRCULATION

Part IV

LARGE-SCALE OCEANIC CIRCULATION

It mounts at sea, a concave wall,
Down-ribbed with shine,
And pushes forward, building tall,
Its steep incline.

Thom Gunn, *From the Wave.*

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